



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.13

2 October 2023

Consideration of funding proposals – Addendum XIII

Funding proposal package for FP222

Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled "Renewable Energy Performance Platform (REPP 2)" submitted by Camco Management Limited;
- b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel's assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- f) Gender documentation of the funding proposal.

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled "Renewable Energy Performance Platform (REPP 2)" submitted by Camco Management Limited. is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

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Funding Proposal

Project/Programme title:	Renewable Energy Performance Platform (REPP 2)
Country(ies):	Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sierra Leone, and Zambia
Accredited Entity:	Camco Management Limited
Date of first submission:	2022/10/20
Date of current submission	2023/09/26
Version number	V7.400



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Programme	A.2. Public or private sector	Private	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <u>overall</u> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.			
		GCF contribution	Co-financers' contribution¹	
	Mitigation total	80 %	80 %	
	☒ Energy generation and access	80 %	80 %	
	☐ Low-emission transport	0 %	0 %	
	☐ Buildings, cities, industries and appliances	0 %	0 %	
	☐ Forestry and land use	0 %	0 %	
	Adaptation total	20 %	20 %	
	☒ Most vulnerable people and communities	10 %	10 %	
	☐ Health and well-being, and food and water security	0 %	0 %	
	☒ Infrastructure and built environment	10 %	10 %	
	☐ Ecosystems and ecosystem services	0 %	0 %	
A.5. Expected mitigation outcome² <i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>	12,720,297 tCO ₂ e	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	Total number of direct and indirect beneficiaries: 729,847	
			Number of direct beneficiaries: 370,244	Number of indirect beneficiaries: 359,603
			Direct beneficiaries / total population 0.04%	Indirect beneficiaries / total population 0.03%
A.7. Total financing (GCF + co-finance³)	250 million USD	A.9. Project size	Medium (Up to USD 250 million)	

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

³ Refer to the Policy of Co-financing of the GCF.

A.8. Total GCF funding requested	50 million USD <i>For multi-country proposals, please fill out annex 17.</i>		
A.10. Financial instrument(s) requested for the GCF funding	☐ Grant <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Guarantee <u>Enter number</u> ☒ Equity <u>USD 50 million</u> ☐ Results-based payment <u>Enter number</u>		
A.11. Implementation period	a) Investment period: Seven (7) years; and b) Repayment period, if applicable: Fifteen (15) years, starting in year 5	A.12. Total lifespan	The technology lifetime varies based on technology type: solar home systems have an average lifetime of 5 years, whereas solar mini-grids and isolated grids have an average lifetime of 20 years, and other on-grid renewable energy technologies have lifetimes of 25 years or more.
A.13. Expected date of AE internal approval	7/31/2023	A.14. ESS category	I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	As further set out in Section B4 there are several executing entities including: <u>GCF REPP Holdings SCSp</u>, a <i>société en commandite spéciale</i> to be established in Luxembourg; <u>Camco REPP SCSp</u>, a <i>société en commandite spéciale</i> to be established in Luxembourg; <u>Camco REPP GP S.a.r.l.</u>, a <i>sociétés à responsabilité limitée</i> to be established in Luxembourg; <u>Camco Management Limited</u>, a limited liability company with its register office at 28 St John's Square, London EC1M 4DN, United Kingdom; <u>Camco REPP Holdings One S.a.r.l.</u>, a <i>sociétés à responsabilité limitée</i> to be established in Luxembourg; <u>Camco REPP Holdings Two S.a.r.l.</u>, a <i>sociétés à responsabilité limitée</i> to be established in Luxembourg; and <u>Alternative Investment Fund Manager and Depository to be appointed by Camco REPP SCSp in Luxembourg.</u>		

A.21. Executive summary

1. Background

The Renewable Energy Performance Platform (REPP Phase 1)⁴ is an existing climate finance facility for small to medium-scale grid-connected and distributed renewable energy projects in sub-Saharan Africa. Operational since 2016, REPP Phase 1 is grant-funded through the United Kingdom (UK) Foreign, Commonwealth & Development Office (FCDO). An initial capitalisation of £48m was followed by a further commitment of £40m from the FCDO due to the success of the programme. Camco Management Limited (Camco) is investment manager for REPP Phase 1.

2. Introducing REPP 2

By launching a new Programme entitled **REPP 2**, Camco is seeking to build on the successful track record of REPP Phase 1 (described further in Annex 28) and scale up support for small-scale renewable energy power generation in key African markets, focusing on Least Developed Countries (LDCs).

REPP 2 is a blended finance facility with a goal to create a paradigm shift for the small-scale renewable energy sector in Sub-Saharan Africa that will scale up and replicate funding for low-emission and climate-resilient power generation, with increased access for local and vulnerable populations to modern renewable energy solutions.

REPP 2 targets funding and support to smaller-scale renewable energy projects and businesses in the LDCs of sub-Saharan Africa. To be eligible for receipt of GCF Proceeds, investments must be located in a jurisdiction which has provided a Letter of No Objection in a form acceptable to the GCF (Host Countries). Nine priority countries, seven of which are LDCs, have been selected for GCF funding, including: **Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sierra Leone and Zambia**.⁵

3. Climate change problem

Many countries in sub-Saharan Africa are highly vulnerable to climate change.⁶ Drivers of this vulnerability include:

- Existing climate hazards (droughts, floods, extreme temperatures, etc.);
- High dependence on rain-fed subsistence agriculture for a largely rural population across the region;
- High levels of poverty and food insecurity; and
- Widespread disregard for human rights (including limited access to water, sanitation, education), which often leads to high levels of national and regional conflict.

As a result, the majority of the region's population (particularly those living in rural areas) have very limited capacity to adapt to climate change. One outcome of this is that fewer than half of the population of the region (estimated at 570 million people) has access to electricity. In rural areas, this proportion falls to 25%.⁷

4. Additional barriers

Smaller-scale and off-grid renewable energy companies and investors face numerous barriers, including:

⁴ Renewable Energy Performance Platform (REPP), <https://repp.energy>. The REPP Phase 1 track record and transformational impact is described in Annex 28.

⁵ Cameroon and Nigeria are not Least Developed Countries (<https://unctad.org/topic/least-developed-countries/list>, last accessed 23 September 2022).

⁶ IPCC, 2022: *Climate Change 2022: Impacts, Adaptation, and Vulnerability*. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. In Press.

⁷ IEA, IRENA, UNSD, World Bank, WHO. 2021. Tracking SDG 7: The Energy Progress Report. World Bank, Washington DC. © World Bank. License: Creative Commons Attribution—NonCommercial 3.0 IGO (CC BYNC 3.0 IGO).

- A **lack of early-stage or development funding** to ensure projects or companies become technically and commercially viable;
- **Limited access to finance** for growth and construction of smaller-scale renewables. In addition to a general lack of funding for smaller-scale renewables, commercial⁸ funding remains scarce at both the funding platform and individual asset levels;
- A **limited appetite for investment in new markets**, with generally higher levels of real and perceived risk;
- **Challenging economics for smaller projects and companies**, due to limited generation and revenue potential, which mean that costs are typically large relative to potential returns for investors. This situation has been exacerbated in the aftermath of the Covid pandemic due to delays, supply chain disruptions and cost escalations;
- **The burden and cost of funder requirements**, which add to delays and project costs;
- **Limited or untested domestic policy and regulatory frameworks**, which further limit the bankability of smaller renewable energy projects and the appetite from private sector investors;
- The potential for negative **climate change impacts on national grids** including the potential for reduced power generation from existing assets such as hydropower plants and the potential for increased grid instability as more variable forms of renewable energy generation are introduced;
- **Grid utility political and bankability risk** which further limit public and private investor appetite;
- **Limited experience and funding for local developers** which can result in delays, higher costs and greater potential for project failure;
- **Under-representation of women** in the sector; and
- **Environmental and social barriers**, with the level of environmental and social safeguards and management systems in many markets in sub-Saharan Africa regularly failing to meet international best-practice standards, leading to environmental hazards which often disproportionately affect women and other vulnerable groups.

5. Climate benefits

REPP 2 will deliver climate impact by:

- Financing renewable energy power generation, which contributes to **reduced GHG emissions from renewable energy installations (Outcome 1)** aligned with Host Country priorities;
- Delivering **improved climate resilience of energy infrastructure (Outcome 2)** through diversification and decentralisation of renewable energy generation;
- Delivering **improved climate resilience of vulnerable people and communities (Outcome 3)** through access to clean energy, which improves livelihoods and builds gender equity;
- Achieving **increased private sector investment (Outcome 4)** for renewable energy generation, particular at the level of the underlying assets funded by REPP 2, which benefits local developers, builds gender equity and establishes a more commercially viable market for smaller-scale renewables; and
- Contributing to **stronger technology deployment, innovation and market transformation (Outcome 5)** for smaller-scale and off-grid renewables in all Host Countries.

All of the targeted climate impacts of REPP 2 are closely-aligned to the national priorities of Host Countries for mitigation and adaptation, including those defined in their Nationally Determined Contributions (NDCs) and National Adaptation Plans (NAPs).

⁸ In this Funding Proposal, a distinction is drawn between 'private' and 'commercial' sources of capital. Private sector capital is sourced from private investors on a commercial basis. Commercial funding may be sourced from public entities including Development Finance Institutions, where such entities source or raise capital on their own balance sheets on a commercial basis including directly from private capital markets and therefore provide this funding on a commercial basis. The objective of leveraging both private and commercial capital, whether provided at either the fund level or the level of the underlying assets, is aligned to the GCF's [objectives](#) on mobilization of capital.

6. REPP 2 financing and support

REPP 2 will further catalyse the growth of sub-Saharan Africa's small-scale renewable energy sector by scaling up from the success of REPP Phase 1, further leveraging private sector capital flows into small-scale and distributed renewable energy markets across the region, with a focus on LDCs. The programme is expected to deliver 330MW of new installed capacity, more than 12.7 million tCO₂e (mtCO₂e) of GHG emission reductions and provide new or improved access to clean energy to more than 7.7 million people and improve the resilience of nearly one million beneficiaries over its 20-year total lifespan.

REPP 2 financing and support mechanisms include:

- **Early-stage and development funding**, which includes early-stage and growth funding for off-grid companies (including solar home system and mini-grid companies) to demonstrate that selected technologies are fit-for-purpose, to acquire first customers and to secure inventory and to demonstrate that innovative new business models are viable and sustainable. Additional development-stage funding will be provided for grid-connected projects;
- **Gap funding for construction and implementation of grid-connected projects** to ensure generation and installed capacity targets in Host Countries are achieved. REPP 2 will also facilitate access to and financial risk mitigation to ensure bankability of projects; and
- **Technology deployment, innovation and market transformation**, designed to build skills and capacity amongst local businesses and developers, improve the ability of businesses and developers to deliver gender equity, and improve their ability to manage environmental and social governance ("ESG") risk and impact.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context

1. Climate change problem statement

Climate change hazards

People and communities across sub-Saharan Africa are exposed to a wide and expanding range of climate change hazards.⁹ A summary of key climate hazards for each REPP 2 Host Country is provided below and discussed in greater detail in Annex 2:

- **Cameroon:** Hazards include drought, heavy rainfall events, flooding, heavy winds, high temperatures, and heatwaves. Energy infrastructure is vulnerable to precipitation, extreme rainfall and flooding, as well as drought, and sea-level rise;
- **DRC:** Hazards include intense rainfall, erosion, floods, heatwaves, and seasonal droughts. Reliance on the use of traditional biomass significantly increases the climate vulnerability of the population;
- **Lesotho:** Hazards include more frequent, severe droughts and increased flooding. The energy infrastructure, particularly hydropower, is highly vulnerable to increasing temperatures. Reliance on subsistence agriculture significantly increases the climate vulnerability of the rural population;
- **Madagascar:** Hazards include intensification of cyclones and associated flooding, sea-level rise, intensification of droughts in the south and west, and a more intense wet season elsewhere. The energy sector is vulnerable to a reduction in precipitation and cyclones destroying infrastructure. Over-reliance on the use of firewood further increases the level of vulnerability;
- **Malawi:** Hazards include floods, droughts, strong winds, intense rainfall, riverine and flash floods and tropical cyclones. Vulnerabilities include poverty, rapid population growth, dependence on the agricultural sector reliance on rain-fed agriculture and biomass as well as lack of financing and weak institutional capacity to manage climate change;
- **Niger:** Hazards include higher temperatures, increased frequency and intensity of extreme events, increased frequency of erratic rainfall and extreme droughts. Vulnerabilities include lack of financing, rapid population growth, and low institutional capacity. The energy sector in Niger is expected to be impacted by the effects of heavy rainfall resulting in damage to energy infrastructure;
- **Nigeria:** Hazards include extreme heatwaves and rainfall events leading to droughts, floods, erosion, as well as rise in sea-level leading to intrusion and inundation of land. The energy system is vulnerable to increased demand for cooling and heating and already experiencing multiple supply shortages;
- **Sierra Leone:** Hazards include floods, droughts, coastal erosion, decreasing river flow, storm surges, flash floods, sea-level rise, and high winds. Vulnerabilities include water resources and a high dependence on commodity price; and
- **Zambia:** Main hazards includes floods and wildfires, with droughts also increasingly common. The frequency and intensity of extreme climate events are expected to rise, posing significant risks to Zambia's economy, its people's livelihoods (nearly 60 % of Zambia's people earn less than the international poverty line, and three quarters of the poor live in rural areas), and the sustainability of its natural resources base.

⁹ Trisos, C.H., I.O. Adelekan, E. Totin, A. Ayanlade, J. Efitre, A. Gameda, K. Kalaba, C. Lennard, C. Masao, Y. Mgaya, G. Ngaruiya, D. Olago, N.P. Simpson, and S. Zakieldein, 2022: Africa. In: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 1285–1455, doi:10.1017/9781009325844.011.

Climate change vulnerabilities

The sub-Saharan Africa region as a whole is generally highly vulnerable to climate change.⁶ There are several mutually reinforcing drivers of this vulnerability, including:

- **Exposure to pre-existing climate hazards**, as outlined above;
- **High levels of poverty and food insecurity**: More than two thirds (417 million people in 2017) of the global population living in extreme poverty live in sub-Saharan Africa.¹⁰ Levels of inequality across the region are also pronounced and have widened this century.¹¹ In terms of food insecurity across Africa, 282 million people (20% of the population) faced hunger in 2019.¹² The recent COVID-19 pandemic and war in Ukraine have exacerbated food insecurity across the region;^{13, 14}
- **High dependence on rainfed subsistence agriculture for a largely rural population**: More than 95% of agricultural production in sub-Saharan Africa is rainfed.¹⁵ More than 70% of the region's total population is dependent on agriculture for their livelihoods, the majority of whom are smallholder farmers who rely on subsistence production.¹⁶ The World Bank estimates that 58% of the region's population lived in rural areas as of 2021;¹⁷
- **Energy poverty** is endemic across all of the Host Countries, with fewer than half of the population of sub-Saharan Africa having access to electricity. Of the 760 million people globally, who do not have access to electricity, 568 million (77%) are located in the region and only 25% of people living in rural areas have access;⁷
- **Vulnerability of energy infrastructure to climate hazards and climate change impacts**: Countries which depend on hydropower generation and are exposed to high rainfall seasonality and the occurrence of droughts are vulnerable to climate change. Projected decreases in annual rainfall and increases in the frequency and intensity of severe droughts exacerbate the vulnerability of energy infrastructure. Increased temperature decreased water availability and extreme weather events are anticipated to lead to reduction in plant efficiencies. Given that many national grids are small and poorly-maintain, power generation as well as transmission and distribution infrastructure are often vulnerable to climate extremes;¹⁸
- **Inability to cater to basic human rights** (including limited access to water, sanitation, education);⁶ and
- High levels of **gender-based violence, as well as national and regional conflict**, which may be exacerbated through the impacts of climate change and competition for resources.⁶

As a result, the majority of the region's population, particularly those living in rural areas, have very limited capacity to adapt to climate change.

¹⁰ World Bank. 2020. Poverty and Shared Prosperity 2020: Reversals of Fortune. Washington, DC: World Bank. doi: 10.1596/978-1-4648-1602-4. License: Creative Commons Attribution CC BY 3.0 IGO

¹¹ United Nations Conference on Trade and Development (UNCTAD), 2021: Economic Development in Africa Report 2021: Reaping the Potential Benefits of the African Continental Free Trade Area for Inclusive Growth, United Nations Publications, New York, ISBN: 978-92-1-113004-1.

¹² UN Food and Agriculture Organization (FAO), ECA and AUC. 2020. Africa Regional Overview of Food Security and Nutrition 2019 - In brief. Rome.

¹³ Food and Agricultural Organisation, FAO, 2022: COVID-19 impacts on agrifood systems in Africa: policy priorities for inclusive and resilient recovery, ARC/22/8, March 2022.

¹⁴ United Nations, 2022: Global Impact of war in Ukraine on food, energy and finance systems, Brief No. 1, 13 April 2022.

¹⁵ Abrams, L., 2018: Unlocking the potential of enhanced rainfed agriculture. Report no. 39. Stockholm International Water Institute, Stockholm.

¹⁶ FAO, 2017: [Unlocking agriculture's potential in Southern Africa](#)

¹⁷ <https://data.worldbank.org/indicator/SP.RUR.TOTL.ZS?locations=ZG>

¹⁸ Johnston, P.C., J.F. Gomez and B. Laplante, 2012: Climate Risk and Adaptation in the Electric Power Sector, Asian Development Bank, <http://www.iadb.org/intal/intalcdi/PE/2012/12152.pdf>

2. Climate rationale for mitigation

Despite being amongst the lowest contributors to global GHG emissions, sub-Saharan Africa has already experienced widespread loss and damage due to climate change, with impacts including biodiversity loss, water shortages, reduced food production, reduced economic growth and even loss of lives.⁶ Power generation across most parts of sub-Saharan Africa is still dominated by fossil fuels. In rural areas not connected to the grid, power generation is largely dependent on biomass and diesel generation.

The potential of the REPP 2 Programme to mitigate GHG emissions lies in:

- **Replacement of fossil fuel-intensive grid electricity with renewable energy power generation** from grid-connected utility-scale projects and large mini-grid projects (isolated grids);
- **Replacement of fossil fuel-intensive off-grid electricity with renewable energy power generation**, through large mini-grid projects (isolated grids); and
- **Provision of new or improved access to clean energy for people and communities** that have historically had little or no electricity access, through off-grid solar home systems and solar photovoltaic mini-grid systems (which typically replace candles, kerosene and diesel power generation).

The REPP 2 Programme will avoid and reduce greenhouse gas emissions (MRA 1) and promote a paradigm shift for climate mitigation in each of the Host Countries. The Programme will be aligned with the principles for demonstrating the climate rationale of mitigation activities, as recently clarified by the GCF.¹⁹ Specifically:

- Emission reduction activities supported by REPP 2 will be strongly aligned with priorities defined in the Nationally Determined Contributions (NDCs) of all Host Countries;
- At the time of investment, REPP 2 will determine and assess project impact boundaries, GHG emission baselines using standard emission factors that are aligned to national GHG reporting, projected emissions avoided or reduced, and additionality using internationally-accepted Clean Development Mechanism (CDM) methodologies (as presented in Annex 22.B);
- The same CDM methodologies will be used to establish monitoring protocols; and
- A robust and comprehensive measurement, reporting and verification (MRV) system for GHG emission reductions has been prepared in accordance with international best practise and GCF requirements.

3. Climate rationale for adaptation

The combination of hazards, exposure, vulnerability to current climate variability and existing and limited adaptive capacity due to existing socioeconomic circumstances described above leave people and communities across sub-Saharan Africa highly vulnerable to climate change. The mutually reinforcing vulnerabilities (induced by climate, socio-economic, cultural and political drivers) result in very limited adaptive capacity and make building climate resilience a priority. Climate adaptation initiatives and the need for significantly scaled-up financing for adaptation are thus a priority in all Host Country NDCs and National Adaptation Plans.

Without increasing adaptation capacity, climate change impacts are anticipated to deteriorate in all Host Countries, including decreased food security and further loss of livelihoods. Climate change hazards are also likely to disproportionately affect rural women, given their culturally-defined roles, and other vulnerable groups. The potential of REPP 2 to increase climate change resilience and adaptive capacity lies in:

- The use of **multiple energy sources and technologies to make energy systems more climate resilient** against supply disruption. For example, hybridising hydropower generation (which is often

¹⁹ GCF, 2022: [Steps to enhance the climate rationale of GCF-supported activities](#)

seasonally variable and may be negatively impacted by reduced rainfall due to climate change) with solar or wind generation can provide a buffer against climate variability;

- **Decentralising and diversifying energy supply with renewable energy systems and off-grid solutions**, which can provide increased energy security through the ability to localise and buffer grid-based supply disruptions;
- **Increasing access to renewable energy in off-grid communities** to provide households and businesses with solutions that help to build resilience and respond to the impacts of climate change. Such solutions include electricity provision for clinics to respond to increased illness, malnutrition, and injury; electricity provision to refugee camps that are accommodating displaced people; electricity provision to educational facilities to improve awareness and knowledge about climate change; improved access to climate information through cellular networks, radios, and internet enabling people and communities to benefit from access to climate early-warning systems for disaster risk reduction and seasonal climate information for planning and decision-making for management of crops and livestock; providing power for increased cooling need; irrigation to increase agricultural productive output; e-mobility to provide increased access to markets as well as providing improved cooking solutions to households;
- **Supporting local private developers** and helping them to become financial sustainable, thereby making the local renewable energy sector (as well as local economies more broadly) more resilient to macroeconomic shocks; and
- **Supporting technological innovations** that build the capacity of households and communities to anticipate and respond to climate impacts, including innovations for the delivery of climate services (e.g., access to mobile phones and internet improves access to meteorological and agrometeorological reports and early warning systems), market information, and other information on climate change, its impacts and the solutions; as well as technological innovations for productive use applications.

Please see Tables 1 and 2 below which present REPP 2 eligible adaptation responses. Detailed country level data is provided in Annex 2.

The REPP 2 Programme responds directly to the urgent and immediate needs of vulnerable countries, as outlined in their NDCs. The Programme focuses on nine Host Countries in sub-Saharan Africa, of which seven are LDCs. The potential of REPP 2 to assist these countries to adapt to the impacts of climate change in the context of promoting sustainable development and a paradigm shift will be confirmed through the following:

- **Identification:** People, communities and energy infrastructure in all Host Countries and across the sub-Saharan African region are highly vulnerable to climate change (see below eligibility criteria);
- **Response:** The Programme will target adaptation impacts through increasing the resilience of the most vulnerable people and communities (ARA1), in addition to energy infrastructure and the built environment (ARA3);
- **Alignment:** REPP 2 will place a strong focus on aligning its investments with national-level climate, energy and sustainable development priorities. The proposed climate adaptation impacts of the Programme (improved resilience of energy infrastructure and local populations) are closely aligned with and strongly support national plans and climate strategies of the Host Countries. This is ensured through document analysis and extensive engagement with key governmental and non-governmental stakeholders, and confirmed through securing Non-Objection Letters from all Host Countries; and
- **Monitoring and evaluation:** A comprehensive and detailed monitoring and evaluation framework has been developed for the Programme as part of the Environmental and Social Management Framework (ESMF) to demonstrate its adaptation impact over time, as is expected to be approved by the general partner of Camco REPP SCSp.

Eligibility criteria for programmatic adaptation projects

The REPP 2 Programme is cross-cutting, targeting a 20% overall contribution to climate change adaptation and 80% contribution to mitigation. Interventions may be classified as contributing to climate change adaptation where they can be demonstrated to be a response to specific impacts and are expected to deliver improved adaptive capacity or climate resilience. The contribution towards the 20% target for adaptation will be based only on those funded interventions which are classified as an adaptation response and may be less than the total funding provided to that investment. In all cases, classification of adaptation responses will be undertaken through application of Camco's Adaptation Decision and Project Tool (ADAPT), and guidance on the eligibility of the investment as an adaptation response will be considered by the REPP 2 Investment Committee. The ADAPT tool will be implemented according to the following high-level decision process:

- **STEP 1: Identify climate hazards and climate change impacts**
- **STEP 2: Define adaptation response**
- **STEP 3: Identify adaptation metrics and monitoring, reporting and evaluation framework**

The ADAPT tool is integrated within the existing analytical framework which drives eligibility, selection and approval of candidate investments for categorisation as a climate change adaptation response. The ADAPT tool will provide analytical guidelines, including clear exclusion criteria and minimum criteria for acceptability. Assessments will be qualitative, rather than via a score, requiring the analyst to provide a rationale to support an investment's eligibility as an adaptation response as well as the nature and extent of that contribution. Metrics, targets and monitoring and evaluation arrangements must be considered as part of the evaluation. Approval for the classification of an investment as cross-cutting, along with the climate change adaptation targets, will be provided by the REPP 2 Investment Committee.

Further details of the process for classifying and investment opportunity as an adaptation response are provided in Annex 24. Examples of the application of the ADAPT tool, based on an ex-post assessment of three case studies from the REPP Phase 1 portfolio are shown in Annex 27.

Most African countries (as illustrated in Annex 2) have low electrification rates, inefficient grid infrastructure and generally rely heavily on hydropower generation (with additional dependence on diesel generation for sites not connected to the grid). As a baseline for the REPP 2 Programme, eligible adaptation responses for improving the climate resilience of existing generation infrastructure are presented in Table 1 (after Johnston et al., 2012²⁰ and World Meteorological Organisation, 2017²¹). Similarly, eligible adaptation responses with improved climate resilience for people and communities are shown in Table 2. Further detail is provided in the Market Analysis (Annex 2) for each Host Country.

Table 1: Adaptation responses for improving climate resilience of energy infrastructure

Energy subsector (baseline)	Climate hazards	Climate change impacts	Eligible adaptation responses	Renewable energy technology	Metric monitored
Hydropower generation	Increasing temperatures and evaporative losses	Reduction in available generation capacity or output	Increase decentralised renewable energy generation	Grid-connected solar PV and wind	Number of hydropower plants hybridised using renewable

²⁰ Johnston, P.C., J.F. Gomez and B. Laplante, 2012: Climate Risk and Adaptation in the Electric Power Sector, Asian Development Bank, <http://www.iadb.org/intal/intalcdi/PE/2012/12152.pdf>

²¹ World Meteorological Organisation, 2017: Energy Exemplar to the User Interface Platform of the Global Framework for Climate Services, Geneva, Switzerland, https://library.wmo.int/doc_num.php?explnum_id=3581

	Changes in mean annual rainfall Changes in rainfall seasonality Changes in frequency and / or intensity of droughts Increasing intensity and frequency of flooding	Changes in operating conditions Increased risk of physical damage to infrastructure	Hybridisation of hydropower generation capacity using renewable energy technology		energy generation USD invested in hybridisation
Transmission and distribution grids	Increasing air temperatures Increasing frequency and/ or intensity of extremes (temperature or wind) Increasing frequency and severity of wildfires Increasing intensity of storm events leading to flooding	Reduction in transmission efficiency and available transmission capacity Increased risk of physical damage and decreased transmission capacity due to storm damage	Increase decentralised renewable energy generation to buffer against disruption of power supply	Solar, wind and hydro: mini-grid and isolated-grids	USD invested which clearly demonstrate increase in resilience of energy system

Table 2: Adaptation responses for improving climate resilience of people and communities.

Energy subsector (baseline)	Climate hazards	Climate change impacts	Eligible adaptation responses	Renewable energy technology	Metric monitored
Consumer energy demand	Increasing air temperatures Reduced rainfall / increase in droughts Increased rainfall / increase in floods Increasing frequency and/ or intensity of	Increased electricity demand for cooling. Increased risk of crop damage and loss of production leading to loss of income/ food insecurity Increased malnutrition, disease, injury	Provide renewable energy power generation for: - Refrigeration and cold storage - Irrigation - Mobility (increased access to markets) - Critical care services (clinics & hospitals)	Off grid solutions using solar, wind and hydropower: mini-grid and isolated-grids, SHS solutions	Number of customers in each subcategory

	extreme weather events Sea-level rise	and death leading to increased need for critical services. Increased need for awareness raising and education. Increased need for climate and weather information. Increase amount of climate-induced migration	• Critical community services (fire & police stations) • Educational facilities • Information systems: phone, TV, radio, cell phone towers, internet • Refugee camps international agency special missions		
Biomass	Increasing air temperatures Decreasing water availability Sea-level rise Increasing intensity and / or frequency of flooding	Reduces biomass availability for cooking.	Provide renewable energy power generation for households to support the use of improved cooking solutions (such as pressure cookers, heat retaining solutions)	Off grid solutions using solar, wind and hydropower: mini-grid and isolated-grids, SHS solutions	Number of customers with improved cooking solutions

4. Business as usual scenario

Given the paucity of funding (particularly in LDCs) for smaller-scale and off-grid renewables, and the catalytic role that GCF funding will play in mobilising funding at both the portfolio and project level, there is high confidence that the projected level of GHG emission reductions will not be achieved without GCF funding.

5. Related projects and interventions

Despite the successes of earlier programmes such as REPP Phase 1, key and persistent gaps in the funding landscape for smaller-scale renewables remain. This finding is reinforced after a detailed analysis of the potential for overlap and complementarity between REPP 2 and other financing programmes, including programmes funded by the GCF. Key gaps that REPP 2 will seek to address in nascent markets across sub-Saharan Africa, including LDCs, include:

- An almost total lack of equity funding available across the spectrum for companies that need development capital for projects;
- A very limited supply of growth equity for off-grid companies; and
- Very limited investment in smaller-scale renewable energy projects in least developed markets, where real and perceived risks remain high.

From a country perspective, investment is lacking in the places that need it most as exemplified by the distinct lack of funding in LDCs, even for more mature technologies, such as utility-scale solar PV. Similarly, there is also a significant and persistent funding gap for companies and projects that use innovative

technologies, focus on decentralised projects, which can improve climate resilience of energy infrastructure, or where some level of business model risk exists, even in middle- and upper middle-income countries.

Of the 32 projects approved by the GCF that focus on energy generation and access in Africa, there is relatively little direct overlap with REPP 2.¹³ Where overlap does exist, there are extensive opportunities for complementarity, particularly because of the focus of REPP 2 on early-stage funding to fill gaps not addressed by other programmes. The potential for overlap and complementarity of the REPP 2 Programme is analysed in further detail in Annex 23.

An update on the status of co-financing for REPP 2 is provided in Annex 25. A summary of the REPP 2 pipeline is provided in Annex 26. The rationale for selection of Host Countries is provided in Section B.3.

B.2 (a). Theory of change narrative and diagram

1. Goal statement

REPP 2 will aim to create a paradigm shift by leveraging additional private sector funding to promote investments in small-scale renewable energy projects in LDCs and other sub-Saharan African countries. As result of this, local populations of some of the world's most vulnerable communities will have improved access to clean energy, improved access to climate resilient solutions and information, and the Host Countries will transition towards low-emission resilient development pathways, meeting their NDC targets. Accordingly, with increased investment, the GHG emissions from electricity generation will be reduced, climate resilience of the energy infrastructure is enhanced, and climate resilience of vulnerable people and communities will be improved.

2. Paradigm shift potential

If the proposed programme receives GCF funding and leverages further private sector and commercial finance, then local populations of some of the world's most vulnerable communities will have improved access to clean energy, improved access to climate resilient solutions and information, and Host Countries will transition towards low-emission development pathways, meeting their NDC targets. Because with the increased investment the GHG emissions from electricity generation will be reduced, climate resilience of energy infrastructure will be enhanced, and climate resilience of vulnerable people and communities will be improved.

The REPP 2 activities are anticipated to lead to more skilled and experienced local businesses and project developers, which in turn will develop with the help of REPP 2 funding more financially sustainable, innovative, environmentally and socially sound climate resilient renewable energy projects. In this way, by building local developer capacities, as well as de-risking and innovating financial solutions which enable renewable energy development, REPP 2 will create a higher volume of projects that will catalyse further private sector finance, contributing to stronger business environment and leading to more technology transfer.

The resulting increases in renewable energy capacity will lead to cleaner production of energy, reduced GHG emissions, and greater energy security due to hybridisation of energy systems that are vulnerable to climate hazards, and by decentralising energy systems, transmission and distribution which is likely to be impacted by the increased risk of physical damage and distribution. REPP 2 investment is anticipated to lead to improved resilience to disasters through an improved ability to provide localised power solutions and to buffer supply disruptions through decentralisation.

Furthermore, project developers are anticipated to provide electricity not only for off-grid household basic electricity use, but also to critical services (e.g. clinics, water pumping stations), cooling services (e.g. refrigeration), information services (e.g. cell phone towers, radio) and productive use of energy opportunities

which increase climate resilience (e.g. irrigation). These opportunities are expected to lead to additional income generation opportunities and especially in off-grid project communities works as a driver for long-term economic growth. Off-grid renewable energy projects also often deliver first-time access to electricity for marginalised and vulnerable communities (including women-headed households) and businesses, placing them on a low-emission, climate-resilient, gender-sensitive and sustainable growth trajectory

3. Activities and outputs delivered by REPP 2

The activities and outputs delivered by REPP 2 are designed to strengthen the market for small-scale renewables in sub-Saharan Africa and crowd in private and commercial capital leading to GHG mitigation and improved resilience of people and energy infrastructure. This is achieved through an integrated suite of interventions, including:

Early-stage development funding

Activity 1: Provide early-stage growth funding for off-grid companies. REPP 2 will fund the early-stage growth and development of off-grid companies, notably solar home system businesses and mini-grid developers with a gender lens. This means funding of feasibility studies, ESIA's, design and implementation of ESMS, design of gender action plans, as well as funding to acquire first customers and secure inventory.

Activity 2: Provide development-stage funding for grid-connected projects. REPP 2 will provide loans to developers of grid-connected projects to co-finance development-stage costs, such as feasibility studies, grid studies, ESIA's, design and implementation of ESMS, legal due diligence, and gender action plans.

All funding activities will be considered from a gender lens perspective, with the view to target that a minimum 30 % of REPP 2 funding is invested in women owned and led companies, as well as companies that provide solutions to women as energy end users in line with the 2X GLI criteria.

The two early-stage development funding activities delivered by REPP 2 will ensure increased technical and commercial viability of small-scale renewable energy projects (**Output 1**), leading to an increase in renewable energy capacity and generation (**Output 2**) as more projects reach financial close, are constructed and become operational due to REPP 2 financing. Furthermore, the funding of environmental and social studies (incl. gender) and management plans will help the renewable energy project developers to understand and better identify risks and impacts and to establish comprehensive management plans to mitigate and minimise such risks and to deliver environmentally and socially sound projects and ensure gender equity through increased awareness and action (**Output 3**).

Activity 3: Finance expansion of SHS businesses. REPP 2 will provide equity or loans to fund scale up of solar home system businesses with a gender lens. This growth-stage capital, in the form of loans or equity will help to ensure that SHS business achieve scale and long-term commercial viability (**Output 1**) and will increase the renewable energy capacity and generation (**Output 2**). All funding activities will be considered from a gender lens, with the view to target that 30% of REPP 2 funding is provided to women owned and led companies, as well as companies that provide solutions to women as energy end users in line with the 2X GLI criteria. This will help to promote women's empowerment and female ownership in REPP investments leading to gender equity (**Output 3**).

Activity 4: Finance construction of mini-grid portfolios. REPP 2 will provide financing for construction of mini-grids. The mini-grids funded can be in both peri-urban and rural areas as well as for isolated grid projects, which are typically large installations (servicing entire towns and tens or hundreds of thousands of beneficiaries) which are isolated from the national grid. This construction finance from REPP 2, which can be provided in the form of loans or equity, will demonstrate that more small-scale renewable energy projects can achieve technical and commercial viability (**Output 1**) and ensure increased renewable energy capacity when more mini-grids and isolated grids are constructed and become operational (**Output 2**). REPP 2 will

be aligned with 2X GLI criteria, which means that it will target 30 % of funding is provided to women owned and led companies, as well as companies that provide solutions to women as energy end users to deliver greater gender equity (**Output 3**).

Gap funding for construction and implementation of grid-connected projects

Activity 5: Provide gap funding for construction of grid-connected projects. REPP 2 will provide gap financing for the construction of on-grid projects with a gender lens. Construction loans from REPP 2 will demonstrate that more grid-connected small-scale renewable energy projects can achieve technical and commercial viability (**Output 1**) and are constructed and become operational leading to increased renewable energy capacity (**Output 2**). The modalities of providing such funding will include innovative and less traditional financing approaches to reduce cost and complexity for developers and sponsors in bringing their projects to financial close. This approach is necessary given the limited capacity of many first-time and/or local project developers when combined with the small scale of REPP 2 projects. In practice this requires Camco to balance the “belts and braces” approach to risk in project finance structures with the barriers to financial close that already prevent the deployment of small-scale renewables. For example, REPP 2 will consider taking a more limited set of direct agreements, proceeding with a limited or even no government guarantee, accepting a scaled-down security package and/or considering a more balanced covenant package. However, it should be noted that Camco will nevertheless ensure that material risks are addressed in documentation in a manner that protects the interests of REPP 2 and its investors. REPP 2 will also provide other type of financing which are needed in the market, such as mezzanine finance, bridge/construction finance, and convertible notes. REPP 2’s attractiveness in the market stems from its flexible approach and the ability to provide the right financing from the large tool box of instruments. As mentioned above, all funding activities will be considered from a gender lens, which means that a portion of REPP 2 funding (a target of 30%) will be provided to women-owned and led companies, as well as companies that provide solutions to women as energy end users in line with the 2X GLI criteria. This will help to promote women’s empowerment and female ownership in REPP investments leading to gender equity (**Output 3**).

Activity 6: Support the provision of financial risk mitigation to ensure appropriate and cost-effective risk mitigation. REPP 2 will innovate in terms of its approach to funding and mitigating risk in SHS businesses (**Activity 3**) mini-grid portfolios (**Activity 4**). This may include additional sources of local currency funding and designing new risk mitigation strategies, including minimum revenue guarantees (in partnership with key risk mitigation providers in the region), to de-risk these investments and attract more commercial and private sector funding into these sectors. In addition, REPP 2 will be promote the provision of and access to connections subsidies in the markets in which it invests, and will encourage its investees to secure such subsidies. For individual transactions REPP 2 will employ a comprehensive risk management framework to ensure that cost-effective financial, technical and political risk mitigation solutions can be offered for projects.

By de-risking investments in smaller-scale renewables, REPP 2 can ensure that more projects become commercially viable (**Output 1**), leading to increased renewable energy capacity (**Output 2**).

Technology deployment, innovation and market transformation

Activity 7: Provide targeted support to developers. REPP 2 will provide targeted support and capacity building to developers and entrepreneurs as part of the investment process. Tailored support and capacity building will be provided for improving financial models, business plans and risk mitigation as well as projects’ environmental and social management. This support will improve skills and expertise of project developers and off-grid companies in the small-scale renewables sector across sub-all target markets, thus supporting market transformation (**Output 3**).

Activity 8: Improve environmental and social risk mitigation. REPP 2 will provide tailored support and capacity building to its investees on environmental and social issues and require all investees to develop

and implement an appropriate environmental and social management system (ESMS). Implementation and maintenance of suitable environmental and social safeguards in accordance with industry best practice will improve skills and capacity for local developers and off-grid companies and improve their ESG practices, making them ready to take on further investment and scale their operations in the future. **(Output 3).**

Activity 9: Non-financial support to promote gender equity. In addition to the investments which will be aligned with the 2X GLI criteria, REPP 2 investee companies will be required to establish gender action plans which will lead to **co-benefits** in terms of more gender balanced job creation, more equitable workplaces, as well as more gender balanced stakeholder and community engagement and land acquisition. Further detail on the gender-disaggregated impacts and indicators are provided in Section D.1 and Annex 8.

Further details on all activities are provided in Section B.3 (item 2, REPP 2 activities).

4. Key barriers and risks for climate outcomes

The following key barriers and risks which may prevent the realisation of REPP 2 outcomes have been identified:

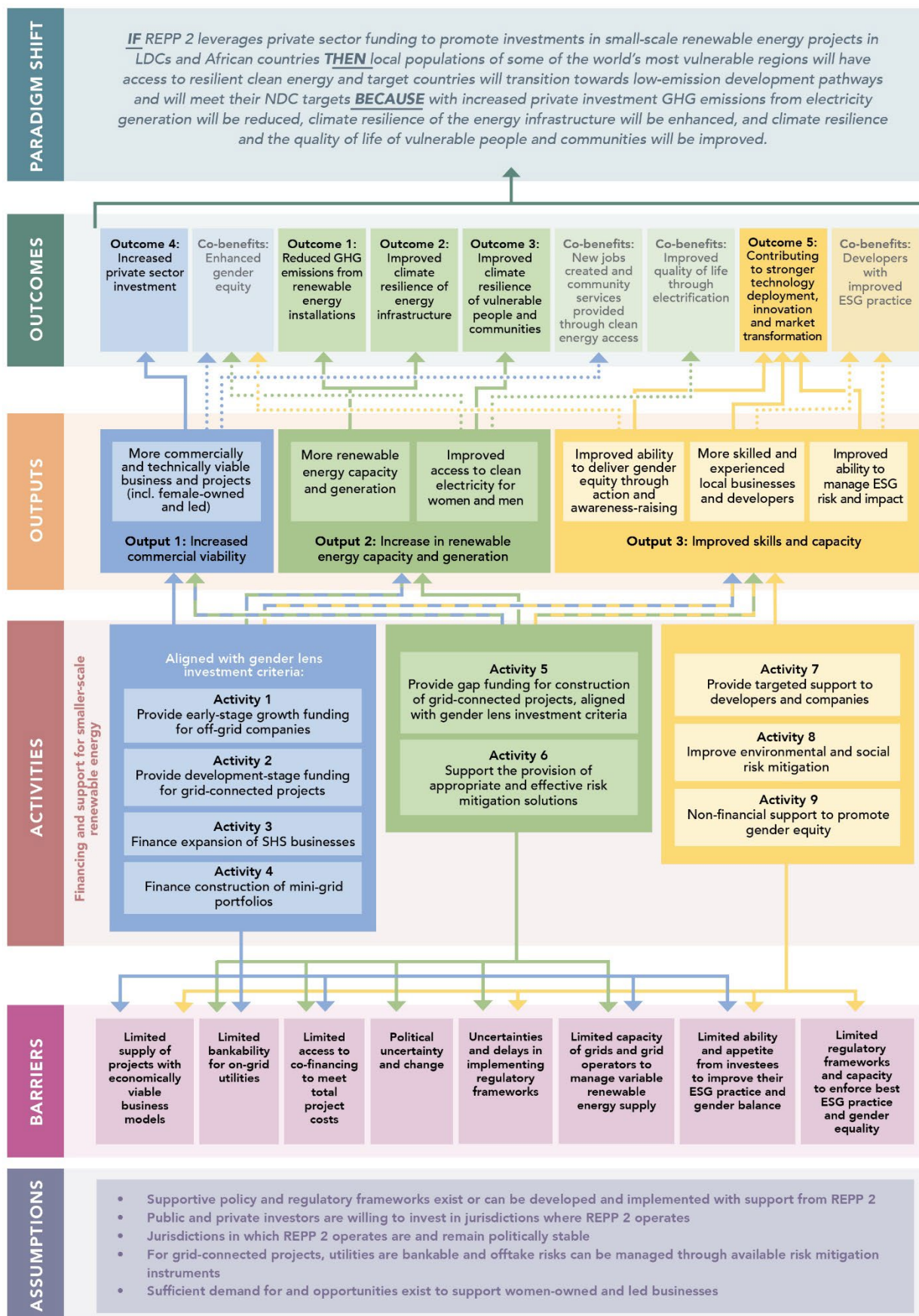
- **Uncertainties and delays in implementing policy and regulatory frameworks** and/or limited experience amongst public sector stakeholders with implementation of new or untested regulatory frameworks and project finance documentation can result in delays in project implementation, additional costs, devolve private sector involvement and slow down or prohibit sector growth.
- **Political uncertainty and change** which may have unexpected influence on projects' financial viability due to change in IPP approval process, grid extension planning, feed in tariffs and mini-grid tariff setting, land-use, currency risk and legal and documentation risk. This may hinder projects from progressing, reaching financial close or receiving anticipated returns.
- **Limited bankability for on-grid utilities** in Host Countries in which the national grid utility is not regarded as creditworthy. This will increase real and perceived risk, which may limit co-founder's appetite in the smaller renewables sector and hinder a project reaching financial close.
- **Limited access to finance** at both the fund level and the level of underlying projects. At the fund level, there is limited appetite from investors (public and private) for smaller-scale renewables. Commercial investors (both public and private) typically seek risk mitigation through the capital structure of a fund, requiring junior investors or guarantee providers to provide first loss protection. Asset-level financing is usually available at financial close for "bankable" projects or off-grid companies of sufficient scale. Smaller ticket sizes may fall below the appetite of public funders such as development finance institutions (DFIs), infrastructure funds as well as private and commercial investors, which means that such investments face the risk of not achieving financial close. As a consequence, REPP 2 may struggle to attract co-financing, particularly private and commercial co-financing, and may also struggle to leverage third-party financing into individual assets that it funds.
- **Limited supply of projects with economically viable business** which provide tailored energy services and innovative financing models for resilient productive uses of electricity and last mile customers. Smaller project sizes limit generation and revenue potential. Costs of development and growth to scale are typically large relative to this potential, resulting in more marginal return expectations. Higher levels of macroeconomic risk have emerged, due to high inflation-low growth conditions in local economies which have followed the Covid- pandemic, as well as higher foreign exchange and balance of payment risks as interest rates are raised to combat inflation and protect growth.
- **Weak/underdeveloped legal frameworks and capacity to enforce best ESG practice and gender equality.** The level of environmental and social safeguards, their operationalisation and monitoring in many markets in sub-Saharan Africa regularly fail to meet international practice standards on pollution prevention, labour rights and occupational health and safety, as well as gender balanced consultation and implementation. This may create an environment where investee is not motivated to uphold the required standards.

- **Limited ability and appetite from investees to improve their ESG practice and gender balance.** Investees may consider this out of their core focus, time-consuming and costly without understanding the importance for risk mitigation and benefit of creating sustainable projects and businesses. Furthermore, amongst local communities, women are often constrained by gender-defined cultural roles and the rights of women – notably with respect to land ownership – are frequently not respected, which is likely to increase the complexity of implementing gender balanced engagement.
- **Limited capacity of grids and grid operators** to manage variable renewable energy supply which may arise due to lack of grid connectivity between high renewable resource regions to high power demand centers, grid flexibility or deficient assessment of available resources. This may lead to increased project costs due as grid infrastructure needs to be upgraded to enable project to feed in.

The following external assumptions, which cannot be controlled by REPP 2, have been made for a successful implementation of REPP 2 mandate:

1. Sufficient supportive policy and regulatory frameworks exist or can be developed and there is political will to do so;
2. Public and private investors are willing to invest in jurisdictions where REPP 2 operates;
3. Jurisdictions in which REPP 2 operates are and remain politically sufficiently stable for renewable energy development;
4. For grid-connected projects, utilities are bankable and offtake risks can be managed through available risk mitigation instruments; and
5. Sufficient demand for and opportunities exist to support women-owned and led businesses.

The theory of change for REPP 2 is summarised in the diagram below.



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1- 4)				GCF Adaptation Results Area (ARA 1- 4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1: Reduced GHG emissions from renewable energy installations	☒	☐	☐	☐	☐	☐	☐	☐
Outcome 2: Improved climate resilience of energy infrastructure	☐	☐	☐	☐	☐	☐	☒	☐
Outcome 3: Improved climate resilience for vulnerable people and communities"	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 4: Increased private sector investment	☒	☐	☐	☐	☐	☐	☐	☐
Outcome 5: Contributing to stronger technology deployment, innovation and market transformation	☒	☐	☐	☐	☐	☐	☒	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Jobs created	☐	☒	☐	☐	☐	☐

Co-benefit 2: Enhanced gender equity	☐	☒	☐	☒	☐	☐
Co-benefit 3: Improved quality of life	☐	☒	☐	☐	☐	☐
Co-benefit 4: Improved ESG practices	☒	☒	☐	☐	☐	☐

B.3. Project/programme description

1. The REPP 2 Programme

REPP 2 will finance and promote the growth of small to medium-sized renewable energy projects across sub-Saharan Africa. Through this process, REPP 2 will catalyse the growth of sub-Saharan Africa's smaller-scale renewable energy sector, mobilising additional public, private and commercial funding to combat climate change and provide build climate resilience for people, communities and energy infrastructure. Building on the success of REPP Phase 1, REPP 2 will focus on support in nascent renewable energy markets, including eight LDCs. The programme is expected to deliver 330MW of new installed capacity, reduce or avoid more than 12.7m tCO₂e of GHG emission, provide new or improved energy access to more than 7.7 million people, improve the quality of life of 3.7 million people and improve the resilience of nearly one million beneficiaries.

In addition to reducing GHG emissions through renewables, REPP 2 will deliver climate adaptation impacts and build climate resilience by providing energy access to vulnerable people and communities, as well as diversifying and decentralising renewable energy infrastructure across sub-Saharan Africa. All of REPP 2's targeted climate impacts are closely aligned with the national priorities of Host Countries for mitigation and adaptation as defined in their NDCs, NAPs and other policy and strategic frameworks.

REPP 2 will fund both on-grid and off-grid renewable energy projects and companies, including isolated grid developments in which renewable energy generation is used as the source of power for entire towns and small cities. REPP 2 investments will have significant transformational potential, with opportunities to provide clean, often first-time affordable power to tens of thousands of beneficiaries through one installation alone. REPP 2 will build on the success and learning from REPP Phase 1 to achieve greater scale and mobilise additional public and private funding into the smaller renewables sector. The Programme will focus on LDCs, where the funding and capacity gaps are greatest.

The REPP 2 strategy is to take a bottom-up approach, working directly with private sector developers and companies to help develop a viable renewable energy sector and a vibrant financing market for smaller renewable energy projects. Furthermore, REPP 2 seeks to be:

- **Additional**, in that it stimulates sectoral development that would otherwise lag or not occur; and
- **Transformational**, in that it helps to establish or build a market ecosystem that allows the ongoing growth of small-scale and distributed renewables in the region.

REPP 2 will deliver on this mandate through providing funding and additional forms of support to developers designed specifically to overcome the barriers identified in Section B.1.

2. REPP 2 activities

All REPP 2 activities fall within a single component, designed to provide funding and support for smaller-scale renewable energy businesses and projects across sub-Saharan Africa. The activities delivered by REPP 2 are designed to strengthen the market for small-scale renewables in sub-Saharan Africa and crowd in private capital leading to GHG mitigation and improved resilience of people and energy infrastructure.

The activities outlined below describe the provision of:

- Early-stage growth funding for off-grid companies and development funding for grid-connected projects;
- Gap financing for growth of SHS businesses as well as construction of mini-grids, isolated grids and grid-connected projects; and
- Support for technology deployment, innovation and market transformation

Early-stage development and growth funding

A range of funding activities for REPP 2 is described below. For all equity investments, REPP 2 will typically take a minority shareholding in accordance with its Investment Policy. REPP 2 will act as an anchor investor and seek to crowd in other public and private sector investors to meet total project capital requirements. For debt investments, REPP 2 will typically require a minimum of 30% contribution from the investee.

Activity 1: Provide early-stage growth funding for off-grid companies

REPP 2 will provide growth capital to off-grid electrification companies, many of whom are small, locally-owned businesses. Growth capital for off-grid companies will be provided in the form of hybrid equity instruments (highly subordinated debt or senior ranked equity, typically convertible notes or redeemable shares), as well as subordinated and senior loans.

Identified needs for SHS businesses include smaller equity investments to demonstrate that selected technologies are fit-for-purpose and the necessity to establish innovative new business models, acquire first customers and secure inventory.

Mini-grid developers require smaller equity investments to demonstrate that selected technologies are fit-for-purpose, as well as the need to establish innovative new business models, acquire first customers and to meet project development expenses.

This funding will help these businesses to achieve scale and sustainability and overcome key barriers, such as a lack of early-stage or development funding, limited access to finance and challenging economics for smaller projects.

Activity 2: Provide development-stage funding for grid-connected projects

REPP 2 will provide development-stage funding in the form of hybrid equity instruments that are typically either repaid at financial close or include an option for REPP 2 to convert the loan into funding for construction at financial close. This funding approach directly addresses the key barriers experienced by developers of smaller-scale projects which are the lack of early-stage or development funding, and limited access to finance to develop bankable projects.

The simplified development loan terms offered by REPP 2 will also help to reduce barriers related to challenging economics for smaller projects because the costs of development are typically large relative to the potential of smaller projects to generate revenues, resulting in more marginal return expectations for developers, sponsors and commercial lenders.

The simplified terms and conditions for REPP 2 development loans will also help to overcome the burden and cost of funder requirements. In accordance with the risk appetite of REPP 2, the development loans it offers will be structured to assume equity-like risk during the development phase and will be priced to ensure

commensurate returns if the development is successful. This structure is generally faster to execute, allowing more rapid deployment of capital for development than standard equity investments.

Development funding for grid-connected projects will be used as a contribution towards the costs of project development (including design and permitting, technical feasibility studies, environmental and social impact assessments, gender actions plans, etc.) and due diligence (including legal fees and other third-party costs) as the projects approach financial close.

Activity 3: Finance expansion of SHS businesses

REPP 2 will provide funding for the expansion of SHS businesses in the form of equity (ordinary and preference shares) and debt. Where REPP 2 provides debt, this will be in the form of mezzanine and senior loans. This funding must be used to scale these businesses to become fully sustainable once their business model is proven. Funding for SHS businesses will be provided in the form of hybrid equity instruments, as well as subordinated and senior loans.

This activity addresses barriers to growth for SHS businesses, including a lack of early-stage or development funding, limited access to finance for growth, and a limited appetite for investment in new markets.

Activity 4: Finance construction of mini-grid portfolios

REPP 2 may fund construction of mini-grid portfolios and isolated grid developments by way of hybrid equity instruments, as well as subordinated and senior loans. This funding will be used for construction and to achieve scale and sustainability once the mini-grid business model is proven.

For mini-grid developers, this activity will help to overcome barriers which include a lack of early-stage or development funding, limited access to finance for growth, and a limited appetite for investment in new markets. Mini-grid developers face additional barriers related to limited or untested domestic policy and regulatory frameworks where regulatory frameworks are new and regulated tariffs are not cost-reflective.

Activities 1-4 all support Output 1 (Increased Commercial Viability) of off-grid businesses and grid-connected projects. Output 1 is described in detail in item 3 below.

Funding for the expansion of SHS businesses (Activity 3) and construction of mini-grid portfolios (Activity 4) also help to deliver improved quality of life for people and communities provided with first-time electricity access for productive use of electricity and critical services (Output 2), described in further detail in item 3, below.

Gap financing for construction and implementation of grid-connected projects

Activity 5: Provide gap funding for construction of grid-connected projects

REPP 2 will seek to crowd in private and public sources of funding and provide additional (gap) funding to fill a confirmed gap in the capital structure and achieve financial close. REPP 2 may provide funding via hybrid equity instruments, subordinated or senior loans. Loans will have a targeted weighted average tenor for the portfolio of 10 years and may also fund on concessional terms based on the needs of each project.

This funding will help to overcome key barriers, including limited access to finance, by providing access to funding for construction for smaller projects, where investment ticket sizes often fall below the appetite of development finance institutions (DFIs), infrastructure funds as well as private and commercial lenders. REPP 2 funding for construction will also help overcome challenging economics for smaller projects and companies, where the smaller project size limits generation capacity and thus revenue and return expectations. Simplified and streamline funding conditions will also help to overcome the burden and cost of funder requirements.

Activity 1-5: To enhance gender equality within renewable energy sector REPP 2 will target 30% of all funding towards companies that meet 2X GLI criteria. These are companies that:

- Are founded by a woman or women or are majority-owned by women; and / or

- Have at least 30% women in senior management; and / or
- Provide products or services designed to disproportionately benefit women; and / or
- Have a workforce that is comprised of at least 30% women, together with a progressive human resource policy.

Activity 6: Support the provision of financial risk mitigation

In nascent markets, developers of grid-connected projects face additional barriers related to limited or untested domestic policy and regulatory frameworks and need to secure power purchase agreements with grid utilities and operators with limited creditworthiness and experience of working with independent power produced and variance energy generation from renewable energy sources such as solar PV (examples of grid utility political and bankability risk).

REPP 2 will seek to ensure appropriate and cost-effective risk mitigation for off-grid business, as well as developers and sponsors of grid-connected projects. In doing so, REPP 2 seeks to reduce generally higher levels of real and perceived risk, particularly when developing smaller renewable energy projects in nascent and LDC markets. Elements of a more innovative approach to mitigating real and perceived risk and attracting more commercial and private sector funding include:

- Attracting additional local currency funding for SHS businesses (Activity 3), mini-grid portfolios (Activity 4) and grid-connected projects (Activity 5). To reduce currency risk for businesses and funders alike, REPP 2 will work with additional sources of commercial as well as public local currency funders (in country as well as regional), to co-finance the roll-out of mini-grid portfolios and expansion of SHS business.
- Securing innovative risk mitigation strategies for mini-grid portfolios, including minimum revenue guarantees. This new form of risk mitigation is being considered in the market but is not yet widely available. REPP 2 will work with key risk mitigation providers for the region, to de-risk these investments and attract more commercial and private sector funding into this sector. REPP 2 will also promote the provision of and access to connections subsidies.
- Employing innovative and less traditional project finance approaches to reduce cost and complexity for developers and sponsors to bring their projects to financial close. This approach may be implement for investments in both off-grid and grid-connected opportunities. Examples (often already utilised in REPP Phase 1) include growth capital for off-grid companies (Activity 1) in the form of hybrid equity instruments, asset finance loans with loan principal growing in line with the asset base of the company, development-stage funding via hybrid equity instruments, typically either repaid at financial close or including an option for REPP 2 to convert the loan into funding for construction at financial close (Activity 2) and use of hybrid equity instruments to fund investments in the SHS sector (Activity 3).

The capital structure of REPP 2 has been designed to address endemic risk factors such as currency risk, which impacts investors and investees. The junior equity/first loss tranche is designed to absorb losses from any cause across the portfolio, including through defaults caused by currency devaluation transfer and inconvertibility risk. This is likely to be more efficient than mitigating risk at the level of each transaction. REPP 2 will also investigate sources of first or second loss portfolio guarantees to boost loss protection for senior equity investors and noteholders. Such guarantees cover the risk of losses due to any form of non-payment, including where the cause of the failure to pay may be related to currency devaluation.

At the transaction level, financial risk mitigation may take the form of political risk insurance, credit enhancement or mitigation of foreign exchange risk to address currency mismatches between revenues derived in local currencies and funding obligations in hard currencies. REPP 2 will work closely with its network of risk mitigation providers to reduce risks for governments, developers and funders alike. As was the case for REPP Phase 1, where a first-of-a-kind innovative risk mitigation solution for offtaker liquidity risk was adopted, REPP 2 will continue to work with its partners to develop and implement innovative risk

mitigation solutions. Existing risk mitigation providers used by REPP 1, and likely to be used by REPP 2, including the African Trade Insurance Corporate (ATI), GuarantCo and World Bank MIGA. Continuous exchange of pipeline information and collaboration with partners are ensured through well-established communication channels. In all cases, a decision to secure risk mitigation (e.g. payment liquidity or payment default cover under a Power Purchase Agreement (PPA), or other forms of political risk cover) will be made on a case-by-case basis, based on the assessed level of risk and availability of risk mitigation instruments. These decisions will be approved by the REPP Investment Committee in every case.

REPP will attempt to secure (as appropriate and as available) the following existing forms of financial risk mitigation, in line with its Investment Policy:

- Political risk insurance (PRI):** PRI will be sourced at a transaction level. Typically, PRI cover will include transfer and convertibility risk; breach of contract; expropriation; and war and civil disturbance. REPP 2 will source PRI cover from existing REPP partners such as the Africa Trade Insurance Corporation (“ATI”) as well as other reputable providers such as the World Bank (through the Multilateral Investment Guarantee Agency, “MIGA”) and the African Development Bank (“AfDB”).

Offtaker credit risk: Various forms of credit enhancement are possible / available to address offtaker credit risk for national utilities. These include a) payment liquidity cover (e.g. standby letters of credit, or fully-funded escrow accounts); and b) payment default cover (e.g. via sovereign guarantees, corporate guarantees or breach of contract cover provided through PRI). REPP 2 may source payment liquidity cover from the ATI’s Regional Liquidity Support Facility (“RLSF”) or seek funding of liquidity facilities directly from offtakers and sponsors (funding for debt service reserve accounts). REPP 2 may source non-payment (default) cover from the ATI as well as GuarantCo and other risk mitigation providers. A further source of protection for offtaker default risk will be via government guarantees (in support of national utilities which are acting as the offtaker for independent power producer projects).
- Currency hedging:** In addition to efforts to secure local currency funding wherever possible and where project economics and counterparty creditworthiness allows, currency risk may be mitigated through denominating loans and equity investments in hard currency, which transfers FX risk from REPP 2 to borrowers, investees and offtakers.

REPP 2 may engage in direct currency hedging at the transaction level (via foreign exchange forwards and cross-currency swaps) which would reduce currency devaluation risk to REPP 2 and also to borrowers / investees but would pass the costs of that hedging on to those borrowers / investees. Relative to the reduced scale of the revenues generated by smaller-scale renewable energy projects, these hedging costs are often prohibitive, even when sourcing hedging solutions from specialist market providers such as TCX. Other (potentially cost-effective) opportunities exist for FX hedging at the portfolio level, which would also reduce FX risk to REPP 2 and help achieve economies of scale in terms of costs which can be passed on to borrowers / investees.

Activities 5 and 6 support achievement of Output 2 (Increase in renewable energy capacity and generation) through the construction of grid-connected projects.

Support for technology deployment, innovation and market transformation

A portion of REPP 2 operating costs will be allocated to build skills and capacity amongst local businesses and developers (Output 3), improve the ability of businesses and developers to deliver gender equity, and improve their ability to manage ESG risk and impact. This support will be offered alongside hybrid equity instruments or as part of subordinated or senior loan commitments at financial close. Activities supported include the following:

Activity 7: Provide targeted support to developers

REPP 2 will work to overcome the limited experience and funding for local developers by engaging with them at their own pace and building local developer capacities by providing technical support to upskill project developers and off-grid companies. This activity, conducted as part of REPP 2 investment approach,

will build project finance and specifically project finance risk management skills for local developers, enabling those developers to mitigate key risks and thereby secure both public and private sector funding. It will also support off-grid companies in strengthening their business plans and fundraising processes. Investees will also be supported in strengthening their internal gender and inclusion policies and designing their Gender Action Plans.

Given the significance of local currency funding in providing long-term sustainable financing solutions for local developers, REPP 2 will also work to upskill and crowd in local (usually local-currency) funders, thereby making an important contribution to market development.

Activity 8: Improve environmental and social risk mitigation

REPP 2 will require that all investees develop and implement an environmental and social management system (ESMS) that is commensurate with the level of risk for that investment and is fit for purpose in accordance with the REPP 2's environment and social policy, which will itself be aligned with GCF and international best practise standards. This will support knowledge transfer to developers in REPP 2 Host Countries.

REPP 2 will fund and proactively support investees in the development of their ESMSs and will establish new – and expand on existing – guidelines and toolkits to build their capacity.

Activity 9: Non-financial support to promote gender equity

REPP 2 will integrate a gender equity perspective into the design, implementation, monitoring and assessment of all renewable energy activities financed by REPP 2. This involves:

- Analysing gender equity in the investee company, country of operation, and targeted sector;
- Establishing an investee-specific GAP by identifying gender performance indicators and gender-disaggregated targets against an established baseline; and
- Integrating the items listed in the two bullet points above into a monitoring plan.

REPP 2 support for technology deployment, innovation and market transformation will build skills and capacity amongst local businesses and developers (Output 3), as described further in item 3.

3. REPP 2 outputs

The above activities will lead to the following REPP 2 outputs

Output 1: Increased commercial viability

All REPP 2 activities, including early-stage growth and development funding, gap funding for construction and support for market development combine to demonstrate the potential for technical and commercial viability of smaller-scale renewable energy businesses and projects including female owned and led in Host Countries and across the region. Over its 20-year lifespan, REPP 2 targets support for thirty-seven (37) new investments. All new investments and the full funding commitment will be disbursed within seven (7) years of the first close for REPP 2.

Output 2: Increased renewable energy capacity and generation

Early-stage and construction funding from REPP 2 ensures that new renewable energy capacity and generation is realised. Renewable energy capacity and generation financed by REPP 2 contributes directly to improved access to clean electricity for women and men and host-country priorities as outlined in their policy frameworks as well as their Nationally-Determined Contributions and National Adaptation Plans.

Output 3: Improved skills and capacity

REPP 2 support for project developers establish a more active and commercially-viable market for smaller-scale renewables, which is better-able to attract both public and private finance in future. These impacts accrue to Host Countries and more broadly throughout sub-Saharan Africa, with a demonstration impact for other emerging markets and with developers being better placed to expand to those markets. REPP 2 will directly contribute to improved gender equity through a focus on gender mainstreaming within investee projects and businesses.

4. REPP 2 outcomes

The activities and outputs delivered by REPP 2 are designed to strengthen the market for small-scale renewables and crowd in private capital leading in sub-Saharan Africa leading to GHG mitigation and improved and strengthen resilience energy infrastructure and vulnerable people and communities, as follows:

Outcome 1: Reduced GHG emissions from renewable energy installations

Early-stage and development funding from REPP 2 will contribute to the achievement of the programme's objectives to reduce GHG emissions and increase renewable energy generation capacity by helping off-grid companies to establish their business models and begin their growth towards scale and sustainability, and by helping grid-connected projects to reach financial close.

REPP 2 funding for construction ensures that renewable energy businesses and projects are constructed and become operational, thereby reducing or avoiding GHG emissions for Host Countries. Targeted mitigation outcomes include 12.7m tCO₂e of avoided or reduced GHG emissions and 330MW of new renewable energy capacity over the 25-year total lifetime n of the technologies.

Outcome 2: Improved climate resilience of energy infrastructure

Funding for construction from REPP 2 contributes to the climate resilience of energy infrastructure in numerous ways (see Annex 2 for country specific details):

- Funding to achieve scale and sustainability of off-grid renewable energy installations which offer lower-cost, climate-resilient solutions and stable sources of clean energy generation for people, communities and business not connected to the grid;
- Decentralisation and diversification of energy generation can lead to greater grid stability and climate resilience, especially in countries where transmission and distribution infrastructure is already constrained, and where an increased risk of physical damage and decreased generation and transmission capacity is expected due to climate change.
- Countries which depend on hydropower generation but are also drought-prone with high rainfall seasonality are vulnerable to current climate variability. Projected decreases in annual rainfall and/or increases in the occurrence and severity of droughts increase climate change vulnerability. Hybrid power solutions (e.g., where solar PV is augmented with existing hydropower infrastructure) can directly contribute to improved grid stability and greater climate resilience;²²

Outcome 3: Improved climate resilience of vulnerable people and communities

Funding for off-grid electrification projects will not only provide first-time and affordable clean energy access and improve quality of life of low-income people who typically live in rural communities and are most vulnerable to climate change, but it will also build climate resilience outcomes including:

²² Note: in instances where REPP 2 will fund run-of-river hydropower assets, careful resource and design assessments will be undertaken to manage climate change risk

- Communities not connected to the grid often rely extensively on wood biomass, kerosene and diesel as a fuel source. Reduced biomass availability and increase in fuel prices represent additional climate change vulnerabilities. Adoption of renewable energy solutions, such as solar lighting, renewable energy powered pressure cookers, reduce these vulnerabilities;
- Improved access to climate information through cellular networks powered by renewable energy improve ability to plan and reach to the impacts of climate change. Local residents and the broader community can then benefit from access to early-warning systems relating to both pending disasters (storm, hail flooding risk) and seasonal climate information (relating to planning and decision-making for management of crops and livestock). Similarly, households and community's access to information can be improved through solar powered cell phone charging, radio and TV;
- Improved and more stable energy supply for critical community services such as health care facilities, emergency response, schools and water pumping stations build ability to respond to climate risks;
- Similarly, e-mobility can assist some of the most vulnerable agricultural communities to better access markets. The need for irrigation as well as cooling will increase, and solar powered solutions can help to build climate resilience and better adaptation response.

Outcome 3 (improved climate resilience for vulnerable people and communities) supports all Host Country adaptation and climate resilience priorities.

Outcome 4: Increased private sector investment

Programme Funding raised: In addition to the GCF Proceeds, REPP 2 aims to raise USD200 million of funding for smaller-scale renewables across sub-Saharan Africa through co-financing of the Programme. A total of USD150 million is expected to be raised from public funding sources and USD50 million is targeted to be raised from private funding sources.

Additional finance mobilised: The current REPP 2 pipeline represents a total estimated project cost of USD1.02 billion. On this basis, the target for additional finance mobilised through co-investment directly into off-grid businesses and grid-connected projects is USD786 million. The target for private and commercial finance mobilised for investment is USD393 million.²³

In addition to the above, REPP 2 intends to target up to 30% of its funding in support of 2X Challenge Gender Lens Investing (2X GLI) opportunities. This represents a target of USD70 million.

Outcome 5: Contributing to stronger technology deployment, innovation and market transformation

REPP 2 will:

- contribute to more and more rapid technology deployment by providing targeted support and finance to renewable energy companies during early and construction stages of the assets and by financing the deployment of smaller-scale and off-grid renewable energy technologies in nascent markets, often financing first-of-a-kind transactions and business models; and
- contribute to market development and transformation through its financing and capacity building support to local developers, based on the application of sound commercial investment principles aimed at delivering commercially viable and bankable projects. This support – directly and through demonstration effect to broader sector stakeholders – will stimulate sectoral development that would otherwise lag or

²³ To date, REPP Phase 1 has mobilised USD460 million in third party funding, representing a 3.8x multiple for third-party funding mobilized into the underlying assets (according to its Quarterly Impact Report for Q2 2023 <https://repp.energy/resource-center/publications/q2-2023/>). Of this, approximately 50% has been leveraged from private or commercial sources (unpublished data). Based on the proven track record of REPP Phase 1, the same proportion has been applied to the pipeline for REPP 2, which seeks to mobilise a total of USD786 million in third party funding. On this basis, the target for private or commercial funding mobilised is USD393 million.

not occur and help to establish or build a market ecosystem that will allow the ongoing growth of small and distributed renewables in Host Countries and across the region.

5. REPP 2 co-benefits

The construction and operation of renewable energy power plants will create **new jobs** during construction and operations for 76,801 people (co-benefit 1).

REPP 2 will also **enhance gender equity** (co-benefit 2) through:

- Anticipated 23,040 jobs for women
- 70 mUSD funding committed to 2X Challenge's Gender Lens Investing criteria-aligned companies
- 11 female led or managed companies funded by REPP 2;
- All REPP 2 investees will be required to establish gender action plans to increase gender equity awareness and performance within investee company and through their project implementation. The REPP 2 target is 37 developers with improved gender equity awareness and practices;
- 1.1 million women with improved quality of life.
- REPP 2 funding and support will lead to **improved quality of life** through community and critical services connected to electricity (co-benefit 3). Co-benefits include improved access to healthcare, lighting, refrigeration, improved access to information and more productive use of energy (PUE) opportunities.

These co-benefits will allow further socioeconomic development of Host Countries and their off-grid communities. The target for this co-benefit is improved quality of life through access to improved lighting for 3.7 million people, improved access to information for 1.7 million people.

REPP 2 will deliver **improved ESG practices** (co-benefit 4) by providing its investees with training and funding to identify environmental and social risk and impacts, and to establish mitigation and management systems in accordance with industry best practices, leading to improved environmental and social performance. Examples include improved waste management practices, mitigating the potential for negative impacts on biodiversity, soil and water resources, improved labour practices and Occupational Health and Safety (OHS) standards. developers.

6. Impact recognition

Individual projects or off-grid renewable energy companies are eligible for multiple forms of support and funding from REPP 2. For example, an off-grid company may receive both growth funding (Activity 1) and gap financing for construction of mini-grids (Activity 4). Similarly, a developer of a grid-connect project may receive funding during the development phase (Activity 2) as well as gap financing for construction (Activity 5). Recognition of targeted impacts and Key Performance Indicator metrics will not be duplicated in these situations. Any decision to provide follow-on financing will typically not alter those metrics, with progress towards achievement of those metrics measured and reported through the project life cycle.

Where a decision by REPP 2 to provide follow-on financing results in a verifiable change to the amount of co-financing leveraged (from either public, whether concessional or commercial, or private sources) or a change to the planned installed capacity for a project (or mini-grid or SHS portfolio), such changes may result in changes to impact and KPI targets.

7. Rationale for selection of interventions

The host country feasibility studies presented in Annex 2 include a country-level needs assessment that highlights the need for funding for on- and off-grid renewables. The demand for the types of interventions outlined above is verified in Annex 2 based on reported impacts from REPP Phase 1 interventions as well as the stated pipeline of projects for REPP 2.

The design of REPP 2 is informed by and builds on the successes of REPP Phase 1 to deliver scale, mobilise climate finance and achieve meaningful climate change impacts through both mitigation and adaptation in each of the Host Countries.

8. Eligibility criteria for REPP 2

Installed capacity: REPP 2 will invest in renewable energy power generation projects with an installed capacity of up to 25MW (and 50MW for wind). Eligible projects can be further categorized as follows:²⁴

- Grid-connected projects (independent power producers or IPPs);
- Large, isolated grids, where power is provided to entire towns (tens of thousands of connections); and
- Off-grid RE power companies, providing mini-grid (MG) and solar home system (SHS) solutions

Eligible technologies: As outlined in Annex 6, REPP 2 may invest in the following renewable energy power generation technologies: solar photovoltaic (PV) with battery backup, onshore wind and run-of-river hydropower.

Eligible countries: REPP 2 targets small scale renewable energy power generation projects in the Least Developed Countries of sub-Saharan Africa. To be eligible for receipt of GCF Proceeds, investments must be located in Host Countries, i.e. a jurisdiction which has provided a Letter of Non-Objection in a form acceptable to the GCF. All such Host Country investments will be funded by Portfolio HoldCo 1 S.à r.l.

REPP 2 may investment in other sub-Saharan African countries (including countries which are not classified as Least-Developed Countries) where a Letter of No Objection has not been provided. Such investments will be funded from Portfolio HoldCo 2 S.à r.l. and will be ineligible to receive GCF funding.

E&S risk categorisation: REPP 2 will only invest in projects that are categorised as IFC risk category B or C. Category A projects are excluded. Please see Annex 6 for further detail.

Investee requirements: Where REPP 2 is making a project finance investment (typically for grid-connected IPPs), repayments and returns to REPP 2 depend on project SPV cashflows only (by definition, there is no recourse to the balance sheets of the sponsors). For this reason, risk assessment focuses on the commercial and technical viability of the project itself.

Where REPP 2 is providing growth capital (either as debt or equity) to off-grid companies, the ability to service debt or generate returns depends on the long-term sustainability of the borrower. Key evaluation criteria include estimates of demand and revenue growth. In some cases, REPP 2 may insist on parent company guarantees prior to investment to boost creditworthiness of the borrower.

Where REPP 2 is funding development for IPPs via development loans, such loans are exposure to the risk of failed development. Investment criteria include a requirement for a discernible pathway to commercial and technical viability for the project. In such cases, REPP 2 typical takes security in the form a pledge of the sponsor's shares in the project SPV as well as legal charge over project assets.

Off-grid companies: Early-stage and growth capital for off-grid companies will be provided in the form of equity (ordinary and preference shares), convertible notes, junior and senior loans.

SHS businesses: SHS businesses which sell SHS kits (typically panels, batteries, radio and/or TV, mobile phone charging and lights) and other appliances, including PUE devices (, cooling solutions, barber kits, irrigation etc.) are eligible for REPP 2 funding. Businesses which do not provide RE generation devices (e.g. sale of pico-solar devices only, or cookstoves), and business which operate as retailers or wholesales are not eligible for REPP 2 funding.

²⁴ As also further detailed in Annex 14 (Term Sheet)

Mini-grid businesses: Mini-grid businesses must engage in the development, construction and / or operation of renewable energy generation facilities (including generation, backup and reticulation) at sites which are not initially connected to the main electricity grid. Typically, first-time connection includes lights. Such business may also provide PUE services (e.g. irrigation, milling) and other appliances (such as pressure cookers, freezing) to their clients.

On-grid projects: IPPs must engage directly in the development, construction and operation of renewable energy power generation facilities. REPP 2.0 may also fund the costs of grid connection.

Skills and capacity development: Skills and capacity building for developers (Activity 7), improved ESG practice (Activity 8) and promotion of gender equity (Activity 9) are targeted towards investee companies. To be eligible for these forms of support, beneficiaries must have an association with those companies (owners, management, staff, third-party advisors).

9. Country prioritisation

Only GCF Host Countries are eligible for investment from Portfolio HoldCo 1 S.à r.l. These Host Countries have been prioritised based on climate change vulnerabilities and impacts outlined in Section B.1, Section D.4 as well as Annex 2. Based on their NDCs and NAPs, further prioritisation is achieved through stakeholder consultation and by aligning their climate change mitigation and adaptation priorities with the objectives of REPP 2, as outlined in Section D.4 as well as Annex 7 and Annex 2.

For Portfolio HoldCo 2 S.à r.l., countries have been prioritized based on LDC status as well as demonstrated demand based on existing pipeline. In all cases, projects selected align with and promote country-level priorities in accordance with government strategic objectives, as reflected in NDCs, NAPs and other policy and strategic frameworks.

Given that REPP 2 is driven by market demand for funding and requested support by project developers and off-grid renewable energy companies, country prioritisation is also a function of that demand, with the indicative pipeline shown in Annex 26 reflecting the current demand.

10. REPP 2 risk management framework

REPP 2 will incorporate a comprehensive enterprise risk management (“ERM”) framework, As outlined in the Operations Manual (Annex 21). Throughout the investment life cycle (selection, analysis, approval, due lifecycle, implementation and exit / repayment), risk will be assessed using a risk management toolkit which includes:

- The ADAPT tool, used to assess a candidate investment’s eligibility to be classified as a climate change adaptation (cross-cutting) investment;
- A project risk assessment tool (“RAT”), which is used to assess a wide range of risk factors in all investments, including market, credit, financial, technical, environmental and social, country and regulatory as well as legal and documentation risks; and
- Additional specialised tools for detailed assessment of counterparty credit risk (for offtakers and borrowers) as well as environmental and social risks (unique to each technology and asset class).

The REPP 2 risk management framework is designed to facilitate robust and dynamic risk identification, analysis, mitigation, reporting and ongoing review of the effectiveness of mitigation actions already taken. As Manager of REPP 2 and REPP Phase 1 before that, Camco Management Limited has extensive experience and expertise in managing a wide range of risk factors, which affect individual technologies, asset classes and countries in unique ways. The ERM framework and proactive risk management approach adopted by an experienced team with on-the-ground presence throughout sub-Saharan Africa means that Camco is well-placed to manage existing and emerging risks effectively.

The ERM framework includes detailed policies and procedures, to be followed by all staff during origination, structuring and ongoing portfolio management (once investments have been made). REPP 2 will appoint a dedicated risk manager. Risk governance and oversight will be provided by AIFM as supported by the Investment Committee, with approval of all policies and procedures provided by the general partner of Camco REPP SCSp.

The ERM framework for REPP 2 will include a risk appetite statement, approved by the AIFM, to designed to ensure that the potential for financial losses within the REPP 2 portfolio remain within approved risk tolerance levels. A detailed limits framework, covering a) counterparty concentration limits at the borrower, sponsor and sponsor group levels, b) limits for countries, technologies and asset classes and c) market risk limits (including foreign exchange and interest rate limits) is defined in the REPP 2 Operations Manual (Annex 21).

The enterprise risk management framework, including policies, procedures, governance arrangements, risk management toolkits and indicative limits framework are described in the Operations Manual (Annex 21). The Operations Manual outlines specific eligibility criteria and risk assessment procedures for dealing with the unique risks associated with each technology (solar PV, wind and hydropower), asset class (SHS, mini-grids and grid-connected projects) and Host Country. Key risks and risk mitigation actions for the successful implementation of REPP 2 and achievement of its goals are outlined in Section F.

B.4. Implementation arrangements

Camco REPP SCSp

The main fund vehicle of REPP 2, and the primary Executing Entity (Camco REPP SCSp), is a *société en commandite spéciale* (“SCSp”) or special limited partnership, established in Luxembourg under the Luxembourg law of 10 August 1915 on commercial companies (the “1915 Law”).²⁵

The advantages of this structure for REPP 2 include:

- The SCSp is a form with unlimited capacity and limited liability for its limited partners, provided they do not participate in the management of the SCSp vis-à-vis third parties (such management instead being carried on by its general partner);
- It enables a flexible structure that can fully accommodate the blended capital structure of REPP 2;
- The structure is market standard, frequently used for similar pooled investment structures and attractive to co-financing investors of REPP 2;
- Governance can be adapted to reflect both the fiduciary standards of the GCF and any requirements of co-financing investors of REPP 2; and
- Based in Luxembourg, Camco REPP SCSp will be an alternative investment fund or “AIF” for the purposes of the EU Alternative Investment Fund Managers Directive (AIFMD) and the UK equivalent of AIFMD. Compliance with AIFMD can be expected to provide additional comfort to EU/UK co-financing investors of Camco REPP SCSp.

The general partner of Camco REPP SCSp will be Camco REPP GP S.a.r.l, a Luxembourg *société à responsabilité limitée* (a Luxembourg limited company) wholly owned by Camco Management Limited.

GCF REPP Holdings SCSp

It is proposed that the GCF participate in Camco REPP SCSp via a special purpose vehicle (“GCF REPP Holdings SCSp”) established as a Luxembourg SCSp under the 1915 Law for which the GCF would be investor of record and no other limited partnership interests would be issued. GCF Proceeds would flow via

²⁵ Subject to ongoing finalisation and further diligence of certain structural discussions with the Secretariat and legal advice from external counsels, which will be concluded and reflected in the Term Sheet. This section may be revised to reflect the above. Further Legal and Tax Due Diligence is also being undertaken will completed prior to submission of the Funding Proposal to the Board.

the AE to GCF REPP Holdings SCSp and then be contributed to Camco REPP SCSp. Camco Management Limited would be the sole shareholder of the general partner of GCF REPP Holdings SCSp. This GP/LP structure would be tax efficient for the GCF for the underlying participation in REPP 2. The funds received by GCF REPP Holdings SCSp would be contributed into Camco REPP SCSp via the Subscription and Partnership Agreement. Other investors would also contribute to Camco REPP SCSp or its feeder vehicle Camco REPP 2 Feeder SCSp as limited partners and Note holders. The Subscription and Partnership Agreement, binding on all limited partners, will be based on the Accreditation Master Agreement (“AMA”), the Term Sheet set out in Annex 14 and the Funded Activity Agreement (“FAA”) and provide for the investment terms, governance structure, policies and procedures, reflow waterfalls and respective rights of the limited partners – including for distributions and the return of capital. A draft will be prepared subsequent to the approval of the FP by the GCF Board.

Other feeder vehicle(s) in Camco REPP SCSp

Holders of senior equity interests and senior note interests in Camco REPP SCSp are expected to participate in Camco REPP SCSp via one or more feeders which are also anticipated to be formed as Luxembourg SCSps under the 1915 Law.

Side Letters

A Side Letter is anticipated to be entered into with the GCF to implement certain aspects of the Term Sheet in Annex 14. Since side letters are likely also to be required by other co-financing investors in Camco REPP, the Term Sheet contains a most favoured nation provision (“MFN”) to protect the position of the GCF in Camco REPP. This MFN right is included in the Term Sheet in Annex 14 and provides that during the term of the FAA, Camco REPP SCSp shall not enter into any additional, or modify any existing agreements with any existing or future investors to the effect of establishing rights or otherwise benefiting such investor in a manner more favourable in any material respect to such investor than the rights and benefits established in favour of the Fund pursuant to the Subsidiary Agreements, unless, in any such case, the Fund has been provided with such rights and/or benefits or waived its entitlement to such rights and/or benefits.

Subsidiaries/Portfolio HoldCos

To ensure that GCF Proceeds are provided only to Host Countries (being countries that have issued a Letter of Non-Objection), Camco REPP SCSp will establish two subsidiaries (in Luxembourg or other tax neutral or tax positive form), likely to be Luxembourg *sociétés à responsabilité limitée* (S.à r.l.) with Camco REPP SCSp, acting by its general partner, as the sole member: Portfolio HoldCo 1 S.à r.l. and Portfolio HoldCo 2 S.à r.l.

Portfolio HoldCo 1 S.à r.l. will only execute investment agreements for projects in Host Countries, i.e. for which the GCF has received Letters of No Objection. These countries are Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sierra Leone and Zambia at the time of submission but this list may expand in future if further Letters of Non-Objection are received and an amendment to the Programme is subsequently approved by the GCF Board.

Portfolio HoldCo 2 S.à r.l. will execute investment agreements in the full range of REPP 2 eligible countries (non-NOL countries in sub-Saharan Africa with a focus on LDCs). According to the REPP 2 eligibility requirements, the list of non-NOL countries may include any other countries in sub-Saharan Africa (provided that in the case of investments in the Host Countries, only after Portfolio HoldCo 1 S.à r.l. is 75% committed for investment and in respect of these investments, the Fund and other Junior Class Interest (as such term is defined in the Term Sheet in Annex 14) holders would be drawn pro rata to their undrawn commitments).

The Subscription and Partnership Agreement, constitutional documents of the HoldCos and intra-group Camco REPP SCSp structuring will ensure that the capital contribution of the GCF will be deployed (and returned) solely in relation to the investments of Portfolio HoldCo 1 S.à r.l. in Host Countries. The anticipated split between the Portfolio HoldCos is set out in the financial model, but for the purpose of clarity, non-GCF investors will also be investing in projects in Host Countries via Portfolio HoldCo 1 S.à r.l. and this could include participation of other Junior Interest Class holders. However, GCF Proceeds may only be invested

in Host Countries through Portfolio Holdco 1 S.à r.l. As such, if an investment of Camco REPP SCSp is located in Host Countries and in other countries, it may only be made through Portfolio Holdco 2 S.à r.l.

The dual Holdco structure provides GCF the protections necessary to ensure separation of funds, while minimizing transaction costs and maintaining a simplified governance agreement (under a single Subscription and Partnership Agreement).

Management

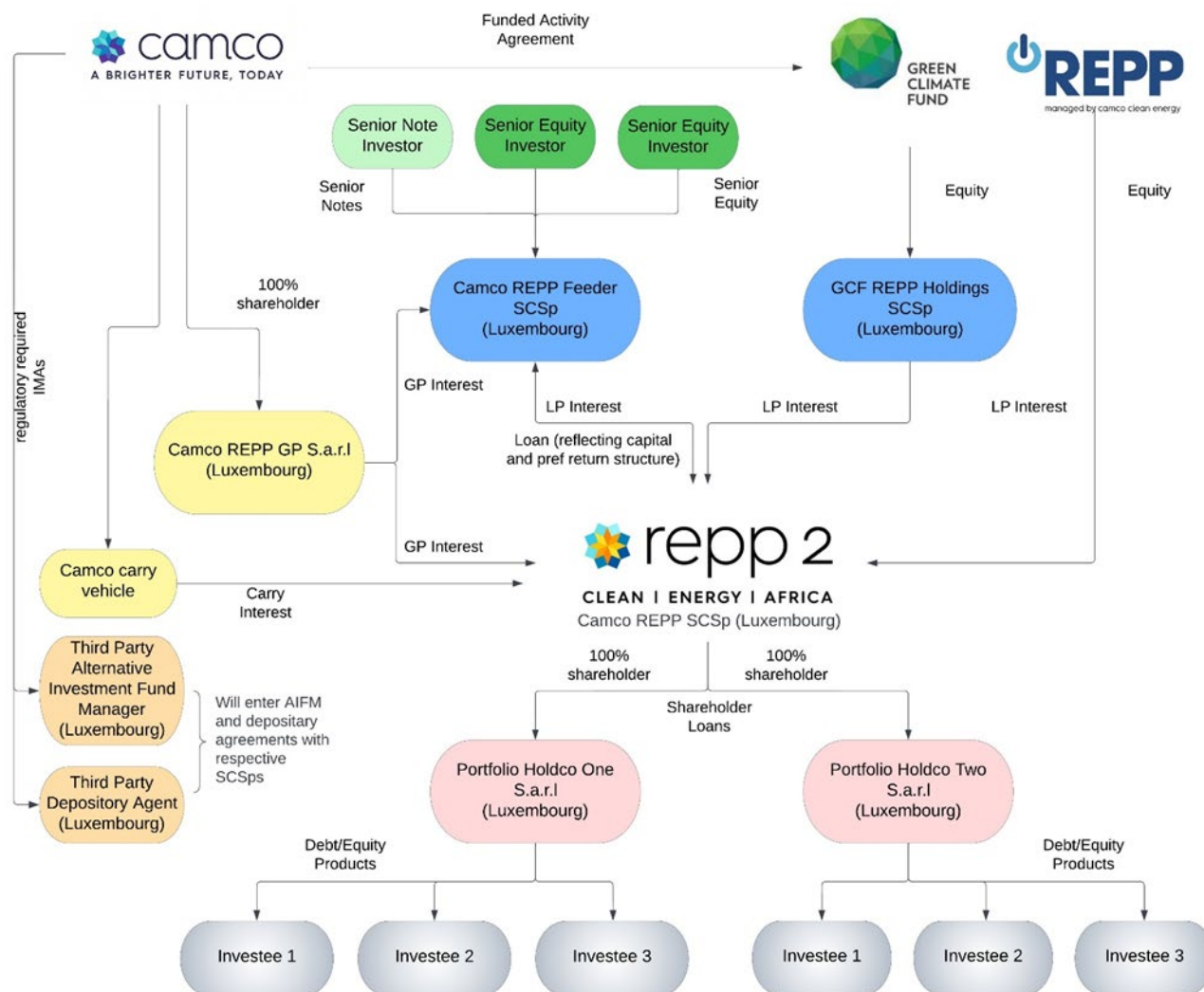
The general partner of Camco REPP SCSp (which will be a Camco-owned entity, Camco REPP GP S.à r.l.) has primary responsibility for management of the partnership. However, as an AIF, Camco REPP SCSp (and, subject to the terms of the long form documentation and further Luxembourg legal review and advice, GCF REPP Holdings SCSp) will appoint an external third-party service provider as its AIFM. Under AIFMD, the AIFM will be responsible for risk management and portfolio management of REPP 2. The AIFM will delegate certain of its duties to Camco Management Limited pursuant to an Investment Management Agreement. We anticipate these duties will include portfolio management.

On the basis that:

- a. The Accredited Entity is a party to the AMA and will execute the FAA with the GCF;
- b. The Accredited Entity will be the sole shareholder of Camco REPP GP S.à r.l. and a party to its LPA;
- c. The third party AIFM will be appointed as alternative investment fund manager by Camco REPP SCSp pursuant to an alternative investment fund management agreement;
- d. The Accredited Entity will be appointed as investment manager by Camco REPP SCSp and will be the sole shareholder of the general partner of Camco REPP SCSp, which will be a party to the Subscription and Partnership Agreement
- e. Portfolio HoldCo 1 S.à r.l. and Portfolio HoldCo 2 S.à r.l. will adhere to the Investment Management Agreement,

each of the above is a Subsidiary Agreement as defined in the AMA (and see “Roles and Responsibilities” below)

Proposed structure



The Accredited Entity will ensure the obligations of the AE under the AMA, and in particular Clause 5 (*Results Management Framework; Monitoring and Accountability*), Clause 7 (*Disbursement of GCF Proceeds to the Accredited Entity; Unused Funds; Reflowed Funds; Investment income*), Clause 8 (*Implementation*), Clause 9 (*Standard of Care*), Clause 10 (*Executing Entities; Affiliates*), Clause 11 (*Administration of Funds by the Accredited Entity*), Clause 13 (*Compliance with Standards, Policies and Procedures*), Clause 14 (*Procurement*), Clause 15 (*Record Keeping and Reporting*), Clause 16 (*Periodic Reviews, Ad Hoc Checks and Evaluations*) and Clause 17 (*Financial Management; Financial Information*) are fully implemented as binding contractual obligations of the Executing Entities in each of the Subsidiary Agreements, subject as varied by the Term Sheet in Annex 14. Furthermore, the Accredited Entity shall ensure that each Investment Agreement executed by either Portfolio HoldCo 1 S.à r.l. or Portfolio HoldCo 2 S.à r.l. contains necessary and sufficient obligations on each Investee for the Accredited Entity to comply with its obligations under the AMA and FAA.

Accredited Entity

Based on the Term Sheet contained in Annex 14, GCF and Camco Management Limited will enter into a Funded Activity Agreement ("FAA") detailing the key implementation arrangements, commercial terms and legal obligations between the parties. The GCF will transfer funds to the AE under the FAA, which will then

be transferred to GCF REPP Holdings SCSp and used as the capital contribution of GCF REPP Holdings SCSp in Camco REPP SCSp as outlined above and in the Term Sheet in Annex 14.

Camco Management Limited, as AE, will oversee the implementation of the programme on behalf of the GCF and will ensure that:

- All GCF policies and procedures (including the Indigenous Peoples Policy, Environmental and Social Policy and the Undated Gender Policy) are implemented at both the Camco REPP SCSp level and in the underlying investment agreements;
- The GCF Account is maintained and operated pursuant to the requirements of the AMA and FAA;
- The GCF Proceeds are only deployed in Host Countries;
- As set out in Annex 14, GCF REPP Holdings SCSp, Camco REPP SCSp and the Portfolio HoldCos comply with the Major Change provisions of the AMA and FAA;
- GCF Holdings REPP SCSp, Camco REPP SCSp and the Portfolio HoldCos comply with Monitoring and Reporting Framework and the Results Management Framework;
- All Funded Activities fall within the Environmental and Social Risk Categories for which the AE has been accredited;
- The administration of GCF Proceeds will be carried out with at least the same standard of care as it uses for the administration of its own funds or funds for which it has management or investment responsibility; and
- GCF is provided access to information relating to the performance of Camco REPP SCSp vis-à-vis the targeted outcomes and results of Camco REPP SCSp.

Portfolio HoldCo Structure

The dual-HoldCo structure is designed to address the twin aims of addressing the broader market demand for funding for smaller-scale renewables and building on the success of REPP Phase 1, while maintaining the strict separation of funds and reflows required under the FAA. The dual HoldCo structure is a simple and cost-effective way of ensuring separation between investments (and resulting cash flows) in Host Countries (Portfolio HoldCo 1 S.à r.l.) and non-Host Countries (Portfolio HoldCo 2 S.à r.l.). GCF Proceeds, as channelled via the AE, through GCF REPP Holdings SCSp into Camco REPP SCSp will be available to Portfolio HoldCo 1 S.à r.l. only.

The capital structure (i.e. same blended finance structure of junior equity, senior equity and senior debt) is broadly duplicated in both HoldCos (subordination is identical), as shown in the indicative table below. Senior debt distributions are made in proportion to funding commitments from the Camco REPP SCSp to each Portfolio HoldCo. Investments and reflows are made independently in each HoldCo.

It is possible that the amount of commitments of different classes could be different in each of Portfolio HoldCo 1 S.à r.l. and Portfolio HoldCo 2 S.à r.l., but with effect from the Final Closing Date Portfolio HoldCo 1 S.à r.l. shall not have a greater ratio of Junior Class Interests to Senior Class Interests as compared to Portfolio HoldCo 2 S.à r.l., subject always to the restriction that Junior Class Interests may not (with effect from the Final Closing Date) exceed 40% of Total Capital Commitments.

Indicative illustration – commitment sizes may vary

	Amount (USDm)	% Total	Amount (USDm)	% Total	Amount (USDm)	% Total
Notes	50	20%	25	20%	25	20%
Senior Equity	100	40%	50	40%	50	40%
Junior Equity	100	40%	50	40%	50	40%
Total	250	100%	125	100%	125	100%

Roles and Responsibilities

For the implementation of the Funded Activity, the Accredited Entity shall enter into the following agreements, which will be considered Subsidiary Agreements under the AMA and FAA:

- i) Limited Partnership Agreement for the establishment of GCF REPP Holdings SCSp between Camco REPP GP S.à r.l and GCF as limited partner, with the GCF being the investor of record in GCF REPP Holdings SCSp;
- ii) Articles of Association establishing the general partner of GCF REPP Holdings SCSp and (if separate) of Camco REPP GP S.à r.l . This limited company will be a subsidiary of the Accredited Entity and have no purpose other than acting as a general partner of REPP 2 partnerships;
- iii) Subscription and Partnership Agreement for setting out the subscription for and the rights and obligations of the limited partners (including GCF REPP Holdings SCSp) in Camco REPP SCSp including the implementation of the Distribution Waterfall, establishment of governance bodies, compliance with GCF policies and procedures, powers enabling establishment of Portfolio Holdco 1 S.à r.l and Portfolio Holdco 2 S.à r.l and adoption of investment policy;
- iv) Alternative Investment Fund Management Agreement appointing a third party Luxembourg AIFM to act as alternative investment fund manager of Camco REPP SCSp;
- v) Investment Management Agreement appointing the Accredited Entity as investment manager of Camco REPP SCSp providing for all relevant management services to be provided to Camco REPP SCSp by the Accredited Entity (subject to applicable law including where a service is required to be provided formally by a third party Luxembourg AIFM);
- vi) Deed of Adherence to the Investment Management Agreement for Portfolio HoldCo 1 S.à r.l;
- vii) Deed of Adherence to the Investment Management Agreement for Portfolio HoldCo 2 S.à r.l;
- viii) Articles of Association establishing Portfolio HoldCo 1 S.à r.l;
- ix) Articles of Association establishing Portfolio HoldCo 2 S.à r.l;
- x) Intra-group loan agreements providing for the disbursement of funds from Camco REPP SCSp to Portfolio Holdco 1 S.à r.l and Portfolio Holdco 2 S.à r.l on a needs basis; and
- xi) Service Agreements with independent members of the Investment Committee of Camco REPP SCSp.

A further description of the roles and responsibilities of the AE and each Executing Entity is set out in the table below:

Role	Name	Responsibilities
Accredited Entity	Camco Management Limited	• Execute FAA with GCF • Establish and manage, as general partner, GCF REPP Holdings SCSp • Receive the GCF Proceeds under the FAA (into the GCF Account) • Transfer the GCF Proceeds from the GCF Account to a dedicated account in the name of GCF REPP Holdings SCSp • Transfer the GCF Proceeds contemporaneously as a capital contribution to Camco REPP SCSp as set out in the Subscription and Partnership Agreement • Enter the Subsidiary Agreements with each of the Executing Entities • Crowd in up to USD200m of co-investors into REPP 2 by way of equity and debt

		• Ensure compliance with GCF fiduciary standards and the obligations of the AMA and FAA • Ensure compliance with applicable local law
Executing Entity	GCF REPP Holdings SCSp	• Act as SPV for GCF funding to Camco REPP SCSp • Contribute capital to Camco REPP SCSp as per the Subscription and Partnership Agreement • Distribute returns to GCF as per the Holdings LPA • Ensure compliance with applicable local law • Pursuant to the Subsidiary Agreement with the AE, comply with GCF policies and procedures
Executing Entity	Camco REPP SCSp,	• Pool capital of GCF and other investors and debt providers • Establish Portfolio HoldCo 1 S.à r.l (for GCF Target jurisdictions) and Portfolio HoldCo 2 S.à r.l (for other sub-Saharan African jurisdictions) • Establish Advisory Board and (unless established by Camco Management Limited) Investment Committee • Distribute capital and returns to investors as per the Subscription and Partnership Agreement (and Notes Agreements to the extent not applicable to any feeder vehicle if that is the contracting party for a Note Agreement) • Provide, or procure the provision of, the reporting and monitoring as required by the AE to enable the AE to comply with its obligations under the AMA and FAA • Ensure compliance with applicable local law • Pursuant to its Subsidiary Agreement with the AE, comply with GCF policies and procedures
Executing Entity	Camco REPP GP S.à r.l	• Serve as general partner of Camco REPP SCSp • Provide, or procure the provision of, the reporting and monitoring as required by the AE to enable the AE to comply with its obligations under the AMA and FAA
Executing Entity	Portfolio HoldCo 1 S.à r.l	• To serve as the contractual counterparty to investees in the GCF Host Countries • Pursuant to its Subsidiary Agreement with the AE, comply with GCF policies and procedures
Executing Entity	Portfolio HoldCo 2 S.à r.l	• To serve as the contractual counterparty to investees in Camco REPP countries other than Host Countries
Executing Entity	Camco Management Limited (or subsidiaries)	• Through its subsidiary, Camco REPP GP S.à r.l act as general partner of Camco REPP SCSp • Through its subsidiary, Camco REPP GP S.à r.l act as general partner of GCF REPP Holdings SCSp • Serve as Investment Manager of Camco REPP SCSp pursuant to the terms of the Investment Manager Agreement • Provide, on behalf of Camco REPP SCSp, the reporting and monitoring as per the Partnership Agreement and Notes Agreements • Lead the due diligence, negotiation, execution, implementation, monitoring and enforcement of investment agreements • Ensure compliance with applicable local law

		Pursuant to its Subsidiary Agreement with the AE, comply with GCF policies and procedures
Executing Entity	AIFM & Depositary Agent	- To have regulatory responsibility for REPP 2 to the <i>Commission de Surveillance du Secteur Financier</i> of Luxembourg - Ensure compliance with applicable local law - Pursuant to its Subsidiary Agreement with the AE, comply with GCF policies and procedures

Track record

Camco is a climate and impact fund manager, leading the clean energy transition in emerging markets. Having spent over thirty years in the renewable energy sector, we offer practical and valuable financing solutions, pairing the discipline of a development bank with the agility of a private company. Camco was formed in Nairobi in 1989, is a regulated small Alternative Investment Fund Manager in the UK and is a Green Climate Fund Accredited Entity.

As a climate and impact fund manager, Camco's track record includes:

- Co-advisor of the USD200 million Green Africa Power (GAP) mandate, 2014-2017.
- Sole manager of the Renewable Energy Performance Platform (REPP Phase 1): GBP48 million first close in 2015, GBP40 million second close in 2019.
- Sponsor and manager of Spark Energy Services, a USD50 million, commercial and industrial energy efficiency and renewable energy facility focused on key African markets. Spark is funded by private and commercial funders and is currently in financial close; and
- Currently fundraising for Transforming Island Development for Electrification and Sustainability (TIDES), a USD100 million renewable energy and energy access facility for Pacific Islands.

Over a 30-year track record, Camco impacts include:

- 60+ projects completed
- 15.5 million people provided with new and improved electricity access
- 600 MW installed capacity realized; and
- USD900 million in third party funding mobilized

Camco's key strengths include:

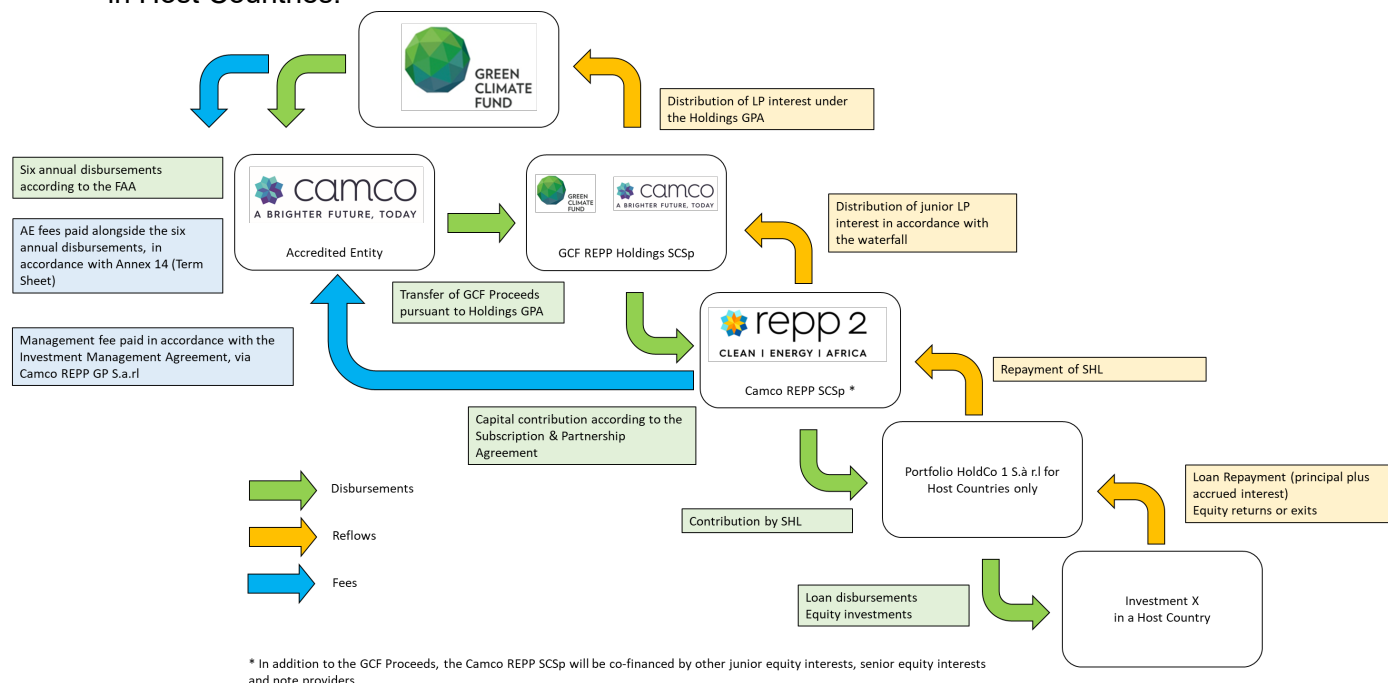
- A long-standing presence in the region, with offices in west, east and southern Africa;
- Detailed local knowledge from the ground up, combined with international reach;
- Extensive, deep relationships with – and understanding of– developers, policymakers, donors, DFIs and private finance; and
- Experience across a range of key disciplines including policy, regulation, market and project development, technical assistance, equity, debt, monitoring & evaluation, asset management and exits.
- Deep experience of financing small-scale renewables in Sub-Saharan Africa.

Flow of Funds

Disbursements

- The GCF will commit USD50 million, with GCF Proceeds provided to Camco Management Limited as AE in six annual disbursements, in accordance with Annex 14 (Term Sheet);
- The GCF will pay fees to the AE in 6 equal instalments in accordance with Annex 14 (Term Sheet);
- The GCF Proceeds received by Camco Management Limited as AE under the FAA will be transferred from the GCF Account to a dedicated account in the name of GCF REPP Holdings SCSp;
- The GCF Proceeds will be transferred contemporaneously as a capital contribution to Camco REPP SCSp as set out in the Subscription and Partnership Agreement;

- Camco REPP SCSp will maintain separate bank and ledger accounts for receipt and transfer of capital contributions in respect of Partnership Expenses; Portfolio HoldCo 1 S.à r.l.; and Portfolio HoldCo 2 S.à r.l.
- Camco REPP SCSp will transfer capital contributions from the designated Portfolio HoldCo1 S.à r.l. account it maintains to Portfolio HoldCo 1 S.à r.l.;
- Camco REPP SCSp will pay fees from the designated Partnership Expenses account to Camco as Investment Manager, in accordance with the Investment Management Agreement;
- Only Camco REPP SCSp will transfer capital contributions to Portfolio HoldCo 1 S.à r.l. and
- Portfolio HoldCo 1 S.à r.l. will disburse loans or make equity investments in investment opportunities in Host Countries.



Reflows

- Individual loan repayments (principal plus accrued interest) from investments in Host Countries will be made to Portfolio HoldCo 1 S.à r.l.;
- Similarly, equity investments in Host Countries will generate dividends or will be repaid via exits, with funds flowing to Portfolio HoldCo 1 S.à r.l.;
- Portfolio HoldCo 1 S.à r.l. will make payments or distributions to Camco REPP SCSp;
- Returns from investments, being realised proceeds from the Portfolio HoldCos will be allocated to investors per the distribution waterfall. Amounts held may be reinvested. Such reinvested amounts shall be deemed as a Capital Contribution for holders of Junior and Senior Class Interest Holders
- Camco REPP SCSp will distribute amounts relating to GCF's junior LP interest in accordance with the waterfall to GCF REPP Holdings SCSp, subject to any necessary or appropriate retentions; and
- GCF REPP Holdings SCSp will distribute amounts under the Holdings Partnership Agreement, subject to any necessary or appropriate retentions.

B.5. Justification for GCF funding request

1. Motivation for GCF funding

The proposed capital structure for REPP 2 is outlined in the table below.

Instrument	Description	Final close (USD million)	tenor / hold (years)	Return target	Funders / Co- funders
Senior debt ²⁶	Impact and capital preservation with debt returns	50			private investors, DFIs
Equity	Impact and equity risk / return	100			Private investors, IFIs, DFIs, Foundations
Junior equity (concessional equity)	Impact and mobilisation of third-party funds	100			GCF, donors
TOTAL		250			

The capital structure for REPP 2 is designed to help to address **key market failures** for smaller-scale renewables across sub-Saharan Africa, including:

- A lack of early-stage or development funding for smaller-scale renewables; and
- A limited risk appetite from both public and private funders for early-stage and smaller-scale projects.

Approved GCF funding programmes for **climate mitigation** through renewables are typically focused on larger-scale projects. These programmes have provided only limited support for early-stage funding, particularly for smaller project sizes (Annex 23). There is also relatively limited geographic overlap between the proposed Host Countries for REPP 2 and programmes which are already funded by the GCF. Climate mitigation funding from GCF for REPP 2 thus has the potential to be strongly complementary to existing funded activities.

In terms of **climate adaptation**, the GCF recognises “the urgent and immediate needs of vulnerable countries, in particular least developed countries (LDCs), small island developing States (SIDS) and African States” and has recently reaffirmed this decision.^{27,28} Despite this, the scale and scope of funding from the GCF for adaptation in sub-Saharan Africa is relatively small. For the African region, adaptation makes up 9% of total approved funding within the GCF portfolio, with cross-cutting programmes making up a further 13%.²⁹ Of the energy programmes reviewed in Annex 23, only two are cross-cutting programmes involving renewables. GCF funding for REPP 2 and a cross-cutting programme will thus contribute to the GCF’s strategic priorities and will make a materially positive impact to build climate resilience across the region.

There is no lack of financing for large-scale, commercially viable renewable energy projects across sub-Saharan Africa. The analysis presented in Annex 23 shows a different picture for smaller-scale projects as well as nascent renewable energy markets in LDCs and middle-income countries, where significant funding gaps exist. The same analysis highlights the paucity of funding available for early-stage growth capital of off-

²⁶ Terms dependent on current market conditions

²⁷ <https://www.greenclimate.fund/document/gcf-b05-23>

²⁸ <https://www.greenclimate.fund/document/gcf-b33-19>

²⁹ <https://data.greenclimate.fund/public/dashboard/data-browser/approved-portfolio/financing> (last accessed 08 September 2022).

grid renewables and development funding for grid-connected projects. In general, private capital providers do not offer funding to cover development costs or do so on strictly commercial terms which tend to discriminate against (“crowd out”) local developers. REPP 2 will seek to support smaller-scale renewables and local developers, providing funding for growth during project development and addressing funding gaps for construction. In the process, REPP 2 will both address a key barrier and have the potential to be highly complementary when working with other sources of public and private capital. REPP 2’s focus on funding to achieve scale and sustainability for off-grid companies, combined with its relatively unique risk appetite for early-stage, smaller-scale renewables, will make its funding strongly additional – acting to crowd in other sources of public and private funding for construction.

REPP Phase 1 is funded through the United Kingdom’s Foreign, Commonwealth and Development Office (FCDO). An initial capitalisation of GBP 48 million was, due to the early success of the programme, followed by a further commitment of GBP 40 million. Despite this, the current funding cycle for REPP Phase 1 is coming to an end and the FCDO has no obligation to provide funding to REPP 2.

Despite this, REPP Phase 1 / FCDO has agreed in principle to provide matching funding of up to USD50 million in junior equity funding, alongside the GCF, for REPP 2. A term sheet is in an advanced stage of discussion.

The AE is in advanced discussions with several private investors (such as Calvert Impact) as well as commercial investors (such as FinnFund, which has indicated that the source of its funding will be from private capital markets, with funding provided to REPP 2 on commercial terms). Both investors have indicated their interest in REPP 2 through Term Sheets and Heads of Terms.

Junior equity funding from both the GCF and REPP Phase 1 / FCDO will be catalytic in securing the fundraising targets for REPP 2. In fundraising conducted to date, private and commercial funders have consistently indicated a desire to secure funding for the junior equity tranche (which effectively de-risks all more senior investors) before considering their own funding decision. The process of securing the initial funding targets for REPP 2 and crowding in private and commercial funders into the structure has been delayed for these reasons. Requirements and commitments for co-investment, including commercial co-investment, are outlined further in Section C.1.

The target size of REPP 2 (\$250m) was set to meet several objectives: being large enough to build a diversified portfolio (37 transactions), thereby minimizing concentration risk, being large enough to attract commercial and private investors (as typically investors don’t want to commit a large portion of fund size), and being able to meet the significant market needs as exemplified by the large pipeline of transactions originated by Camco (\$1bn+).

2. Platform financing instruments

REPP 2 is a blended finance facility featuring a junior equity tranche, equity and debt. The proposed capital structure for REPP 2 (and the GCF’s position within it), along with the selection of financing instruments used to complete the capital stack have been designed to crowd in additional sources of both public and private capital.

The junior equity tranche, which will act as a first loss buffer, is designed to deliver incrementally greater impact and further mobilisation of third-party funds (both public and private) from more risk-averse investors. The sizing (at least 30% of total available capital) and subordinated position of this tranche in the REPP 2 capital stack will provide protection for all investors in more senior tranches. It is proposed that the GCF invest junior equity on a *pari passu* basis alongside other donors and DFIs. For clarity, the protection provided by the GCF junior equity investment will be limited to other investors in the NOL portfolio (Portfolio HoldCo 1 S.à r.l). The junior equity tranche will target low, single-digit returns in the base case.

The senior equity tranche is designed to provide more commercially oriented equity returns. Camco is targeting 40% of the total capital requirement in the form of senior equity from a combination of private investors, international financial institutions (IFIs) and development finance institutions (DFIs), all of whom will effectively be crowded into the capital structure of REPP 2 through the protection of returns provided by the junior equity tranche.

The senior debt tranche will enjoy maximum credit enhancement from the equity tranches. The senior debt tranche is expected to be financed by commercial debt providers.

3. Level of concessionality

The GCF's strategic objective is to catalyse climate finance for mitigation and adaptation which is new, additional (in the sense that a project or programme would not have occurred in the absence of GCF funding), adequate and predictable.³⁰ Accordingly, REPP 2 will ensure that its investments are additional and contribute to the establishment of a more viable and active market for smaller-scale renewables in Host Countries.

In financing smaller-scale renewable energy projects and businesses, REPP 2 recognises the need to strike a balance between the need for concessionality, the levels of real and perceived risk and the desire to grow a sustainable market for renewables. In the African context, these risks are higher, given that the market is as yet unproven, and the perception of risks is also higher than other markets. In order to mobilize capital in the sector the balance of risk and return provided needs to match the risk appetite and requirements for all investors.

Small-scale renewable energy businesses in African LDCs struggle to attract funding in general, and particularly private and commercial funding. The fact that barriers and significant funding gaps persist is demonstrated through the experience of REPP Phase 1, the broader market barriers identified in Section B.2(a) and the need to develop a tranching capital structure for REPP 2 with a layer of junior equity below senior equity and notes layers to attract private and commercial investors as much as possible, given limited appetite of such investors for the sector.

At the level of the underlying assets funded by REPP 2, and in providing gap financing to smaller-scale grid-connected projects, REPP 2 will help developers and sponsors of these projects to overcome a range of barriers, notably limited access to finance and to financing on terms that allow these projects to become commercially viable. Traditional investors, such as banks and utilities, often consider small-scale projects to be too risky or too expensive to finance. In addition, development finance institutions and other typical investors in the sector typically require minimum ticket sizes of USD10-20 million, which is often larger than the financing needed by small-scale projects. The economics of small-scale projects are often marginal, and the project development and construction risks are often higher than for large projects, with the potential for revenue and profit generation limited by the smaller installed capacity of the plant. For these reasons, REPP 2 must take a more flexible approach to address the financing gap, with a risk-adjusted return expectation that is lower than what is typically required by the market.

To address these challenges, REPP 2 offers flexible financing with a risk-adjusted return that is lower than what is typically required by the market, in order to help reduce the barriers to financing small-scale renewable energy projects in Sub-Saharan Africa and promote the development of these projects in the region.

Targets for leveraging additional private and commercial capital at the level of the underlying assets funded by REPP 2 are also commensurate to the level of appetite from such investors. The target of leveraging USD393 million in private and commercial funding in the underlying investments represents 50% of the overall target for third party funding mobilised (USD786 million). The target is derived from the experience

³⁰ GCF 2020: Updated Strategic Plan for the Green Climate Fund: 2020-2023, GCF document GCF/B.27/21
<https://www.greenclimate.fund/document/gcf-b27-21>

of REPP Phase 1 and is set conservatively in that the proportion of third party funding from private and commercial sources is no higher than that already achieved in REPP Phase 1. While the AE is motivated to increase this proportion over time, the target is set conservatively in recognition that the risk appetite from private and commercial funders remains low and it remains a difficult task to leverage additional private investment.

In this context, GCF funding is highly catalytic, in attracting the existing fund-level co-financing targets from private and commercial investors and in leveraging additional co-financing at the level of the underlying investments.

REPP 2 acknowledges that it cannot price for risk on a purely commercial basis. If commercially viable projects and a ready supply of commercial funders already existed, the funding gaps identified in Section B.2(a) and Annex 23 would not exist and there would be no need for a facility such as REPP 2. The learnings from REPP Phase 1 and other related facilities active across sub-Saharan Africa is that small-scale renewable energy businesses struggle to access private capital on terms which leave their projects commercially viable and that public financing in this space is still required. In this context, funding for REPP 2 will contribute to the continued growth and increasing sustainability of commercial renewable energy sectors in Host Countries.

If REPP 2 adopted a purely concessional approach to pricing, the effect would be to negatively disrupt a nascent market for renewables in Host Countries. A purely concessional pricing approach would lack financial rigour and have the unintended consequence of crowding out private sector funding. The approach to concessionality for REPP 2 will thus strive to achieve a balance between the need for a considered level of concessionality, in order to build the market and crowd in private sector funding, and the need for rigour and more commercial pricing approaches, where the market, investor appetite and project economics allow for it.

The concessionality of the GCF junior equity funding is defined to protect other more senior investors in the capital stack and provide an appropriate level of return for the GCF, while simultaneously crowding in (catalysing) other private and commercial investors. The junior equity tranches in Portfolio HoldCo 1 S.à r.l and Portfolio HoldCo 2 S.à r.l (sized and priced identically) are designed to provide first loss protection to other investors in the capital stack. The financial model demonstrates that the sizing (40% of total capital) and pricing of the junior equity tranche is met while also facilitating targeted returns for the equity providers. For senior debt providers, the structure and pricing of the capital stack also supports current market pricing.

4. Risk-sharing

The proportion of GCF and other public funding to private funding sources targeted for the REPP 2 fundraise (a ratio of 1:4) represents a realistic goal given the scale of private financing that Camco expects can be mobilised into this sector, for the activities proposed in this Programme.

For the GCF, where an investment in the junior equity tranche is proposed, risk will be shared *pari passu* with other investors.

B.6. Exit strategy

1. Overall exit strategy

REPP 2 is a fixed-term (20-year) fund. The limited partners of Camco REPP SCSp may agree to extend the objects of the REPP 2 as further set out in Annex 14 (Term Sheet).

Junior equity investors will earn a targeted return over the 20-year term of REPP 2. Senior equity investors will earn a targeted return in the base case. Senior debt will be repaid in full (principal plus accrued interest) over a weighted average tenor of 7 years. The Subscription and Partnership Agreement will, on customary terms, contain a range of exit scenarios in re limited members including insolvency of the limited member or transfer for value by the limited member. The dissolution mechanics for GCF REPP Holdings and Camco REPP SCSp will be set out in Holdings GPA and Subscription and Partnership Agreement, respectively. Following the completion of the activities of REPP 2 (including the dissolution of the Portfolio HoldCos), the final disbursements under the cash waterfall and the completion of any residual obligations of any Executing Entity pursuant to a Subsidiary Agreement, a recommendation would be made by Camco REPP GP S.a.r.l, as general partner of Camco REPP SCSp to the limited partners of Camco REPP SCSp for dissolution. In the absence of a voluntary dissolution, Camco REPP SCSp will be dissolved at the end of its term as set out in its deed of incorporation. Pursuant to, GCF REPP Holdings will be dissolved in a similar manner upon the satisfaction of all the obligations of the Holdings GP under the Holdings GPA.

For debt instruments, the baseline scenario is repayment in full (principal plus accrued interest) over the loan tenor. Where possible, REPP 2 will seek to reduce or exit its debt investments through refinancing opportunities, either for individual transactions or at the portfolio level.

The baseline scenario for equity investments is realisation in full of principal repayment plus returns over the targeted investment period. Where possible and profitable, REPP will seek an early exit from equity investments through new fundraising rounds and / or sell-down of existing shareholding to new investors.

The portfolio of active transactions for REPP 2 will be actively managed in order ensure orderly repayment of debt and equity returns and to proactively identify and execute opportunities for new investment, refinancing or sale.

2. Pricing approach

From a pricing perspective, the REPP 2 approach ensures an appropriate blend of concessional and commercial pricing. This approach accounts for its additionality mandate and seeks to minimize market distortion as far as possible – crowding in other sources of private and public funding rather than crowding other funding out through a pricing approach which is too concessional.

Where debt and equity investments are priced on a more concessional basis, the goal will be explicitly to support business growth in order to achieve long-term commercial sustainability.

Where REPP 2 makes equity investments in off-grid companies, opportunities for co-investment and sell-down exist as these businesses grow (later fundraises) – which may represent opportunities for REPP 2 to achieve partial or full exit or continue as a more-diluted equity sponsors.

Where REPP 2 provides development loans, these loans will include convertibility options which are designed to compensate the facility for the additional risk taken during development. This approach also enforces appropriate rigour among developers and sponsors.

Where business or project economics allow, REPP 2 will invest on purely commercial terms. This approach ensures minimum market distortion, maximises pricing rigour in line with market benchmarks and ensures that funds can be returned or repaid to REPP 2 and can achieve approach return for risk.

3. Paradigm shift

REPP 2 helps to innovate and transform the market for smaller-scale renewable in each of the Host Countries. Benefits including skills and technology transfer, development of an active market with direct private sector participation are all permanent. These elements achieve the targeted paradigm shift potential.

With the expectation of SHS systems, all technologies will have a lifetime over 20 years, therefore impacting countries transformation to resilient low carbon world. SHS companies are expected to succeed in their business mode, offering affordable energy to vulnerable communities, and hence providing long-term service beyond technical lifetime of the system. Monitoring and verification of the effectiveness of the projects is conducted in accordance with industry-best practices as described in Section E.7. REPP 2 performance will be reported to its stakeholders, through annual reports on its key KPIs, which are published on the REPP 2 website. Through organic growth of host country financial markets and with REPP 2 demonstrational effect it is expected that other developers and funds will adopt a similar approach.

C. FINANCING INFORMATION

C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	50			million USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
(i) Senior loans	0				As per Term Sheet	
(ii) Subordinated loans	0					
(iii) Junior Equity	50					
(iv) Guarantees	0					
(v) Reimbursable grants	0					
(vi) Grants	0					
(vii) Results-based payments	0					
(b) Co-financing information	Total amount			Currency		
	200			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace ³¹	Pricing	Seniority
DFIs, private investors, commercial banks	Senior Loans	50	million USD (\$)	10 years 5 years	As per Term Sheet	senior
DFIs, foundations, private investors, donors	Equity	100	million USD (\$)	13 years 5 years	As per Term Sheet	senior
DFIs, donors	Equity	50	million USD (\$)	20 years 12 years	As per Term Sheet	junior
(c) Total financing I = (a)+(b)	Amount			Currency		
	250			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	For the first close of REPP 2, there will be a requirement for co-investment on commercial terms from at least one additional investor, with the funding from third party investors matched by REPP Phase 1 / FCDO. A minimum of USD60 million has been agreed for a first close, in which the GCF would provide USD20 million, REPP Phase 1 / FCDO would provide a further USD20m, and additional commercial investor(s) would provide a further USD20 million, with one commercial investor in each senior equity and notes tranches.					
	Further disbursement from the GCF will be subject to the aggregate amount of commitments held by GCF REPP Holdings in the Camco REPP SCSp (including the requested disbursement) not exceeding 40% of the total equity commitments in the Camco					

³¹ For senior loans, reference to tenor reflects weighted average tenor for the portfolio. Reference to grace reflects a maximum grace period for principal and interest repayments. For equity investments, reference to tenor reflects the expected period over which full equity returns (principal plus preferred return) are realized. Reference to grace reflects the expected delay before dividends begin to be paid or exit is expected to be achieved.

	REPP SCSp (including the requested GCF disbursement) and no more than 20% of the total capital commitments in the Camco REPP SCSp (including the requested GCF disbursement). To meet investor expectations, REPP2 will offer a minimum internal rate of return to senior equity investors³². As the asset class is relatively new and considered very risky by investors, co-financing from private investors isn't expected to be raised at first close. The sizing of the notes and senior equity tranches in the REPP 2 capital stack may be adjusted prior to financial close to maximum returns to private and commercial investors. The Investment Manager is working to secure a portfolio guarantee to provide credit enhancement through the tranching structure, thereby protecting returns for more senior investors (including private and commercial investors) and enhancing the potential to attract commercial refinancing of the portfolio at a later date. The AE is working with the Swedish International Development Cooperation Agency (SIDA) to secure a second-loss portfolio guarantee, which will provide additional loss protection for senior equity providers as well as noteholders. The AE will conduct high-level KYC on prospective funders and undertake a simplified pre-classification exercise to identify the suitability of each party as a possible investor into REPP2. Detailed client classification and take-on procedures will be undertaken upon signing non-binding term sheets or letters of intent with respective parties. All of the potential co-financiers are well known financial institutions, so KYC risk is regarded as low. KYC will be conducted before the Subscription and Partnership Agreement is signed by the investor. Camco Management Limited or its Affiliates, as limited partner to the Camco REPP SCSp, will commit a Senior Interest amounting to 1% of the total commitments of the Camco REPP SCSp.
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C.2. Financing by component

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Financing and Support for Smaller-Scale Renewable Energy	Output 1: Increased commercial Viability	154.94	30.98	Junior Equity	123.96	Senior loans, Equity, Junior Equity	DFIs; commercial bank; foundations; private investors; donors
	Output 2: Increased renewable energy capacity and generation	94.16	18.83	Junior Equity	75.33	Senior loans, Equity, Junior Equity	DFIs; commercial bank; foundations; private investors; donors
	Output 3: Improved skills and capacity	0.90	0.19	Junior Equity	0.71	Junior Equity	DFIs, donors
Indicative total cost (million USD)		250.00	50.00		200.00		

The GCF contribution to Output 3 (Skills and Capacity Development) will be funded out if its contribution to operational expenditure, alongside all other investors.

³² Up to 15% to commercial private investors.

C.3 Capacity building and technology development/transfer	
C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐
A technical assistance component of REPP 2 will be funded separately to the Programme on a grant-funded basis through the Technical Assistance Facility (to be established as a separate legal entity). Until the TAF is established and operational, a portion of REPP 2 operating costs will be allocated to support market development, skills and capacity building and technology transfer.³³ REPP 2 funding for skills development and capacity building within Host Countries as well as for the smaller-scale renewables sector as a whole will be used to: • Build the capacity, skills and experience of renewable energy project developers, both local and international, as well as management and staff of early-stage off-grid companies; • Contribute to a stronger enabling environment for renewables through building institutional and regulatory capacity and strengthening policy and regulatory frameworks; and • Ensure business model innovation and technology transfer occur. These activities will be funded out of normal operating costs of REPP 2 (with contributions from all investors, including the GCF), based on a final budget that is expected to be approved by the general partner of Camco REPP SCSp.	

³³ A term sheet for initial funding of the TAF from REPP Phase 1 / FCDO is in an advanced stage of discussion.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

1. Programme targets aligned to GCF indicators for reduced emissions and increased resilience

Targets for the REPP 2 are as follows:³⁴

GHG emissions reduced, avoided or removed/sequestered

- Total GHG emissions reduced, avoided or removed/sequestered (Core Indicator 1): The target for GHG emissions reduced or avoided over the lifetime is 12,720,297 tCO₂e.³⁵
- Installed renewable energy capacity (Supplementary Indicator 1.3): The installed capacity target for the Programme is 330MW.
- Renewable energy generated (Supplementary Indicator 1.4): The renewable energy generation target for the Programme is 812,771 MWh/year.

Direct and indirect beneficiaries reached

- Total number of direct and indirect beneficiaries reached (Core Indicator 2): The target for the total number of direct beneficiaries, which is the number critical community services customers people benefiting of productive use of energy equipment that builds resilience, as well as people benefiting from isolated grids and mini-grids that improve the resilience of energy infrastructure is estimated ex-ante at 370,244.
- The numbers of direct female beneficiaries are estimated to be 111,073 and the number of direct male beneficiaries is estimated to be 259,171
- The target for the total number indirect beneficiaries, which is the number of people with improved energy connections through energy systems made resilient, reached through the Programme is estimated ex-ante at 359,603. The numbers of indirect female beneficiaries are estimated to be 179,801 and the same amount of indirect male beneficiaries is expected.
- Beneficiaries adopting improved and/or new climate-resilient livelihood options (Supplementary Indicator 2.1): The target for the number of beneficiaries adopting improved and/or new climate-resilient livelihood options through the Programme is 27,652 people. The numbers of female beneficiaries are estimated to be 8,295 and the estimated number of male beneficiaries is 19,356.

Infrastructure and built environment

- Value of physical assets made more resilient to the effects of climate change and more able to reduce GHG emissions: The target for improved climate resilience of energy infrastructure USD37 million.

2. Mitigation outcomes

Outcome 1: Reduced GHG emissions from renewable energy installations

The mitigation targets will be estimated using a technology split based on REPP 2's transaction pipeline, average technology-specific electricity generation estimates, UNFCCC-approved grid emission factors and average technology lifetimes. The energy baseline is the total annual electricity generated or renewable energy supplied by the project activity in line with the Clean Development Mechanism (CDM) methods: AMS-I-A - Electricity generation by the user (version 19), AMS-I.D – Grid-connected renewable electricity generation (version 18), and AMS-I.L – Electrification of rural communities using renewable energy (version 4). All mitigation targets will be aligned with host-country targets and priorities.

³⁴ All figures shown here reflect targets for the full REPP 2 Programme, not merely the portion attributable to GCF funding.

³⁵ Lifetime applied for each technology is as follows: wind = 20 years, grid-connected solar = 25 years, hydropower = 25 years, solar PV mini-grids = 20 years, and solar home systems (SHS) = 5 years.

3. Adaptation outcomes

Outcome 2: Improved climate resilience of energy infrastructure

The value of the energy infrastructure made resilient is estimated based on the estimated value of the pipeline of projects in countries in which the energy system is currently vulnerable due to 1) a reliance on hydropower in countries which are prone to drought and 2) grid instability due to widespread power outages and poor grid maintenance and management and exposure of grid to climate impacts. This methodology is consistent with GCF core indicator 3 (value of physical assets made more resilient to the effects of climate change). The Annex 2 provides country specific vulnerabilities of the energy sector.

Currently, in average 20 % of all REPP 2 on-grid investments are anticipated to assist in building resilience of energy infrastructure. Resulting to USD 37 million value of assets made more climate-resilient. Please see Annex 22 for detailed calculation and assumptions.

Outcome 3: Improved climate resilience of vulnerable people and communities

Improved climate resilience of vulnerable people and communities will be achieved through the provision of access to clean energy, which improves the ability of off-grid communities to respond to climate impacts and build resilience.

The number of people who will adopt improved livelihood options has been estimated with reference to the REPP Phase 1 experience, specifically relating to the number of productive uses of energy (PUE) activities and microbusinesses operated by off-grid companies. Using this data, it is estimated that 5% of off-grid customers will be microbusinesses and approximately half of these microbusinesses are anticipated to implement PUE activities that build resilience i.e. include solar water pumping and irrigation, produce refrigeration, electric milling, improved access to markets through mobility etc. Similarly based on REPP Phase 1.0 data it is estimated that 1% of off-grid customers will be providers of critical services e.g., clinics and hospitals, fire stations, schools, drinking water stations, cell phone towers etc.

4. Enabling outcomes

Outcome 4: Increased private sector investment

Finance mobilised for renewable energy generation benefits local developers, builds a stronger local market for smaller-scale renewables and helps to build gender equity.

REPP 2 will work to mobilise the private and public finance necessary to grow a robust and viable market for renewables by applying innovative climate funding solutions to overcome historic barriers to finance and demonstrate the bankability of projects to crowd in other investors. In addition, REPP 2 will facilitate knowledge transfer and innovation through demand-driven capacity building efforts to unlock investments and support successful project implementation. REPP 2 is anticipated to mobilise USD786 million additional finance towards climate-resilient renewable energy infrastructure.

USD70 million – or 30% – of REPP 2 funding will be targeted to assist the USD70 billion funding gap facing female entrepreneurs in Africa.³⁶ REPP 2 will aim to dedicate a portion of its funding to renewable energy companies in line with 2X Challenge's Gender Lens Investment (2X GLI) criteria. This means the company must:

- be founded by a woman or women or be majority-owned by women; and / or
- have at least 30% women in senior management; and / or
- provide products or services designed to disproportionately benefit women; and / or
- have a workforce that is at least 30% women, together with a progressive human resource policy.

Outcome 5: Contributing to stronger technology deployment, innovation and market transformation

REPP 2 will contribute to stronger enabling market environment for renewables through building capacity and financing a stronger level and faster technology deployment, innovation and market development in Host Countries. To achieve this, REPP 2 will:

- contribute to technology deployment, dissemination, development or transfer and innovation (Core Indicator 6): REPP 2 will contribute to more and more rapid technology deployment by providing targeted support to renewable energy companies and local financial institutions during early and construction stages of the assets. REPP 2 will also finance the deployment of renewable energy technology in nascent markets, often financing first-of-a-kind transactions and business models. The *innovative* and *transformational* nature of each transaction will be assessed during the project approval process based on approved selection criteria defined in the REPP 2. Operations Manual (Annex 21) and
- contribute to market development and transformation at the sectoral, local, or national level (Core Indicator 7): REPP 2 will stimulate sectoral development that would otherwise lag or not occur (*additionality*) and help to establish or build a market ecosystem that will allow the ongoing growth of small and distributed renewables in Host Countries and across the region (*transformational*). REPP 2 will take a bottom-up approach, working closely with its investees and by extension, at a deal level, Host Country governments and regulators to help to develop a viable renewable energy sector, and facilitating the development of a sustainable financing market for smaller-scale and off-grid renewable energy projects. REPP 2 will help to build skills and establish disciplines which underpin a commercially-viable market for smaller-scale renewables through its support for local developers and based on its application of sound commercial investment principles.

Progress towards this outcome will be monitored on an ongoing basis and reported via the mid-term and final reports to the GCF.

D.2. Paradigm shift potential

1. Potential for scale, replicability and sustainability at the Programme and market level

The REPP 2 Programme seeks to mobilise additional public and private funding and achieve scale and sustainability for the smaller-scale renewable energy market across sub-Saharan Africa.

REPP 2 aims to deliver a paradigm shift for the small-scale renewables sector within Host Countries and across the region as a whole, with demonstration potential more broadly across emerging markets. The Programme also aims to deliver a paradigm shift at the level of individual projects and renewable energy businesses, where additional funding is needed to achieve scale and sustainability for off-grid companies and where local developers of grid-connected projects need funding and support to develop technically- and commercially-viable projects.

In both senses, REPP 2 builds upon and scales up the early successes of REPP Phase 1 by mobilising finance from public, private and other commercial sources, using its blended finance approach.

The potential of REPP 2 to deliver this paradigm shift can be demonstrated through the following:

- The market study presented in Annex 2 identifies the scale of demand currently unmet for funding of smaller-scale renewables across the region. A *prior* assessment identifies a total funding demand and potential to scale up financing in excess of USD900 million, with more than USD565 million of that target located in Host Countries identified in this Funding Proposal;

³⁶ IFC 2019. [MSME Finance Gap: Assessment of the shortfalls and opportunities in financing micro, small, and medium enterprises in emerging markets](#)

- REPP 2 support for innovative³⁷ technologies, services and business models in nascent markets provides an opportunity to demonstrate technical and commercial viability, implement new regulatory frameworks and crowd in public and private finance;
In this regard, REPP 2 will scale up and replicate existing successes achieved by REPP Phase 1, such as in Burundi, where REPP Phase 1 support for the first solar PV IPP in the country demonstrated the viability of a renewable energy technology in new market and led to further expansion of its market for renewable energy IPPs.^{38,39} Another example can be given from Sierra Leone, where REPP Phase-1 investee provides electrification and access to PuE equipment through an innovative rental model, in which customers rent out solar-charged lithium ion batteries in 24-hour increments to service low income customers;⁴⁰
- By providing early-stage funding and technical support to local developers in Host Countries, REPP 2 will develop a stronger enabling market environment for smaller-scale renewables.
- The design of the REPP 2 facility ensures that GCF and other more senior funders can be repaid in full, inclusive of targeted returns. The blended finance approach of REPP 2 is designed to de-risk investments. With an expected aggregate portfolio loss of 20%, the junior equity providers provide protection to the equity investors, while their investment is still fully returned at the end of the fund. Senior debt providers, including private debt providers, enjoy most risk-reduction through the blended finance structure. The REPP 2 risk appetite, broader risk management procedures, project selection, deal structuring, and pricing and portfolio management approach are also designed to ensure sustainable returns for investors; and
- The platform approach of REPP 2 helps to deliver specific climate change mitigation and adaptation outcomes which contribute to climate-resilient development pathways consistent with national priorities and plans for Host Countries. Such support, and specifically the demonstration impact of a more commercial approach (which directly targets crowding in of private capital), works as a blueprint for other markets, including LDCs, across sub-Saharan Africa and further within emerging markets globally.

2. Potential for scale, replicability and sustainability at the project and investment level

- For REPP 2, investments in off-grid companies help to expand to nascent markets, achieve growth, commercial scale and long-term sustainability;
- For grid connected projects, funding and support from REPP 2 supports the development and implementation of technically- and commercially viable projects which operate for design lifetimes of 20 years and longer, often first-of-its-kind in the host country. The demonstration impact from such funding combined with the knowledge, skills and technology transfer to local developers, funders, institutions and regulatory bodies help to build the enabling environment for small-scale renewables, based on commercial best practise;
- At a community level, REPP 2 especially for off-grid electrification builds climate resilience for women, children and other vulnerable groups; and
- By the providing funding at scale for both off-grid and grid-connected renewables, REPP 2 helps to reduce GHG emissions and build climate resilience and reduce vulnerability to climate change for people and communities in rural areas, as well as improve the climate resilience of energy infrastructure. These represent durable and significant contributions to climate-resilient development pathways which are consistent with relevant national climate change adaptation strategies and plans.

³⁷ Innovative is defined as a project which is first-of-its-kind or among the first in the target country or regions based on technology, service, product or business model which supports the uptake of renewable energy or resilience-building

³⁸ REPP, 2021, [REPP invests in landmark hydro project in Burundi](#)

³⁹ Afrik21, 2021, [BURUNDI: Gitega approves the construction of 2 hydroelectric power stations](#)

⁴⁰ ESIA Africa, 2021, [Talking Points: The Power of pay per use battery sharing model](#)

D.3. Sustainable development

1. Sustainable Development Goals

REPP 2 contributes to the following SDGs:

- SDG 1: No Poverty;
- SDG 5: Gender Equality;
- SDG 7: Affordable, Clean Energy;
- SDG 8: Decent Work and Economic Growth;
- SDG 11: Sustainable Cities and Communities
- SDG 13: Climate Action
- SDG 17: Partnership for the Global Goals

The ultimate beneficiaries of REPP 2 funding for off-grid projects are the communities and businesses who will gain often first-time access to a clean, reliable and affordable source of electricity, which will contribute adaptive capacity and build climate resilience at a local level. Grid-connected projects supported by REPP 2 will provide clean electricity to the grid that will contribute to national GHG emission reductions helping to build the resilience of national energy infrastructure. Local private sector companies developing small-scale renewable energy projects will also benefit from REPP 2 funding and related capacity building leading to quality jobs. REPP 2 will mobilised additional funding towards to climate action.

2. Environmental co-benefits (co-benefit 4)

The potential environmental and social impacts of all projects will be managed according to the REPP 2 Environmental and Social Policy and Procedures (to be approved by the general partner of Camco REPP SCSp) and the Environmental and Social Management (ESMS) Framework (Annex 6). Both the Policy and Procedures and the ESMS will adhere to IFC Performance Standards. At the project / investment level, a specific environmental and social management system (ESMS) will be established to mitigate and manage any negative impacts. REPP 2 capacity building activities and early-stage funding will improve environmental and social performance within portfolio companies (co-benefit 4). The improved performance will be measured through the establishment and implementation of high quality ESMS and will lead to benefits such as improved waste management practices, mitigating negative impact to biodiversity, soil and water resources. In addition to GHG mitigation, additional co-benefits, such as improved ambient air quality are expected to occur. However, these are not quantified.

3. Social co-benefits including health co-benefit

At the household level, the small renewable energy projects that REPP 2 will offer multiple environmental and social co-benefits. REPP 2 funding and support will lead to improved quality of life through communities and critical services connected to electricity (co-benefit 3). Co-benefits include improved access to lighting, healthcare, lighting, refrigeration, improved access to information and more productive use of energy (PUE) opportunities. These co-benefits will allow further socioeconomic development of Host Countries and their off-grid communities. There are particular co-benefits for women and girls, where greater energy access allows them to participate in new income generation and educational activities. In general, improved access to renewable energy through REPP 2 will act as a driver for long-term economic growth in local communities, increasing their resilience to climate change. These co-benefits will contribute towards the achievement of SDG 1.

4. Economic co-benefits

Renewable energy projects have the potential to create employment opportunities, both during development and operation and across numerous skill sets, from engineering and construction to sales and customer services (co-benefits 1). As above, first-time access to energy also generates economic co-benefits through new opportunities for social and economic development through extended working hours and greater efficiency in the workplace. REPP 2 will work to ensure all jobs created through the projects it supports provide decent work, respect labour rights, and improve skills in line with the REPP 2 Environmental and Social Policy and Procedures and the ESMS Framework, and supporting the achievement of SDG 8.

5. Gender-sensitive development co-benefits (co-benefit 2)

The positive impacts identified above will specifically benefit women, since across rural communities in Africa in particular women and girls are the primary energy users in the household, responsible for sourcing fuel, paying for it and managing its use. REPP 2 will be committed to building capacity on gender equality in line with the gender action plan of REPP 2 (see Annex 8). This will be achieved by assisting investee companies in establishing and implementing gender action plans. All companies are required to establish action plans which will help to increase awareness and improve the equality situation at company and project implementation level.

In addition, a target of 30% of REPP 2 funding will be directed to assist the USD42 billion funding gap female entrepreneurs face in Africa in line with 2X GLI criteria. These co-benefits will contribute towards the achievement of SDG 5 – Gender Equality.

D.4. Needs of recipient

6. Vulnerability of the country and beneficiary groups

Country-specific climate change hazards, vulnerabilities, and impacts include:⁴¹

- **Cameroon:** Hazards include drought, heavy rainfall events, flooding, heavy winds, high temperatures, and heatwaves. Energy infrastructure is vulnerable to precipitation, extreme rainfall and flooding, as well as drought, and sea-level rise. Rural electricity access is low, 25% and 26 % of population lives in poverty making them specifically vulnerable to climate impacts;
- **DRC:** Hazards include intense rainfall, erosion, floods, heatwaves, and seasonal droughts. Reliance on the use of traditional biomass significantly increases the climate vulnerability of the population Rural electricity access is extremely low, 1% and majority (70 %) of population lives in poverty making them extremely vulnerable to climate impacts;
- **Lesotho:** Hazards include more frequent, severe droughts and increased flooding. The energy infrastructure, particularly hydropower, is highly vulnerable to increasing temperatures. Reliance on subsistence agriculture significantly increases the climate vulnerability of the rural population. Rural electricity access is low 35% and 32 % of population lives in poverty making them vulnerable to climate impacts;
- **Madagascar:** Hazards include intensification of cyclones and associated flooding, sea-level rise, intensification of droughts in the south and west, and a more intense wet season elsewhere. The energy sector is vulnerable to a reduction in precipitation and cyclones destroying infrastructure. Over-reliance on the use of firewood further increases the level of vulnerability. Rural electricity access is low extremely low 11% and majority (81 %) of population lives in poverty making them extremely y vulnerable to climate impacts;
- **Malawi:** Hazards include floods, droughts, strong winds, intense rainfall, riverine and flash floods and tropical cyclones. Vulnerabilities include poverty, rapid population growth, dependence on the agricultural sector reliance on rain-fed agriculture and biomass as well as lack of financing and weak institutional

⁴¹ Detailed references for the summaries provided here and elsewhere in Section B.1 can be found in Annex 2.

capacity to manage climate change. Rural electricity access is extremely low 6.6% and majority (70 %) of population lives in poverty making them very vulnerable to climate impacts;

- **Niger:** Hazards include higher temperatures, increased frequency and intensity of extreme events, increased frequency of erratic rainfall and extreme droughts. Vulnerabilities include lack of financing, rapid population growth, and low institutional capacity. The energy sector in Niger is expected to be impacted by the effects of heavy rainfall resulting in damage to energy infrastructure. Rural electricity access is extremely low 13% and half of the (51 %) of population lives in poverty making them extremely vulnerable to climate impacts;
- **Nigeria:** Hazards include extreme heatwaves and rainfall events leading to droughts, floods, erosion, as well as rise in sea-level leading to intrusion and inundation of land. The energy system is vulnerable to increased demand for cooling and heating and already experiencing multiple supply shortages. Rural electricity access is low 26% and 31 % of population lives in poverty making them specifically vulnerable to climate impacts;
- **Sierra Leone:** Hazards include floods, droughts, coastal erosion, decreasing river flow, storm surges, flash floods, sea-level rise, and high winds. Vulnerabilities include water resources and a high dependence on commodity prices. Rural electricity access is extremely low 4.2% and 26 % of population lives in poverty making them specifically vulnerable to climate impacts; and
- **Zambia:** Hazards include flooding, including flash floods, extreme temperatures and droughts. Vulnerabilities include poverty, food and water insecurity and dependence on natural resources. Rural electricity access is extremely low 14% and majority (61 %) of population lives in poverty making them extremely vulnerable to climate impacts.

There is a clear need in all REPP 2 target countries for affordable access to energy and adaptation solutions which help to build the resilience of most vulnerable people's resilience.

7. Economic and social development level of the country and affected population

Eight of the REPP 2 Host Countries are classified as LDCs. The economic and social development status summarised below, shows that development status is low, confirming populations vulnerability to climate change.⁴²

Country	GDP (USDbn)	Gross national income per capita (USD)	National poverty ratio (%)	Life expectancy at birth, total (years)	Gini index 0 = max. equality; 100 = max. inequality
Cameroon	25.24 (2021)	1,662 (2021)	25.7 (2014)	60 (2020)	46.6 (2014)
DRC	53.96 (2021)	584 (2021)	69.7 (2012)	61 (2020)	42.1 (2012)
Lesotho	2.52 (2021)	1,167 (2021)	32.4 (2017)	55 (2020)	44.9 (2017)
Madagascar	14.64 (2021)	516 (2021)	80.7 (2012)	67 (2020)	42.6 (2012)
Malawi	12.63 (2021)	643 (2021)	70.1 (2019)	65 (2020)	38.5 (2019)
Niger	14.95 (2021)	595 (2021)	50.6 (2018)	63 (2020)	37.3 (2018)
Nigeria	440.78 (2021)	2,085 (2020)	30.9 (2018)	55 (2020)	35.1 (2018)
Sierra Leone	4.2 (2021)	516 (2021)	26.1 (2018)	55 (2020)	35.7 (2018)
Zambia	21.2 (2021)	1,120.6 (2021)	61.4 (2015)	64 (2020)	57.1 (2021)

Please refer to Annex 8 for country-specific gender baselines.

8. Absence of alternative sources of financing

Adoption of renewable energy technologies is a priority under all target country NDCs, with additional funding needed to address climate change mitigation and adaption challenges. Despite this, the analysis presented in Sections B.1 and B.2 demonstrates the significant funding and regulatory barriers faced by these countries

⁴² World Bank, [World Bank Open Data](https://data.worldbank.org/), last accessed 24/09/2022

to achieve these ambitions. Commercial funding is not readily available, and most investors look to invest in more developed and stable markets.

9. Needs for strengthening institutions and implementation capacity

REPP 2 contributes to the development of an enabling environment in Host Countries by supporting technology transfer and capacity development of local developers. Through its investments, which are often first-of-a-kind, REPP 2 can also support institutional knowledge development in the Host Countries by engaging with government stakeholders – directly or through its investees – to discuss the regulatory frameworks and standard project documentation, sharing a financier’s perspective on what is required to ensure ‘bankability’ of these documents from an international standard. This could subsequently help attract private sector investors to fund the renewable energy priorities which are key to target country climate change responses. Examples include: a) providing a private sector perspective on the current policy and regulatory framework for mini-grids in Nigeria to further strengthen the successful and timely implementation of the Nigeria Electrification Programme; b) advising the Zambian government on the design of a micro-generation programme which would successfully attract private investment; supporting the Government of Niger in strengthening private sector engagement in renewable energy development. As it further develops its relationship with national stakeholders, Camco will provide advice and support for national stakeholders going through the GCF accreditation process (such assistance is already being discussed in Nigeria and Lesotho).

D.5. Country ownership

REPP 2 will place a strong focus on ensuring country ownership of its activities. While it will apply a bottom-up approach to its operations, driven by demand from renewable energy project developers and companies, the Programme will also place considerable emphasis on the alignment of its investments with national-level climate, energy and sustainable development priorities and ensuring that effective consultations with key stakeholders (including both government and non-government entities) take place to inform the Programme’s design.

1. Alignment with and contribution to national priorities and climate strategies

In 2020, Camco undertook an analysis of the key policies in all REPP Phase 1 countries of operation and assessed the extent to which the portfolio was aligned with key national policy targets and priorities. The assessment identified a high level of alignment overall.⁴³ It has also helped to improve how REPP Phase 1 aligns its investment decisions with national policies. The same approach will be critical to ensuring country ownership of REPP 2.

REPP 2’s diverse technology focus and support for the increase of renewable energy generation capacity and diversification of energy systems is strongly aligned with all Host Countries’ NDCs. Most of these countries also consider renewable energy development as a strong adaptation measure through the improvement of energy infrastructure resilience and – by improving livelihoods, with particular attention given to women and vulnerable groups - the resilience of local populations.

Through its investments, REPP 2 will support the numerous national priorities outlined below. A more detailed analysis is provided in Annex 2.

Cameroon

- Reduce emissions by 35% by 2030 through increased renewable energy capacity and expanded energy access (conditional, Updated NDC).
- Reduce energy sector’s vulnerability through focus on small-scale and distributed generation (National Climate Change Adaptation Plan).

⁴³ REPP, 2021: [Advancing national policy agendas through responsible investing](#)

- Support gender goals through focus on the empowerment of women (National Gender Policy; National Development Strategy 2020-2030).

Democratic Republic of the Congo

- Reduce emissions by 21% compared to the 2030 BAU scenario and strengthen resilience of infrastructure (adaptation goal) through investments in renewable energy (Updated NDC; National Adaptation Plan to Climate Change).
- Expand access to modern energy sources through funding for off-grid projects.
- Optimise related adaptation and resilience benefits, including reduced use of traditional biomass (Updated NDC; DRC GCF Country Programme; National Development Strategic Plan).

Lesotho

- Reduce emissions by 25% by 2030 (conditional, Updated NDC)
- Increase power generation using solar PV and wind technologies, as well as the expansion of rural electrification through mini-grid development (conditional, Updated NDC).
- Reduce reliance on hydropower and biomass-based generation, thereby strengthening climate resilience of energy supply (National Climate Change Policy 2017-2027; National Adaptation Programme of Action 2007).

Madagascar

- Reduce emissions by 14% by 2030 (conditional, Updated NDC).
- Increase clean energy supply and reduce reliance on costly fuel imports (Government's Emergence Initiative for Madagascar 2018-2023).
- Build resilience of energy infrastructure and reduce biodiversity loss by diverting energy generation away from fuelwood (National Adaptation Plan).
- REPP 2's focus on gender equity is also well-aligned with the framework of the Government's Emergence Initiative.

Malawi

- Prioritise solar PV, hydropower, wind and off-grid solar (conditional, Updated NDC target).
- Increase renewable energy generation capacity to strengthen energy security (National Adaptation Plan Framework (NAPF).
- Increase climate resilience of infrastructure through support for small-scale solar and run-of-river hydro (National Climate Change Management Policy (NCCMP).
- Increase resilience of rural communities through investments in off-grid solar (Updated NDC; NAPF; NCCMP).
- Support women's economic empowerment through focus on gender equity (National Gender Policy).

Niger

- Reduce emissions by 45% by 2030 (conditional, Updated NDC)
- Promote renewable energy and related 'green' job creation (National Climate Change Policy).
- Support private sector development (Sustainable Development and Inclusive Growth Strategy).
- Improve women's lives through rural electrification and related job creation (Updated NDC).

Nigeria

- Reduce emissions by 47% by 2030 (conditional, Updated NDC target).
- Deliver energy access through a combination of centralised and decentralised approaches (Rural Electrification Strategy and Implementation Plan; Rural Electrification Agency's Off-grid Electrification Strategy).
- REPP 2 will also closely monitor the development of the National Adaptation Plan (as envisaged in the National Adaptation Plan Framework) to ensure its alignment with national priorities.

Sierra Leone

- Reduce emissions by 10% by 2030 and 25% by 2050 (Updated NDC).
- Increase renewable energy capacity and energy access (National Renewable Energy Policy and Action Plan).
- Increase the participation of women in the sustainable energy sector, particularly through off-grid development, thereby building adaptive capacity (Updated NDC).

Zambia

- Reduce emissions by 47% by 2040 (Updated NDC).
- Increase the proportion of renewable energy in the energy mix and promote investments in climate resilient and low carbon development pathways (National Policy on Climate Change 2016).
- Increase equitable access to affordable renewable energy (Zambia's National Gender Policy).

2. Engagement with NDAs and other stakeholders

In 2021, Camco began the process of obtaining no objection letters (NOLs) from the respective National Designated Authority (NDA) in most of the countries that will be supported through REPP 2 GCF's contribution. This involved introductory letters and calls in which the concept of REPP 2 and its planned activities and operational model were explained. The REPP 2 draft Concept Note was shared with key stakeholders in February 2022, followed by presentations and/or email exchanges where specific points raised were discussed, with feedback integrated into the final version of the Concept Note and this Funding Proposal. The Funding Proposal has also been shared and discussed with the NDAs. The ESMF was shared in July 2023.

As part of the engagement process, Camco consulted respective NDAs and other relevant sectoral stakeholders, particularly ministries of energy and rural electrification agencies, virtually and/or in-person (in-person meetings were held with NDAs and/or NDA advisors of Cameroon, DRC, Lesotho, Madagascar, Niger, Nigeria, Sierra Leone and Zambia, as well as several sectoral stakeholders). The aim of these consultations was to ensure an alignment of interests and develop a strong basis for continuous engagement during the approval process of this Funding proposal and throughout the REPP 2 operational timeframe, which will include annual progress updates and ad hoc engagement as required to discuss specific investments. NOLs have been secured from all Host Countries. Additional countries may be added to the programme after initial GCF Board approval, provided they submit an NOL, and the addition is approved by the GCF Board. In addition to public sector engagement, Camco has engaged with multiple private sector stakeholders and civil society representatives.

Country ownership of REPP 2, once operational, will be ensured through regular dialogue and long-term relationship building with national governments and other relevant national stakeholders (identified by the NDAs). This will include:

- Annual updates to NDAs and sectoral stakeholders on the progress of REPP 2 activities and broader discussion on market development;
- Ad hoc engagement with relevant government agencies, management units of sector support programmes, and/or non-governmental institutions and civil society organisations to ensure alignment and inform the requirements outlined in the REPP 2 E&S and gender-mainstreaming policies; and
- Ad hoc engagement as part of investment due diligence (DD) process.

Once the Programme is operational, REPP 2 will promote its activities internationally and in Host Countries through industry associations and its on-the-ground network and partners to ensure local companies in those markets are aware of - and have the opportunity to benefit from - the REPP 2 offering.

Engagement of ultimate beneficiaries at the project site will be led by REPP 2 investees. REPP 2 has a responsibility of ensuring all its investees continually engage with all stakeholders identified within the project area of influence (both directly and indirectly), to fully understand the shortfalls against its standards and international best practice, and identify potential risks, which can effectively be managed. All REPP 2's investees will be required to design and effectively implement a Stakeholder Engagement Plan, as part of their ESMS (see REPP 2's Environmental and Social Management Framework (ESMF) in Annex 6 for further information).

3. Suitability of the Accredited Entity

As a climate and impact fund manager, Camco seeks to deliver on its vision for "A brighter future, today" by supporting the clean energy transition in emerging markets. With local offices and staff in West, East and Southern Africa, Camco leverages its on-the-ground presence, sector expertise and more than 30 years of experience to deliver value as both the REPP Phase 1 manager and as a GCF Accredited Entity.

As the REPP Phase 1 manager, Camco has been integral to the programme's design and implementation and is leading the re-design and fundraising activities for REPP 2. Internally, Camco resources the REPP Phase 1 management role separately from delivering its obligations as a GCF Accredited Entity. From a governance perspective, this separation of roles and responsibilities is important to maintain independence and a focus on delivery.

Camco (as a GCF Accredited Entity), REPP 2 and the GCF, leveraging their respective skills, experience and expertise, all seek to achieve meaningful and lasting climate change mitigation and adaptation impacts. Camco, which is a signatory to the UN Global Compact, recognises its responsibility and opportunity to support both the Paris climate targets and the Sustainable Development Goals. Both remain core objectives of our work.

D.6. Efficiency and effectiveness

1. Rationale for a blended finance approach

The supply of private (and public) capital is scarce for smaller-scale renewables across sub-Saharan Africa, and the limited risk appetite of (particularly) private commercial funders exacerbates this situation. In order to deliver REPP 2's targeted mitigation (via renewable energy) and adaptation (via climate resilience) impacts, the finance mobilisation must expand beyond the single grant-based donor approach of REPP Phase 1 to the blended finance approach of REPP 2. This paradigm shift in funding structure is necessary to provide finance at scale and assist the targeted countries in achieving their climate goals.

2. Adequacy and reasonableness

As outlined in detail in Section B.5, the proposed capital structure for REPP 2 - and the GCF's position within it - have been designed to crowd in additional sources of both public and private capital. The GCF investment in the junior equity tranche would also be shared by another investor with these investments mobilising additional public and private investment in more senior equity and debt tranches.

REPP 2 acknowledges that it cannot price for risk on a purely commercial basis. Since small-scale developers often struggle to access private capital on terms that leave their projects commercially viable, public financing in this space is still required. In this context, funding for REPP 2 will contribute to the continued growth and increasing sustainability of commercial renewable energy sectors, as well as the delivery of climate mitigation and adaptation impacts in Host Countries.

On the other hand, if REPP 2 adopted a purely concessional approach to pricing, it may have the unintended consequence of negatively disrupting a nascent market for renewables in Host Countries crowding out private sector funding. REPP 2's approach to concessionality will thus strive to achieve a considered balance between the need for concessionality, in order to build the market and crowd in private sector funding, and

the need for rigour and more commercial pricing approaches, where the market, investor appetite and project economics allow for it.

3. Efficiency and effectiveness

At the fund level, GCF's funding would mobilise an additional USD200 million from public and private sources in the same or in more senior tranches of the capital stack. Expressed as a ratio of funding expected to be mobilised in more senior tranches (USD150 million), this would represent a **funding mobilisation ratio** of 3:1. Including funding expected to be mobilised into the junior equity tranche alongside the GCF, this ratio would be 4:1.

At the level of investments planned to be made by REPP 2, the GCF investment would be expected to mobilise total finance (total project costs less REPP2 investments) in excess of USD786 million. This would represent an **investment leverage ratio** (total project costs / GCF funding) of 20:1.

REPP 2 will aim to mobilise an additional USD50 million from private investors into both the equity and senior debt tranches. This would represent a **private sector leverage ratio** of 1:1. The analysis presented in Section B.2 identifies limited access to finance in general, and very limited access to private sector capital in particular, as key barriers for the small-scale renewable energy sector in sub-Saharan Africa. Given the limited risk appetite from private and commercial funders, especially for smaller-scale renewables where funding quantum are lower and the potential for returns relative to the level of costs are constrained, it appears unrealistic to target a significantly higher private sector leverage ratio.

The capital structure of REPP 2 is designed to address these key market failures by catalysing funding into the sector. This goal is strongly aligned to the GCF's strategic objective to catalyse climate finance for mitigation and adaptation which is new, additional, adequate and predictable.

The indicative REPP 2 pipeline can be used to estimate cost efficiencies (total installed project cost, including fixed financing costs, per MW of installed capacity), per technology. In general, cost efficiencies for small-scale grid-connected projects and off-grid electrification programmes will be lower than global benchmarks for large utility-scale projects, due to the effects of economies of scale as well as additional costs relating to supply chain and logistics, when purchasing equipment in smaller quantities and needing to maintain operations over a diverse range of sites, often in rural areas. The higher total installed cost for small-scale renewables also reflects the higher cost of capital for such businesses and projects, another financing barrier which REPP 2 seeks to overcome. Total installed cost estimates from the REPP 2 indicative pipeline and relevant benchmarks are shown below.

Generation technology	Installed cost – REPP 2 pipeline (USDm/MW)	Installed cost – global benchmark (USDm/MW)	Source and comments
Solar PV (grid-connected)	1.732	0.857	IRENA, 2022 ⁴⁴ , for global (larger) utility-scale projects
Onshore wind	2.934	1.325	IRENA, 2022
Hydropower	590	2.135	IRENA 2022 (not restricted to small run-of-river hydropower)
Solar mini-grids	4.908	3.659	ESMAP, 2022 ⁴⁵ (the ESMAP benchmark covers capital costs only)
Solar home systems	3.174	1600	IRENA, 2016 ⁴⁶

⁴⁴ IRENA, 2022: Renewable Power Generation Costs in 2021, International Renewable Energy Agency, Abu Dhabi.

⁴⁵ Energy Sector Management Assistance Program (ESMAP), 2022: Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers. Washington, DC: World Bank. www.esmap.org/mini_grids_for_half_a_billion_people.
License: Creative Commons Attribution CC BY 3.0 IGO

⁴⁶ IRENA, 2016: Solar PV in Africa: Costs and Markets

Overall	394	
Other efficiency and effectiveness benefits of REPP 2 - For the equity tranche, targeted returns in the base case are appropriate given the experience from REPP Phase 1 and relevant market benchmarks. - For the GCF and other investors in the junior equity tranche, targeted returns are appropriate in the base case. These returns are aligned with a blended finance approach to mobilise additional public and private capital in more senior tranches. - For the senior debt tranche, targeted returns in the base case are appropriate given representative pricing and the level of protection provided by the blended finance structure of REPP 2. - The estimated cost per tonne of carbon dioxide equivalent (PMF-M core indicator 2) is USD18.35 per tCO_{2e}.⁴⁷ The lower installed capacity of small-scale projects limits their generation potential and consequently the potential to generate emission reductions. In general, total project costs (capital and operating expenditure) are unavoidably high relative to the emission reduction potential of the plant. - Based on a target of 729,847 direct and indirect beneficiaries, the estimate cost per beneficiary over the Programme's lifespan is USD 320 (using total funding of USD250 million as the numerator in the ratio calculation).		

⁴⁷ PMF-M core indicator 2: Total investment cost (USD250 million)/expected lifetime emission reductions (12.7mtCO_{2e}) = USD 18.35 per tCO_{2e}

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

- ☒ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential

Assessment dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	In the target markets as well as most LDCs and other emerging markets across sub-Saharan Africa, there is limited funding available and limited appetite for investment into the smaller-scale renewable energy sector. The limited supply of funding affects off-grid companies, which require growth capital to achieve scale. Funding limitations also affect grid-connected projects which lack capital during the	<u>Medium</u>	The target scenario is a situation in which, through REPP 2, innovative and resilient renewable energy projects in Host Countries and other markets can attract funding to become technically and commercially viable at scale.	REPP 2 will originate and provide early-stage growth funding for off-grid companies as well as development-stage for grid-connected projects. REPP 2 funding and support will help to demonstrate technical and commercial viability prior to financial close, thereby de-risking investments in the renewable energy sector and crowding in additional finance. In the off-grid sector, REPP 2 will also provide gap funding to finance the expansion of solar home system (SHS) companies once their business models are proven and construct mini-grid portfolio as well as grid-connected projects. In doing so, REPP 2 may take an anchor investment position and / or help access risk mitigation instructions to unlock capital markets and crowd in other forms of capital.

	development phase as well as for construction.			The net impact of REPP 2 funding is to increase the scale of renewable energy power capacity, generation and access to clean energy, access to climate resilient energy solutions, all of through which the quality of life in beneficiary communities is improved and the quantum of finance mobilised into the renewable energy sector in host markets.
Replicability	Nascent markets for renewable energy are characterised by limited availability and access to finance, inflexible and high costs of finance from those willing to provide it, limited skills and experience of local developers, and untested markets where policy and regulatory frameworks are unclear, untested or uncertain. All of these factors limit the potential to replicate examples of successful renewable energy investments	<u>Medium</u>	The target scenario is a situation in which REPP 2 demonstrates the successful application of blended finance and innovative financing and support mechanisms in a resilient renewable energy sector. Renewable energy markets in Host Countries and across the region are viewed as bankable, with private lenders and investors more willing to participate to on more standard commercial terms.	REPP 2 support for a company or project in a nascent market will be based, in part, on the project's potential for replication. Its financial and non-financial support will: • Promote first-time applications of technologies and innovative funding instruments in new markets; • Crowd in other sources of both public and private finance, and actively promote investment in smaller-scale supporting efforts to replicate its successes within Host Countries, across the region and in other emerging markets.

	within Host Countries, across sub-Saharan Africa and beyond to other emerging markets.			
Sustainability	Lack of early-stage and development funding in LDCs for small-scale renewable energy projects. Limited experience and commercial backing for local developers and a limited appetite for investment in new markets means that off-grid companies cannot achieve sustainable levels of growth and that the market for on-grid renewables cannot grow without continued interventions from facilities like REPP 2 and from DFIs.	<u>Medium</u>	The target scenario is a situation in which off-grid companies can achieve scale and long-term sustainability for their businesses. For on-grid projects the target scenario is one in which developers and sponsors can attract sufficient volumes of commercial funding, on commercially viable terms and can operate the installations in a manner which can services debt obligations and generate commercially sustainable returns for investors. For the smaller-scale renewable energy sector as a whole, the target scenario is one in which the appetite for risk is increased as real and perceived risks are addressed and the supply of commercial funding is increased, ensuring that the market can operate on more commercial terms without the need for ongoing interventions.	The REPP 2 facility is designed to be 'semi-evergreen,' to ensure that GCF and other commercial funding can be repaid. REPP 2's capital stack is designed to effectively de-risk investments by commercial funders, through the provision of a junior equity tranche provided by the GCF, REPP Phase 1 reflows and other donors. REPP 2's risk appetite, broader risk management procedures, project selection, deal structuring, and pricing and portfolio management approach are designed to ensure sustainability for investors. Furthermore, the ongoing support provided by REPP 2 will also help to build capacity, establish a track record, and attract other finance to achieve sustainability by providing growth capital in nascent markets.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁴⁸	
<u>MRA1 Energy generation and access</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Calculated ex-post based on renewable energy generation data from portfolio companies and reported in annual reports. An independent verification will be conducted before interim and final evaluations.	0	1,666,584 tCO ₂ e	4,166,460 tCO ₂ e (proportional allocation to fund lifespan ⁴⁹) and 12,720,297 tCO ₂ e (full REPP 2 Programme over technology lifetime ⁵⁰)	The baseline emissions have been calculated based on the fuel consumption of the technology that would have been used to generate the equivalent quantity of energy in the absence of REPP 2's intervention, in line with the CDM methods AMS-I.A – Electricity generation by the user (version 18), AMS-I.D – Grid-connected renewable electricity generation (version 18), and AMS-I.L – Electrification of rural communities using renewable energy (version 4).

⁴⁸ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

⁴⁹ Refers to total emission reduction potential calculated at the end of programme implementation period i.e., 7 years.

⁵⁰ Refers to total emission reduction potential at the end of the technology lifetime (25 years for on-grid PV, wind and hydro; 20 years for PV mini-grids; 5 years for SHS).

<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.3: Installed renewable energy capacity</u>	To be measured ex-post based on installed renewable energy capacity data from portfolio companies and reported in annual reports. An independent verification will be conducted before interim and final evaluations.	0	132 MW	330 MW	The baseline is the current energy generation capacity in the host country based on CDM methods AMS-I.A – Electricity generation by the user (version 18) and AMS-I.D – Grid connected renewable. Installed capacity refers to the rated power output when operational in megawatts (MW) of the clean energy technology, either in the output of electrical power (Mwe) or thermal power (MWt).
<u>MRA1 Energy generation and access</u>	<u>Supplementary 1.4: Renewable energy generated</u>	To be measured ex-post based on renewable energy generation data from portfolio companies and reported in annual reports. An independent verification will be conducted before	0	325,108 MWh	812,771MWh	The energy baseline is the total annual electricity generated or renewable energy supplied by the project activity in line with the CDM methods AMS-I.A – Electricity generation by the user (version 18), AMS-I.D – Grid connected renewable electricity generation (version 18),

		interim and final evaluations.				and AMS-I.L – Electrification of rural communities using renewable energy (version 4).
<u>All adaption results areas</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Direct beneficiaries are calculated and disaggregated ex-post based on critical services customers and resilient PUE customers, as well as estimated number of people within households benefiting from climate resilient infrastructure (mini-grids). Indirect beneficiaries are counted ex-ante as the number of people benefiting from energy assets made climate resilient (on-grid, isolated grids).	0	Total direct and indirect beneficiaries 291,939	Total direct and indirect beneficiaries 729,847	Direct beneficiaries are calculated based on the number of active SHS and mini-grid customers that are anticipated to build resilience and average household size of Host Countries. Indirect based on electricity production from resilient energy infrastructure divided by average electricity consumption in the country. Ex-post reporting disaggregated by gender
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	To be calculated and disaggregated ex-post based on critical services customers and resilient PUE	0	Total direct beneficiaries: 148,098	Total direct beneficiaries: 370,244	Calculated based on the number of active SHS and mini-grid customers that are anticipated to build resilience and

		customers, as well as estimated number of people within households benefiting from climate resilient infrastructure (mini-grids).		Direct female beneficiaries: 44,429 Direct male beneficiaries: 103,668	Direct female beneficiaries: 111,073 Direct male beneficiaries: 259,171	average household size of Host Countries Ex-post reporting disaggregated by gender
<u>ARA3 Infrastructure and built environment</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Indirect beneficiaries are counted ex-ante as the number of people benefiting from the energy assets made climate resilient (on-grid, isolated grids).	0	Indirect beneficiaries: 143,841 Indirect female beneficiaries: 71,921 Indirect male beneficiaries: 71,921	Indirect beneficiaries: 359,603 Indirect female beneficiaries: 179,801 Indirect male beneficiaries: 179,801	Indirect based on electricity production from resilient energy infrastructure divided by average electricity consumption in the country. Ex-post reporting disaggregated by gender

<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	To be calculated and disaggregated ex-post based on critical services customers and resilient PUE customers, as well as estimated number of people within households benefiting from the PUE solutions.	0	People benefiting from resilient PUE: 55,303 Women benefiting from resilient PUE: 16,591	People benefiting from resilient PUE: 138,258 Women benefiting from resilient PUE: 41,447	Calculated based on the number of active PUE customers that are anticipated to build resilience and average household size of Host Countries. These numbers relate to direct beneficiaries only; indirect beneficiaries have not been defined. Ex-post reporting disaggregated by gender.
<u>ARA3 Infrastructure and built environment</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	Calculated ex-post based on portfolio investments in countries in which renewable energy generation will improve the resilience of the energy infrastructure. Reported in annual reports and with an independent verification conducted before interim and final evaluations.	0	USD 15million	USD 37 million	The value of the energy assets made resilient was estimated based on the value of pipeline of projects in countries in which the energy system is currently vulnerable due to reliance on hydropower in countries which are prone to drought. The % investment in energy infrastructure that improve resilience is estimated based on technology and country.

						Details provided in Annex 2 and 22. The resilience of each investment will be assessed with the help of ADAPT tool at the time of investment decision.
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 6: <u>Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Local developers often lack the skills and experience needed to develop, construct and operate projects. In under-developed markets, local commercial lenders often have limited or no ⁵¹ experience of lending to the renewable energy sector.	<u>Low</u>	Improved renewable energy generation capacity, knowledge and technical skills at the local renewable energy company and funder level measured through: Operational renewable energy capacity in MW: 330 Private sector funding leveraged in USDm:393	REPP 2 will contribute to enabling environment through technology transfer by providing targeted support to renewable energy companies during early and construction stages of the assets. REPP 2 will also finance the deployment of renewable energy technology in nascent markets.	<u>Multi-countries</u>

<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	Generally higher than average levels of real and perceived risk and a limited appetite for investment in new markets, especially in small-scale renewable energy sector.	<u>Low</u>	Improved market ecosystem that will allow growth of small and distributed renewables in the region, measured through: Number of renewable energy transactions funded: 37 Number of innovative renewable energy transactions funded⁵²: 26	Through its investment strategy, REPP 2 will seek to be: <i>Additional</i>, in that it will stimulate sectoral development that would otherwise lag or not occur, and <i>transformational</i>, in that it will help to establish or build a market ecosystem that will allow the ongoing growth of small and distributed renewables in the region. REPP 2 aims to unlock investments in renewable energy and by often supporting first-mover projects, to test and further inform the development of the enabling environment. This will be measured through the transformative or additional nature of	<u>Multi-countries</u>
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⁵² Innovative is defined as a project which is first-of-its-kind or among the first in the target country or regions based on technology, service, product or business model which supports the uptake of renewable energy or resilience-building.

				the investment at the time of investment decision in accordance with evaluation criteria approved by the general partner of Camco REPP SCSp.	
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E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 4: Increased private sector investment / Output 1: Increased commercial viability	USD funding leveraged from private and public sources	Measured and monitored ex-post based on portfolio investments through the amount specified in the funding agreements by other parties. Reported in annual reports, with an independent verification conducted before interim and final evaluations.	0	USD315 million	USD786 million	As per funding agreements means that the amount is monitored through the amount specified in the funding agreements by other parties.

Outcome 3: Improved climate resilience of vulnerable people and communities / Output 2: Increase in renewable energy capacity and generation	MWh of electricity sold to customers / year	Measured ex-post based on resilient portfolio investments through the amount of electricity sold to customers.	0	52,233 MWh	130,580MWh	In case of off grid customers, driven by customer affordability.
Outcome 5: Contributing to stronger technology deployment, innovation and market transformation/ Output 3: Improved skills and capacity	Operational renewable energy capacity in MW	Measured ex-post based on portfolio investments through MWs commissioned by investee through technical report and for SHS sales records	0	132 MW	330 MW	As defined in technical commissioning report and for SHS companies sales report to define capacity.
	Private sector funding leveraged in USDm	Measured ex-post based on portfolio investments through third party funding agreements	0	157 mUSD	393 USDm	As per funding agreements means that the amount is monitored through the amount specified in the funding agreements by other parties.
	Number of renewable energy transactions funded	Measured ex-post based on portfolio investments through funding agreement	0	15 number of transactions	37 transactions	
	Number of innovative renewable energy transactions funded	Measured ex-post based on portfolio investments as defined in the concept note and/or structure papers	0	10 number of innovative transactions	26 innovative transactions	

Project/programme co-benefit indicators						
Co-benefit 1: New jobs generated, and community services provided through clean energy	Number of jobs disaggregated by type	Measured ex-post based on portfolio investments through employment records. Reported in annual reports, with an independent verification conducted before interim and final evaluations. Type should include, as a minimum, differentiation between construction and operations jobs. Reported as full-time jobs in each year.	0	30,720 jobs	76,801 jobs	
Co-benefit 2: Enhanced gender equity	2.1 Number of jobs disaggregated by gender	Measured ex-post based on portfolio investments through employment records. Reported in annual reports, with an independent verification conducted before interim and final evaluations. Type should include, as a minimum,	0	9,216 female jobs	23,040 female jobs	

		differentiation between construction and operations jobs. Reported as full-time jobs in each year				
	2.2 USD funding committed to 2X Challenge's Gender Lens Investing criteria-aligned companies	Measured ex-post based on portfolio investments through funding agreements.	0	USD28 million	USD70 million	
	2.3 Number of female-led or managed companies funded	Measured ex-post based on portfolio investments through funding agreements to female owned and led companies In line with 2X criteria. Reported in annual reports, with an independent verification conducted before interim and final evaluations.	0	4	11	Ex-ante estimate based on 30 % assumption.
	2.4 Number of women with improved quality of life	Measured ex-post based on portfolio investments through customer records.	0	Number women with improved lighting: 447,101	Number of women with improved lighting: 1,117,752	Calculated based on the number of active customers that are women.

		Reported in annual reports, with an independent verification conducted before interim and final evaluations.		Number of women with improved access to information: 219,075	Number of women with improved access to information: 547,687	
Co-benefit 3: Improved quality of life through electrification	Number of people with first-time clean energy access to energy disaggregated by gender and type of user/ use	Measured ex-post based on portfolio investments through number of customers and type of customers (off-grid companies, only). In addition, ex-post the average energy usage by customer can be assessed based on the amount of electricity sold by the off-grid investees and the number of customers. Reported in annual reports, with an independent verification conducted before interim and final evaluations. Reported based on customer records and energy sales.	0	Number of people with improved lighting: 1,490,336 Number of people with improved access to information: 730,260	Number of people with improved lighting: 3,725,840 Number of people with improved access to information: 1,825,625	

		These numbers relate to direct beneficiaries only; indirect beneficiaries have not been defined.				
Co-benefit 4: Developers with improved ESG practice	Number of projects/renewable energy companies with improved ESG practices.	Measured ex-post based on portfolio investments number of investees with improved quality of their ESIA and/or ESMS as result of REPP 2 intervention. Number of investees who maintain high quality of E&S implementation - assessed annually during investee annual review. Reported in annual reports, with an independent verification conducted before interim and final evaluations.	26 (70%) of portfolio companies are anticipated to have good ESG practices, which are improved through REPP 2 intervention.	10	26	Measured as a YOY comparison between a project's initial ESG status, ESG status and performance after REPP investment and its status 12 months later. Ex-ante assumption is that in 70% of the projects, ES practices will be improved as a result of REPP 2 funding.

E.6. Project/programme activities and deliverables

Activities	Description	sub-activities	Deliverables
		<u>Sub-activity 1.1:</u> Provide growth funding to SHS businesses	The REPP Manager will provide an annual impact report to its investors.

<u>Activity 1:</u> Provide early-stage growth funding for off-grid companies	REPP 2 will provide early-stage growth capital to off-grid electrification companies.	<u>Sub-activity 1.2:</u> Provide growth funding to mini-grid and isolated mini-grid businesses	The REPP Manger will provide an annual impact report to it investors.
		<u>Sub-activity 1.3:</u> Provide capital for environmental, social and gender studies and plans	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 2:</u> Provide development funding for grid-connected projects	REPP 2 will provide development loans prior to financial close for grid-connected projects.	<u>Sub-activity 2.1:</u> Partial funding of project development costs (including design and permitting, technical feasibility studies, environmental and social impact assessments, gender action plans, etc.)	The REPP Manger will provide an annual impact report to it investors.
		<u>Sub-activity 2.2:</u> Partial funding of due diligence costs as projects approach financial close (covering legal fees and other third-party technical, commercial and financial due diligence costs)	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 3:</u> Finance expansion of SHS businesses	REPP 2 will provide mezzanine and senior debt funding for SHS businesses.	<u>Sub-activity 3.1:</u> Growth funding for SHS businesses	The REPP Manger will provide an annual impact report to it investors.
		<u>Sub-activity 3.2:</u> Provide funding with a gender lens	The REPP Manger will provide an annual impact report to it investors.

<u>Activity 4:</u> Finance construction of mini-grid portfolios	REPP 2 will provide junior and senior debt funding to mini-grid and isolated mini-grid projects.	<u>Sub-activity 4.1:</u> Construction funding for mini-grid portfolios and isolated mini-grid projects	The REPP Manger will provide an annual impact report to it investors.
		<u>Sub-activity 4.2:</u> Provide funding with a gender lens	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 5:</u> Provide gap funding for construction of grid-connected projects	REPP 2 will provide equity or debt funding to fill gaps in the capital structure of grid-connected renewable energy projects.	<u>Sub-activity 5.1:</u> Construction funding for grid-connected renewable energy projects	The REPP Manger will provide an annual impact report to it investors.
		<u>Sub-activity 5.2:</u> Provide funding with a gender lens	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 6:</u> Support the provision of financial risk mitigation	REPP 2 will work with risk mitigation providers to de-risk renewable energy projects.	<u>Sub-activity 6.1:</u> Support the provision of appropriate and effective risk mitigation solutions	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 7:</u> Provide targeted support to developers and companies	REPP 2 funding and capacity building will be directed towards local project developers and renewable energy companies.	<u>Sub-activity 7.1:</u> Support and capacity building for project developers	The REPP Manger will provide an annual impact report to it investors.
<u>Activity 8:</u> Improve environmental and social risk mitigation.	Comprehensive assessment of projects environmental and social risks, development and implementation of environmental and social management systems (ESMSs) to avoid and mitigate risk and to improve	<u>Sub-activity 8.1:</u> Funding and support for implementation of ESAs and ESMSs	The REPP Manger will provide an annual impact report to it investors.

	performance in accordance with REPP 2 policy, approved by the general partner of Camco REPP SCSp.		
Activity 9: Non-financial support to promote gender equity	REPP 2 will provide technical support and train to mainstreaming gender into investee company's operations equity.	<u>Sub-activity 9.1:</u> Integration of a gender equality perspective into the design, implementation, monitoring and assessment of all renewable energy activities.	The REPP Manger will provide an annual impact report to it investors.

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

1. Monitoring and reporting

Monitoring and reporting will be conducted in accordance with industry best practice, following the CDM standards mentioned above and the guidance from Global GHG Accounting and Reporting Standard for the Financial Industry. The monitoring methodology to estimate ex-ante and to calculate ex-post has been presented in the Annex 22B.

In addition to the arrangements laid out in the Accreditation Master Agreement and the Funded Activity Agreement between Camco and the GCF, the implementation of REPP 2 will involve:

- Monitoring of programme's key performance indicators (KPIs) as per section E5;
- Monitoring of key environmental and social (E&S) indicators to ensure project-level compliance with the REPP 2 ESMS Framework, to be approved by the general partner of Camco REPP SCSp;
- Monitor the implementation of gender action plans (GAP) and gender disaggregated data;
- Monitoring of portfolio companies' financial performance; and
- Periodic monitoring and reporting of KPIs and E&S indicators.

REPP 2's monitoring and evaluation (M&E) process and requirements are outlined in the REPP 2 Environmental and Social Policy and Procedures and the REPP 2 Operations Manual, to be approved by the general partner of Camco REPP SCSp. The M&E process will include:

- Training investees in REPP 2's M&E process and requirements by the time of signing a Term Sheet;
- Ex-ante estimates of investees' KPIs included in the Support Agreement;
- Agreeing a monitoring plan with investees as part of their ESMS, which will be part of the obligations pursuant to the Support Agreement with each investee. This will detail the indicators each investee will monitor and report to REPP 2 on quarterly basis;

- Quarterly monitoring of portfolio companies' financial performance;
- Collecting KPI and E&S data and supporting documents on a quarterly basis with each investee via the REPP 2 Impact Data Management System (presented in Annex 11B), to be approved by the general partner of Camco REPP SCSp. The investees will report on site-specific data to REPP, with the exception of SHS companies, which will report country specific numbers. This allows aggregation of monitored data and indicators by investee, country and technology. An annual investee review which is used to assess the fitness of investees ESMS implementation and to verify KPIs reported. This review is conducted by REPP 2 Manager.

REPP 2 will provide KPI updates to the AIFM, REPP 2 Investment Committee and REPP 2 Advisory Board on a quarterly basis and a full M&E report in the form of a logframe on an annual basis.

An important aspect of REPP 2's risk management process and impact verification will be the annual review of investments. The annual review will assess fitness of ESMS implementation and verify the reported impact indicators. The annual reviews will be based on document reviews (As an example, the review will review customer records to verify number of customers and employment records and contracts to verify number of employees reported. The records will be provided by the investee company and reviewed by REPP managers impact team. If the review raises any discrepancies, concerns or material issues a site visit is conducted.) and interviews and will measure, compare and analyse a project's performance against the baseline and targets set during the due diligence process. This will include review of investees GAP against the set targets. If substantial material issues are discovered, a site visit may be conducted as an additional part of the annual review. The results of the annual review will confirm whether support can be continued in its current form, if modifications of the REPP 2 support are needed, or if withdrawal of REPP 2 support is required.

2. Evaluation

REPP 2's evaluation process is outlined in the Independent Evaluation of Camco Investment Mandates policy. The evaluation is an independent review commissioned by REPP 2 halfway through the investment period (mid-term review) as well as at the end of the investment mandate (final review) to ensure activities compliance with Camco 2 policies and procedures. An open tender procurement procedure will be followed to select the appropriate service provider to conduct the independent evaluation. Evaluations will be conducted based on following principles:

- **Independence:** Evaluations must be conducted independently from the delivery and management of financial assistance and project implementation.
- **Credibility:** Evaluations must be credible and based on reliable data and observations. Evaluation reports should reflect consistency and dependability in data, findings and lessons learned, with reference to the quality of the procedures and analysis used to collect and interpret information.
- **Impartiality:** Evaluations must give a comprehensive and balanced presentation of the strengths and weaknesses of the activity being evaluated. The evaluation process should reflect impartiality at all stages and consider the views of all stakeholders.

- **Transparency:** Transparency involves clear communication on the criteria applied, the evaluation approach and methods, and the intended use of the findings. Documentation related to evaluations must be freely available, easily accessible and readable for transparency and legitimacy.
- **Integrity:** Evaluators must honour the dignity, well-being and self-worth of individuals and acknowledge the influence of culture within and across groups. Evaluators must respect the right of institutions and individuals to provide information on the facts confidentially.
- **Gender equity:** Evaluations must assess whether – and if so, how - men and women are affected by changes to natural resource use and decision-making resulting from the funded activity.
- **Competencies and capacities:** Evaluations require a range of expertise that may be technical, environmental, cultural, ethical or within a social science. Evaluators must have sufficient experience and skills in the appropriate fields.

3. Evaluation reporting of Camco (Accredited Entity) to GCF

Camco will provide the GCF with:

- Annual activity performance reports on the status of the Programme throughout the relevant reporting period;
- A mid-term evaluation report at the midpoint of the implementation period of the Programme; and
- A final evaluation report at the end of the implementation period of the Programme as per the funded activity agreement.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures

Selected Risk Factor 1 Co-Financing Risk

Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>High</u>

Description

The risk that an insufficient amount of co-financing is raised to meet GCF co-financing requirements or deliver the targeted impacts for REPP 2. There is an additional risk that the terms on which co-financing can be secured, particularly in the current high-inflation, rising interest-rate environment, limit the potential for REPP 2 to be commercially viable and deliver targeted impacts and returns for investors.

Mitigation Measure(s)

- Fundraising efforts on behalf of REPP 2 are ongoing and at an advanced stage
- In the event that the amount of co-financing raised is below the targeted level, the capital structure for REPP 2 is sufficiently flexible to ensure that all GCF requirements in terms of co-funding ratios will be met
- The REPP 2 financial model incorporates an expectation of higher pricing and return expectations in light of current macro-economic trends
- Discussions are ongoing with potential investors (public and private) to secure financing on terms which reflect the REPP 2 mandate and goal to build a more viable market for small-scale renewables in Africa – particularly when macroeconomic conditions exacerbate existing barriers such as access to finance.

Mitigation measures are expected to manage and reduce the probability of occurrence of this risk factor.

Selected Risk Factor 2 Investment Performance Risk

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

The risk that assets in the REPP 2 portfolio underperform investment expectations and that targeted returns for investors are not achieved as a result.

Mitigation Measure(s)

- Robust asset selection processes in accordance with defined investment criteria, including REPP 2 IC approval
- Comprehensive risk assessment and due diligence procedures in line with REPP 2 risk management policies and procedures
- Ongoing assessment of risk management framework and procedures as well as operations and maintenance at asset level
- REPP 2 oversight and input on key strategic decision-making for equity investees through representation on project company boards
- Ongoing monitoring of asset performance and revenue-drivers
- Ongoing communication with investees, combined with annual review cycle for investments, including site visits

Mitigation measures are expected to manage and reduce the probability of occurrence of this risk factor.		
Selected Risk Factor 3 Developer / Investee Performance Risk		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Project developer or investee company management fail to execute on company growth plans (off-grid investments), achieve financial close or complete construction, achieve commercial operations within targeted timeframes and against agreed budgets.		
Mitigation Measure(s)		
• Milestone-based disbursement of funds linked to achievement of pre-agreed growth targets (for off-grid investments) and pre-defined contractual obligations (for development-stage funding and construction of grid-connected projects) • Ongoing communication, oversight and review of investment progress, including via company board representation where an equity investment has been made • Support and capacity building to company management and local project developers • Support and capacity building (as appropriate) to regulators and other government institutions crucial for project or business development The milestone-based disbursement approach is expected to reduce the risk of REPP 2 losses. Capacity building initiatives are expected to help to build institutional, regulatory and business skills and experience, which reduce the risk of project / investment failure and enhance potential for commercial viability and long-term sustainability.		
Selected Risk Factor 4 Exit / Refinancing Risk		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk that REPP 2 is unable to exit or refinance a transaction according to original expectations		
Mitigation Measure(s)		
• Explicit consideration of exit and refinancing plan as part of all investment proposals, with REPP 2 IC approval • Pricing and deal structuring approach aligned to market benchmarks wherever commercially possible • Refinancing plans agreed and approved by refinancing parties (including approvals, wherever possible) before disbursement • Active portfolio and asset management with dedicated role for this purpose • Ongoing oversight and review of performance at an asset and portfolio level to maximise profitable exit, sell-down and refinancing opportunities Mitigation measures described above are expected to reduce the probability of occurrence of this risk factor.		

Selected Risk Factor 5 Conflict of Interest Risk		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>
Description		
The risk that a conflict of interest occurs for Camco between its obligations as the manager of REPP 2 and its obligations as Accredited Entity on behalf of the GCF. Similar potential for conflict of interest in terms of management services rendered to any other platform by Camco and the REPP 2 platform, particularly where these have the potential to result in losses to investors participating in REPP 2 as well as one or more other facilities managed by Camco.		
Mitigation Measure(s)		
- Camco maintains a comprehensive Conflicts of Interest Policy, governing the risks outlined above - Where a potential conflict is identified, appropriate measures will be implemented to manage and mitigate the conflict - The REPP 2 Conflicts of Interest of Policy will be approved by its the general partner of Camco REPP SCSp and supervised by the Advisory Board and AIFM, once constituted and maintained on an ongoing basis - Camco, as fund manager, will appoint and maintain an independent REPP 2 coordinator and (separately) a GCF coordinator - Camco REPP SCSp, as Executing Entity, will be supervised by the AIFM who will ensure compliance with the Luxembourg Law of 10 August 195 on Commercial Companies The measures outlined above are expected to reduce the probability of occurrence and the potential impact of this risk factor.		
Selected Risk Factor 6 General Operational Risk		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Risk of loss or other damage resulting from insufficient or failed policies, processes and procedures, or a failure to implement and maintain existing ones.		
Mitigation Measure(s)		
- Develop detailed and comprehensive policies, processes and procedures covering all 2 REPP activities and maintain these on an ongoing basis - Ensure that policies, processes and procedures are maintained and recorded in the REPP 2 Operations Manual (Annex 21) - Provide ongoing training to REPP 2 staff, IC and Advisory Board - Ensure systems and IT infrastructure are appropriate to meet the operational, monitoring and reporting needs of REPP 2 on an ongoing basis - Minimise key person dependencies and maintain a detailed business continuity plan for REPP 2 - Ensure appropriate segregation of duties amongst REPP 2 staff.		

Selected Risk Factor 7 Offtaker Credit Risk (grid-connected projects)		
Category	Probability	Impact
<u>Credit</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk of payment delay or default associated with utility offtakers for grid-connected projects.		
Mitigation Measure(s)		
- Rigorous analysis credit risk during project selection and deal structuring with approval provided by the REPP 2 IC. - Structure and ensure both innovative and traditional forms of credit risk mitigation for payment liquidity and default, sourced from partner organisations which provide risk mitigation products. Mitigation measures can minimise and transfer offtaker credit risk to parties which are better-positioned to manage it. Where cover is purchased, this provided effective risk mitigation to both project companies and REPP 2 investees.		
Selected Risk Factor 8 Demand Risk (off-grid companies)		
Category	Probability	Impact
<u>Credit</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk that estimates of demand and revenue growth achieved for off-grid companies are lower than initially projected, resulting in delays or failure to achieve growth targets and become sustainable. Persistent failure to grow demand and revenue will lead to below-par returns and losses for REPP 2 in a worst-case scenario.		
Mitigation Measure(s)		
- Ensure rigorous assessment is undertaken by the off-grid company of customer ability- and willingness-to-pay, prior to investment decision - Ensure that drivers of demand and revenue growth are well-understood and that growth projections are realistic - Seek additional / alternative revenue sources to meet projected payment obligations, through product and business model innovation, productive use of energy applications, securing anchor consumers to boost revenues, etc. - Provide capacity building support for off-grid companies to develop skills and expertise in estimating future demand and managing demand / revenue growth risk - Provide strategic oversight and input (via board positions) to ensure costs are managed carefully and that business is focused appropriately on demand and revenue growth - Provide support for development and implementation of policy and regulatory frameworks and well as donor support programmes which provide tariff or result-based subsidies for off-grid electrification in Host Countries Mitigation measures can be effective in reducing demand risk.		

Selected Risk Factor 9 Currency Risk		
Category	Probability	Impact
<u>Forex</u>	<u>High</u>	<u>Medium</u>
Description		
Where mismatches exist between revenues received by investees or borrowers in local currency and their hard currency funding obligations, there is a risk that they will be unable to meet those obligations (in part or in full) as a result of currency devaluation. The consequences for REPP 2 may include a failure to achieve targeted returns for its investors and, in a worst-case scenario, financial losses at the portfolio level. The risk is exacerbated by the high cost and lack of liquidity of hedging for emerging market currencies, especially in comparison to the lower revenues generated by smaller renewable energy projects.		
Mitigation Measure(s)		
• In general – and particularly for grid-connected projects – REPP 2 will invest and provide loans denominated in hard currency, effectively transferring currency risk to investee companies and borrowers. For grid-connected projects, where project companies can negotiate power purchase agreements denominated in hard currency, this risk can be passed on to the offtaker. Where the PPA is denominated in local currency or where off-grid companies generate revenues in local currency, currency risk lies with the investee / borrower. • In rate cases where project economics allow, currency risk may be reduced through FX hedging, with hedging costs passed on to borrowers / investees. • REPP 2 will typically only consider offering investments in local currency where the investment or project can afford the cost of FX hedging or where alternative mechanisms to facilitate local currency lending are available (such as on-lending facilities via local financing partners). • REPP 2 may also consider implementation of portfolio hedging solutions, effectively hedging FX risk for investors and a reduction in the costs of hedging for individual investees or borrowers through portfolio aggregation and scale.		
Selected Risk Factor 10 Currency Transfer and Inconvertibility Risk		
Category	Probability	Impact
<u>Forex</u>	<u>Medium</u>	<u>Medium</u>
Description		
In relation to currency risk, the additional risk that funds received by companies in local currency in a host country cannot be converted into foreign currency and transferred internationally in a timely manner. Delays in this process may result in losses for the investee or borrower due to currency movements, breach of contractual obligations. For REPP 2, the result may be lower-than-expected returns and losses in a worst-case scenario.		
Mitigation Measure(s)		
• Secure government guarantees or assurance regarding access to foreign currency and transfer procedures in Host Countries • Where available, secure transfer and inconvertibility protection as part of a political risk insurance package • Work with local authorities (including central banks) to develop, implement and familiarise local staff procedures for the processing (preparation, approval, etc.) of currency conversion and transfer requests The measures can reduce the probability of occurrence of delays and losses.		

Selected Risk Factor 11 Environmental and Social Safeguards		
Category	Probability	Impact
<u>Reputational</u>	<u>Low</u>	<u>High</u>
Description		
The risk that environmental and social safeguards (ESS) fail to address relevant environmental and social risks, or that failures in the implementation of those safeguards lead to negative and / or unintended environmental or social impacts.		
Mitigation Measure(s)		
- Development of an Environmental and Social Management System (ESMS) framework at the facility level, as well as an ESMS and Environmental and Social Action Plan (ESAP) at investee level. All related policies and procedures to be in accordance with international best practice standards as well as REPP 2 and GCF policies - Preparation of detailed environmental and social impact assessments for each Invested by qualified independent consultants - Careful assessment of environmental and social risk throughout the transaction approval process in accordance with the REPP 2 Environmental and Social Policy and ESMS Framework, from pre-selection through to approval by the REPP 2 IC of a transaction funding proposal - Undertaking careful due diligence, including a site visit prior to execution and disbursement of funds - Extensive and ongoing stakeholder engagement at investee level – particularly for vulnerable people such as women, children and the elderly - Extensive and ongoing training and capacity building for developers as well as management and staff of off-grid companies to build skills and expertise in relation to managing environmental and social risk - Engagement and training provided to institutions in Host Countries responsible for the oversight and compliance with national policy, to develop and implement environmental and social safeguards and policy requirements which are appropriate for the scale and risk associated with each technology asset type - Early-stage development funding to prepare project level ESAs and ESMS - Ongoing monitoring, reporting and evaluation on environmental and social risks during construction and once plants become operational in accordance with the Section E. The measures described above can adequately minimise and manage environmental and social risks and the implementation of environmental and social safeguards.		
Selected Risk Factor 12 Forced and child labour		
Category	Probability	Impact
<u>Reputational</u>	<u>Medium</u>	<u>High</u>
Description		
There is a risk that forced and / or child labour is used in the supply chain of silicon-based solar panels solar and lithium-ion-batteries.		
Mitigation Measure(s)		
- REPP 2 is committed to following the conventions of the International Labour Organisation (“ILO”), as well as the IFC Performance Standards with regard to forced labour and child labour. - REPP 2 will adhere to GCF environmental and social policy and local relevant laws in terms of the prohibition of forced labour in any form, including slavery and human trafficking. - REPP 2 takes a firm stand against all forms of forced labour and expects the same of its primary suppliers, partners, contractors and subcontractors.		

- REPP 2 will not tolerate child labour under circumstances. A clear distinction between child labour and utilisation of young workers shall be established in accordance with internationally-recognized standards. Only people who are 18 years or older will be permitted to be employed on REPP 2 funded projects. These commitments will be absorbed by all REPP 2 investees and projects funded under REPP 2 through the funding agreements which each investee. - REPP 2 investees will be required to prepare, implement and maintain a clear and transparent Supply Chain Procurement Plan (“SCPP”), setting out the measures which must be taken to prevent forced and child labour. The SCPP should clarify that the policies preventing forced and child labour apply to all subsidiaries of a company and their respective supply chains. These policies will also stipulate any training needs of human resource and/or procurement personnel on how to identify forced labour in practice and seek appropriate remedies where it is identified. - REPP 2 and the AE will participate in and contribute to the establishment of supplier code of conduct principles, covering compliance with international labour and human rights laws, including the prohibition of forced and child labour, and occupational health and safety. - REPP 2 will develop, implement and maintain supply chain risk assessment procedures and a due diligence screening mechanism to mitigate the risk of forced and child labour. These procedures will be implemented in the project approval, due diligence and review procedures which are applicable to all investments.		
Selected Risk Factor 13 Compliance Risk		
Category	Probability	Impact
<u>Reputational</u>	<u>Low</u>	<u>Medium</u>
Description		
The risk is that compliance obligations (regulatory or internal) are not met by REPP 2 or Camco, including reporting requirements.		
Mitigation Measure(s)		
- REPP 2 staffing will include dedicated legal and compliance staff - REPP 2 will require quarterly compliance reporting and regular testing - Comprehensively-documented policies and procedures maintained on an ongoing basis - Formal staff induction training and training with respect to compliance requirements during onboarding provided to REPP 2 investees - Ongoing compliance training for REPP 2 staff - Ongoing oversight and review by Camco and REPP 2 governance structures of compliance requirements		
The measures outlined above are expected to provide adequate risk mitigation for compliance risk.		
Selected Risk Factor 14 Governance / Integrity Risk		
Category	Probability	Impact
<u>ML/FT</u>	<u>Low</u>	<u>High</u>
Description		
Risk of failed governance resulting in losses and reputational damage among investees or project companies as well as for REPP 2. Risk of any counterparty or one of its employees, or any Camco employee, acting corruptly or fraudulently or in breach of the law, wilfully or otherwise. And or failure to comply with REPP 2 policies, especially Anti-Corruption and Integrity Policy and Investing with Integrity Policy, Know-Your-Client		

Mitigation Measure(s)		
- Development and implementation of comprehensive and robust policies and procedures governing Know-Your-Client, Anti-Corruption and Anti Money Laundering, Counter-Terrorist Financing, fraud, integrity and tax and domicile requirements in accordance with international law, as reflected in GCF requirements - Implementation and maintenance of good governance practices for REPP 2, in accordance with the requirements of its AIFM and the GCF - Implementation and maintenance of good governance practices for REPP 2 investees in accordance with REPP 2 policies - Ongoing training for REPP 2 staff as well as training for staff of investee companies The measures outlined above are designed to establish and maintain high standards of good governance for investee companies and for REPP 2, thereby reducing the risk of failed governance.		
Selected Risk Factor 15 Policy and Regulatory Risk		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk of a failure to align or comply with host country policy and regulatory requirements, resulting in delays during project development or a failure to obtain necessary permits and licences to proceed with construction or begin commercial operations. Additional risks to project or business development associated with policy and regulatory uncertainty, including gaps in policy and regulatory frameworks as well as slow or inconsistent application of policies and regulations.		
Mitigation Measure(s)		
- Detailed analysis of policy and regulatory frameworks in all Host Countries conducted on an ongoing basis, informed through engagement with government counterparties and their development partners. - Careful assessment of risks posed by policy or regulatory uncertainty to project or business development, construction or operations – prior to decision to invest, with a clear mitigation plan if the investment is made - The broad scope of REPP 2 mandate (SHS, mini/isolated grids, on-grid) will allow it to select investments where policy and regulatory risk is acceptable, while continuing to monitor the enabling environment for other market segment to position its future investments. - Ongoing monitoring of policy and regulatory risks throughout the investment lifecycle - Training and skills development for business owners and project companies to manage policy and regulatory uncertainty - Targeted interventions within Host Countries to support development and implementation of policies and regulations, where Camco and REPP 2 experience can add value. Mitigation measures are designed to reduce policy and regulatory uncertainty, which contributes to broader market development.		
Selected Risk Factor 16 Political Uncertainty		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Low</u>
Description		
Risk of political uncertainty (and in worst cases unrest and violence) resulting in delays, additional costs and / or failure of REPP 2 investments.		

Mitigation Measure(s)		
- Detailed analysis of political uncertainty in all Host Countries conducted on an ongoing basis - Careful assessment of political risks to project or business development, construction or operations – prior to decision to invest - Ongoing monitoring of policy and regulatory risks throughout the investment lifecycle - Continuous and inclusive engagement with public sector stakeholders Mitigation measures do not reduce the probabilities of occurrence of political risk events but may help to minimise delays and losses in the REPP 2 portfolio.		
Selected Risk Factor 17 Climate Risk		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk that funding provided by REPP 2 fails to deliver the intended climate change mitigation and adaptation / climate resilience impacts. Related risks include resource risk and reputational risk to REPP 2 and its investors where unforeseen climate change impacts to infrastructure as well as vulnerable people and communities result in unintended consequences, including reduced climate resilience and maladaptation.		
Mitigation Measure(s)		
- Incorporate comprehensive climate risk assessment in the REPP 2 Risk Management Policy and Procedures, to be approved by the REPP 2 - Internal (desktop) climate risk assessment to be conducted prior to investment approved by the REPP 2 IC and confirmed during due diligence, including via site visit - Climate risk assessment to be conducted as part of all ESIA's to assess the impacts of climate change in aggravating environmental and social risk - Appropriate levels of ongoing monitoring, reporting and evaluation of climate risk for all assets in the REPP 2 portfolio - Training and skills development for off-grid companies and project companies in identification, assessment and management of climate risk (to infrastructure, environment as well as vulnerable people and communities). Particularly attention to be focused on avoiding maladaptation. REPP 2 interventions are designed to contribute to efforts by Host Countries to mitigate and adapt to climate change, building climate resilience for energy infrastructure and for vulnerable people and communities in the process. Mitigation measures identified here can enhance the potential for positive climate change impacts and minimise potential for unintended outcomes such as maladaptation.		
Selected Risk Factor 18 Technical Assistance Facility		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	<u>Medium</u>
Description		
Risk of a limited contribution to building a stronger enabling environment for smaller-scale renewables through a) support for policy and regulatory frameworks, b) support to developers, improved capacity to management environmental and social risk or d) promote gender equity. Potential impacts of this risk include a failure to develop a sufficient pipeline of viable opportunities for investment, leading to a failure to achieve stated climate mitigation and adaptation outcomes (Outcomes 1-3), mobilisation of private sector investment (Outcome 4), contributing to stronger technology deployment, innovation and market transformation (Outcome 5) and numerous co-benefits.		

Mitigation Measure(s)
• Experience in delivering REPP Phase 1 has built networks and expertise in delivery of targeted skills and capacity development which will be applied in REPP 2. • The market feasibility studies in each Host Country identify specific opportunities to provide targeted support for development and implementation of policy and regulations. • The indicative pipeline (Annex 26) demonstrates demand for early-stage and development support. Camco will leverage experience from delivery of REPP Phase 1 to ensure skills and capacity building occurs for developers. • The REPP 2 ESMF requires that all investees must develop and implement an ESMS, which will ensure improved capacity to manage E&S risks • GLI-aligned investment targets to promote gender reduces the risk of a failure to achieve impact. • Explicit plans for skills and capacity development will be approved by the general partner of Camco REPP SCSp following consultation with Advisory Board and updated at least annually. • Camco to focus on establishment and roll-out of the TAF. REPP 2 interventions are designed to minimise effectively the probability and likely impacts of this risk factor.

F. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

An Environmental and Social Policy and Procedures (E&S Policy) and an Environmental and Social Management Framework (Framework) have been developed for REPP 2 to manage the environmental and social (E&S) impacts and risks of investments made by REPP 2. The E&S Policy and Framework are in line with IFC Performance Standards, and the sustainability principles advocated by the UN Global Compact. The Framework is presented in Annex 6.

E&S risk and impacts of smaller renewable energy projects funded by REPP 2 are anticipated to be limited. The magnitude of these impacts is expected to be low and contained within the footprint of the projects. The impacts should be largely reversible and readily mitigated through the mitigation measures implemented by investees as identified in their project-specific Environmental and Social Management System (ESMS). REPP 2 will not support investments if material negative E&S impacts cannot be satisfactorily mitigated.

The E&S risk screening process for REPP 2 (elaborated in the Framework) is:

- **Step 1:** The initial E&S risk screening occurs prior to investment approval and involves a Risk Assessment Tool and an E&S Initial Screening Checklist to pre-define risk category and eligibility of the project;
- **Step 2:** An E&S onboarding with the investee company takes place if the project is approved for further consideration and once the initial E&S risk screening has been completed to ensure the company understands E&S requirements and processes for REPP 2;
- **Step 3:** An E&S Gap Analysis (red flag report) will be completed by the REPP 2 Manager. This will entail a review of the project's documentation (i.e., ESIA and ESMS to the extent that it is available). In case of early-stage growth and development funding, the funding may be used for the development of the ESIA or the ESMS. In this case REPP 2 will assist in reviewing the Terms of Reference for EIAs/ESMS or GAP work and the selection of consultants. In addition, key milestones for disbursement of early-stage growth and development funding are agreed; and
- **Step 4:** Once the ESIA and ESMS have been finalised, REPP 2 conducts a due diligence site visit to verify E&S risks. E&S risks outlined in the ESIA, such as identification of indigenous peoples, potential or necessary resettlement, security risks, protected biodiversity impacts and human rights assessments, will be highlighted in the E&S Red Flag Report, and the extent of the risk will be verified during the due diligence trip. The project risk categorisation will also be reassessed to ensure that all project activities align within their allocated category. All projects funded by REPP 2 will undertake a due diligence trip. The due diligence trip will also serve to assess the investee's internal E&S resources, competence and aptitude for managing E&S risks in accordance with REPP 2 E&S standards.

REPP 2 will be responsible for the implementation of the E&S Policy and the Framework and shall apply an E&S risk categorisation to each potential investment. REPP 2 will have a dedicated impact team which will review each individual investment's E&S risk, ensure the sufficient quality of the ESIA and the ESMS, assess the investee's capacity and competency to manage E&S risk, and assist them with capacity building, if required. The roles and responsibilities related to the implementation of the Framework are described in greater detail in Annex 6.

The monitoring of REPP 2 will involve:

- Monitoring of key performance indicators;
- Monitoring of key E&S indicators to ensure project-level compliance with the Framework;
- Periodic reporting; and
- Annual investee performance reviews to assess the fitness of its ESMS.

The monitoring and reporting obligations will be included in each Investment Agreement between REPP 2 and the investee, who will submit monitoring reports through an impact data management system on a quarterly basis. Reports will be reviewed by REPP 2, and corrective action measures discussed with the investee where relevant.

In its capacity as an AE, Camco has been undertaking extensive stakeholder engagement with NDAs, energy ministries, agencies and regulators, as well as the private sector and relevant NGOs. This engagement will continue throughout the Programme's implementation. All REPP 2 investees will be required to conduct appropriate site-based stakeholder consultations with communities and project beneficiaries in line with the Framework. Stakeholder engagement outcomes and strategy are further outlined in Section D5 and Annex 7.

REPP 2 will require that a grievance mechanism for stakeholders is established to receive and facilitate resolution of affected communities' concerns and grievances about a project's E&S performance. The process shall be easy, accessible and free of charge, and conducted in a confidential manner should this be preferred. The investee will provide guidelines on how grievances, claims or suggestions from the community are promptly managed and effectively resolved to all stakeholders (inclusive of employees, community members, service providers, and suppliers). In addition, Camco in its capacity as an AE, provides a whistle-blower hotline and grievance form on its website that will allow submission of grievances in an anonymous manner.

REPP 2 will recognise the critical importance of respecting the rights of Indigenous Peoples and of avoiding any adverse impacts on them. The Framework includes an Indigenous Peoples Policy (IPP), which will be applicable in all REPP 2 investments where Indigenous Peoples are present and/or have or had collective attachment or rights to the land. The IPP has been guided by the UN Declaration on the Rights of Indigenous, IFC Environmental and Social Performance Standards 7, and the Green Climate Fund Indigenous People Policy and is presented in Annex 6. As per the IPP, a stakeholder engagement will take place with Indigenous Peoples, and in the case of negative impacts the principle of free, prior and informed consent will be applied.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

1. Gender assessment

A gender assessment for REPP 2 was undertaken to explore the relationships between men and women, their access to resources and the constraints they face relative to one another in REPP 2 Host Countries within the context of climate change and energy access.

The REPP 2 investment mandate will extend towards some of the most economically and socially vulnerable countries in Africa, with the majority (>50%) of the population living below the International Poverty Line in more than half of the Host Countries. However, there is some variation in how REPP 2 Host Countries rank in terms of different gender indexes, as outlined below:

- **Gender Development Index (GDI)**, which focuses on health, education and economic resources: Two-thirds of the countries show low levels of gender equity, whereas Lesotho, Madagascar and Malawi all score higher on GDI value, approaching or exceeding the world average.
- **Gender Inequality Index (GII)**, which measures gender inequalities in reproductive health, empowerment and the labour force: All Host Countries experience higher level of gender inequality than the world average and DRC, Niger, Nigeria and Sierra Leone are all below the sub-Saharan African average.
- **Global Gender Gap Index (GGGI)**, which measures gender parity across economic participation and opportunity, educational attainment, health and survival, and political empowerment.⁵³

⁵³ World Economic Forum (WEF), 2022. [Global Gender Gap Report 2022](#).

Compared with the sub-Saharan Africa as a whole (which ranks in the middle ahead of Middle East and North Africa, and South Asia), Cameroon, Lesotho and Madagascar rank above average, while most Host Countries remain on the lower spectrum on gender equity values.

According to a 2022 World Bank et al report on progress towards SDG 7, sub-Saharan Africa remained the least electrified region in 2020. While availability of country-level gender-disaggregated data on electricity access in the region is limited, thereby making it difficult to assess how this lack of electricity access relates to gender, research by IRENA has shown that women-owned households and enterprises find it harder to access modern energy.⁵⁴

While regional and country-level gender-disaggregated data for the energy sector remains scarce, research at the global level shows that the share of women employed in the renewable energy sector (32%) is higher than in the oil and gas industry (22%).⁵⁵ However, the share of female employees across sub-Saharan Africa is substantially higher in non-technical jobs (30-38%) than in STEM jobs (13%). While women's participation in company leadership globally is substantially lower than men, it is positive to see that there are more women on the boards of renewable energy companies in Africa (27%) than on boards in general (Africa's average is 18%).

At the same time, female renewable energy entrepreneurs are struggling to access the financing and business support they need to grow their companies. Although Africa has a higher share of female-owned businesses than anywhere else - which demonstrates the active and important role women play in the continent's commercial sector - the African Development Bank has estimated that there is a USD42 billion financing gap for African women entrepreneurs across business value chains.⁵⁶

2. Gender action plan (GAP)

Access to energy can help individuals, households and business to complete certain time-consuming tasks in more efficient manner. Access to electricity may free up time from fuel collection and household tasks to enable longer periods of study time or other productive use of time, and thereby decrease gender gaps in key educational and economic areas. Additionally, access to energy, for example access to lighting, information technologies or connectivity, are linked to an increase in safety and security and the perception of personal wellbeing.⁵⁷ Further, in REPP 2 we believe that the funding of women-led and focused businesses, empowers women to invest and build companies, teams, products and markets that will produce transformation change for generations to come. This transformational change, driven by gender focused financing, is crucial to overcoming the gender equality gap.

The REPP 2 GAP includes two main actions that work towards bridging the gender equity gap:

Action 1: Integrate a gender equity perspective into the design, implementation, monitoring and assessment of all renewable energy activities financed by REPP 2. This means requesting all investees to put in place sub-project level GAPs (Activity 9: Non-financial support to promote gender equity) with actionable targets to improve gender equality within their operations and project implementation will positively impact the industry and its beneficiaries. This involves:

1. Analysing gender equality in the investee company, country of operation, and targeted sector;

⁵⁴ IRENA, 2022. [Renewable Energy Market Analysis. Africa and Its Regions](#).

⁵⁵ AfDB and UN Women, 2021. [Green Jobs for Women in Africa](#); IRENA, 2020. [Global Landscape of Renewable Energy Finance 2020](#).

⁵⁶ Global Entrepreneurship Monitoring. 2018/2019 Women's Entrepreneurship Report. AfDB, 2017. Unleashing women's entrepreneurship through strategic partnerships.

⁵⁷ Supported by an initial project-level data collection conducted in Goma, DRC:
https://static1.squarespace.com/static/5cc327c55acb9b00019d816a/t/63d16fc81da3f71c853b105a/1674670025093/Assessing+the+Peace+effects+of+PRECS_Final_Goma_June+2022_updated.pdf

2. Establishing an investee-specific GAP by identifying gender performance indicators and gender-disaggregated targets against an established baseline; and
3. Integrating the outcomes of (1) and (2) into a monitoring plan.

REPP 2 will provide investee companies with supporting in assessing gender baseline and establishing action plan. Such support includes:

- i. Gender equality toolkit for renewable energy companies - an example, provided via the REPP 2 [gender toolkit](#);
- ii. Gender mainstreaming training provided in form of webinars and REPP Phase 1 Academy (i.e. closed training provided to REPP investee only - as an example can be provided previous [webinars](#) and REPP Phase 1 Academy training (see appendix 2); and
- iii. Individual support to investee from the REPP 2 Manager's impact team in from of review of gender baseline and action plan.

Action 2: Dedicate a target of 30% of all funding towards companies that meet 2X GLI criteria. These are companies that:

- Are founded by a woman or women or are majority-owned by women; and / or
- Have at least 30% women in senior management; and / or
- Provide products or services designed to disproportionately benefit women; and / or
- Have a workforce that is at least 30% women, together with a progressive human resource policy.

The funding can be in from of early-stage growth funding for off-grid companies (Activity 1), development-stage funding for grid-connected projects (activity 2), finance for expansion of SHS businesses (Activity 3), finance for construction of mini-grid portfolios (activity 4) or gap funding for construction of grid-connected projects (Activity 5) and can be in from of hybrid equity instruments or as senior or subordinated loans, subject to the company's funding needs.

These actions are anticipated to lead to the following impacts:

- Total female beneficiaries reached: 290,875
- Total female beneficiaries adopting improved livelihoods and/or new climate resilient livelihood options, through resilient PUE options: 8,295
- Promotion of gender equity within renewable energy development companies by working towards investing USD70 million in companies meeting the 2X GLI criteria.
- Achievement of numerous co-benefits, including 23,040 female jobs generated, improved lighting 1.12 million women and girls, improved access to information for 547,687 women and girls, and 26 developers with improved ESG practices (including GAPs).

The detailed gender assessment and action plan are presented in Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

REPP 2 will adopt and build upon the policy framework, governance and system architecture that REPP Phase 1 has implemented and refined since its inception in 2015. Where necessary, this will be expanded and further refined to align with the GCF's governance standards and will be overseen by Camco in its role as Accredited Entity.

REPP 2 will produce an annual business plan outlining investment and portfolio targets for the year, as well as an operational budget produced on an activity basis. This will allow the budget-setting committee to determine the activities required to execute the strategy for the coming year, leaving REPP 2 to effectively

undertake the activities and deliver value within the budgetary constraints set. The business plan will also summarise the previous year's performance and any actions resulting from analysis of the results.

REPP 2 will prepare monthly portfolio reports that include:

- Pipeline, deal-flow and progress against origination targets for the platform
- Deals in documentation and updates on transactions as they move closer to being contracted and ready to disburse
- Portfolio valuation and movements in disbursements, accrued income, revaluations, impairments and disposals
- Metrics showing portfolio weightings and performance against targets and risk thresholds, as required (e.g., technology, country, off-taker etc.)

The portfolio reports will allow effective monitoring of the platform's performance by the governing committees and stakeholders, ensuring corrective action is taken and assisting in developing strategy which steers the platform towards its long-term objectives.

REPP 2 will also prepare quarterly management accounts, comprising a profit and loss account, balance sheet and cashflow statement. These will be prepared in accordance with the relevant Generally Accepted Accounting Principles, which are the same accounting standards under which the audited annual report and financial statements will be prepared.

Accompanying the Cashflow Statement will be a Cash Forecast, prepared quarterly and providing a detailed overview of the platform's liquidity requirements during the investment period. It will also include a forecast of cash returns from the portfolio over the life of the platform.

REPP 2 will engage a firm of accountants to undertake an external audit of the platform's annual financial statements. This will confirm independently that the financial statements provide a true and fair view and provide transparency to governing committees and stakeholders, highlighting areas of importance and anything which may require attention.

REPP 2 will open and control bank accounts into which it will receive funds and make disbursements. The bank accounts will be reconciled in the REPP 2 accounting system daily or on an ad-hoc basis.

Disbursements will be controlled through a process that ensures all appropriate milestones have been completed. For example:

- Conditions prior to disbursement in applicable finance agreements have been met
- For payments to suppliers or service providers, service level agreements or valid contracts exist between the platform and beneficiary
- Projects have fulfilled monitoring obligations and investees are compliant with any platform policies and training requirements
- Beneficiaries have undergone appropriate KYC / AML and sanctions checks

Cash controls will be established which include dual authority for setting up and approving payments, and which have payment limits for approving transactions of different sizes. Material transactions will require sign-off from the Depositary Agent.

REPP 2 will adopt and apply the Accredited Entity's approved procurement policy, which is already in use by REPP Phase 1. The policy will be applied throughout the Programme's activities. Please refer to Annex 10: Procurement Policy & Plan for further details.

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

G. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☒ Environmental and Social Management System (ESMS)
 - ☐ Others (Indigenous People's Plan)
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement Policy and Procurement Plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☒ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)
- ☒ Annex 23 Analysis of potential for overlap and complementarity of GCF funding
- ☒ Annex 24 Camco Adaptation and Project Decision Tool (classification of adaptation responses)
- ☒ Annex 25 Progress update for REPP 2 Co-financing
- ☒ Annex 26 REPP 2 indicative pipeline

No-objection letter(s) issued by the national designated authority(ies) or focal point(s)

REPUBLIQUE DU CAMEROUN
Paix – Travail – Patrie
**MINISTRE DE L'ENVIRONNEMENT,
 DE LA PROTECTION DE LA NATURE
 ET DU DEVELOPPEMENT DURABLE**



REPUBLIC OF CAMEROON
Peace – Work – Fatherland
**MINISTRY OF ENVIRONMENT,
 PROTECTION OF NATURE AND
 SUSTAINABLE DEVELOPMENT**

Yaoundé, 09 December 2022

Prof. Paul TCHAWA (NDA)

**To
 The Green Climate Fund (“GCF”)**

Re: Funding proposal for the GCF by Camco Management Limited regarding Renewable Energy Performance Platform (REPP) 2.0

Dear Madam, Sir,

We refer to the programme titled *Renewable Energy Performance Platform (REPP) 2.0* in multiple countries including Cameroon, as included in the funding proposal submitted by Camco Management Limited to us on 24 October 2022.

The undersigned is the duly authorized representative of Ministry of Environment, Protection of Nature and Sustainable Development (MINEPDED), the National Designated Authority of Cameroon.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Cameroon has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Cameroon;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Prof. Paul TCHAWA

Secretary General

Ministry of Environment, Protection of Nature and Sustainable Development
 Cameroon Tél 2 22 23 60 16 – 6 96 93 90 11



République Démocratique du Congo
Ministère de l'Environnement et Développement
Durable



Secrétariat Général à l'Environnement et Développement
Durable

Direction de Développement Durable
Coordination Nationale du Mécanisme de Développement
Propre et du Fonds Vert pour le Climat

L'Autorité Nationale Désignée

To: The Green Climate Fund (“GCF”)

Kinshasa, 28 April 2022

Re: Funding proposal for the GCF by Camco Management Limited (“Camco”) regarding the programme – **“Renewable Energy Performance Platform (REPP) 2.0”**

Dear Madam, Sir,

We refer to the programme titled “Renewable Energy Performance Platform (REPP) 2.0”, in multiple countries, including the Democratic Republic of Congo (DRC), as included in the funding proposal, and as represented in the concept note submitted by Camco Management Limited to us on 25 February 2022.

The undersigned is the duly authorized representative of the National Coordination of the Green Climate Fund, the National Designated Authority of DRC.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal and as represented in the concept note.

By communicating our no-objection, it is implied that:

- (a) The government of DRC has no-objection to the programme as included in the funding proposal and as represented in the concept note.
- (b) The programme as included in the funding proposal and as represented in the concept note is in conformity with the national priorities, strategies and plans of DRC;
- (c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal and as represented in the concept note is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal and as represented in the concept note has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Hans André Lohayo Djamba
Coordinator
National Coordination of the Green Climate Fund
Democratic Republic of the Congo



Ministry of Energy and Meteorology
Private Bag A91
Maseru 100
Lesotho
Tel. 22322334

To: The Green Climate Fund ("GCF")

Maseru, 8 March 2023

Re: Funding proposal for the GCF by Camco Management Limited regarding Renewable Energy Performance Platform (REPP 2)

Dear Madam, Sir,

We refer to the programme titled Renewable Energy Performance Platform (REPP 2) in multiple sub-Saharan African countries, including Lesotho, as included in the funding proposal submitted by Camco Management Limited to us on 24 October 2022.

The undersigned is the duly authorized representative of the Ministry of Energy and Meteorology, the National Designated Authority of Lesotho.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Lesotho has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Lesotho;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. MOKOENA FRANCE
Director (a.i)
Lesotho Meteorological Services
Ministry of Energy and Meteorology
GCF National Designated Authority for Lesotho



Antananarivo, 24th July 2023

SECRETARIAT GENERAL

**BUREAU NATIONAL DES CHANGEMENTS
CLIMATIQUES ET DE LA REDD+**

AUTORITE NATIONALE DESIGNEE

N° **09** -23/MEDD/SG/BNCCREDD+/AND

**To: The Green Climate Fund ("GCF") Secretariat
Song Do, South Korea**

**Re: Funding proposal for the GCF by Camco Management Limited regarding
Renewable Energy Performance Platform (REPP 2.0).**

Dear Madam, Sir,

We refer to the programme titled *Renewable Energy Performance Platform (REPP 2.0)* in Madagascar as included in the funding proposal submitted by Camco Management Limited to us on 14th July 2023 .

The undersigned is the duly authorized representative of the Ministry of Environment and Sustainable Development, the Focal Point of Madagascar.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Madagascar has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Madagascar;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Lovakanto RAVELOMANANA

Director of the National Office of Climate Change and REDD+

GCF FOCAL POINT

MADAGASCAR

Copy to:

The Minister of Environment and Sustainable Development

The General Secretary of MEDD

Telephone: +265 1 771 111
 Telefax: +265 1 773 379

Our Reference: EAD/99/06/04N
 Your Reference:.....



Environmental Affairs Department
 Private Bag 394
 Capital City
 Lilongwe 3

21st December, 2022

The Green Climate Fund ("GCF")
 Executive Director
 Green Climate Fund
 Song Do Business District
 175 Art Center-Daero
 Yeonsu-Gu, Incheon 22004
Republic of Korea

Dear Madam, Sir,

Re: Funding proposal for the GCF by Camco Management Limited regarding Renewable Energy Performance Platform (REPP 2.0)

We refer to the programme titled *Renewable Energy Performance Platform (REPP 2.0)* in multiple countries including Malawi, as included in the funding proposal submitted by Camco Management Limited to us on 24 October 2022.

The undersigned is the duly authorized representative of the Environmental Affairs Department, the National Designated Authority of Malawi.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Malawi has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Malawi;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mrs. Tawonga Mbale-Luka
Director of Environmental Affairs
GCF Nationally Designated Authority for Malawi

To: The Green Climate Fund ("GCF")

Niamey, 08 November 2022

Re: Funding proposal for the GCF by Camco Management Limited regarding Renewable Energy Performance Platform (REPP) 2.0

Dear Madam, Sir,

We refer to the programme titled *Renewable Energy Performance Platform (REPP) 2.0* in multiple countries including Niger as included in the funding proposal, submitted by Camco Management Limited to us on 24 October 2022.

The undersigned is the duly authorized representative of National Council of the Environment for Sustainable Development (CNEDD), the National Designated Authority of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

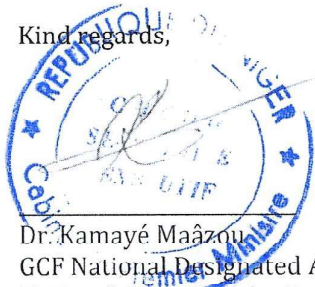
- (a) The government of Niger has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Niger;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Dr. Kamayé Maâzou
GCF National Designated Authority
National Council of the Environment for Sustainable Development (CNEDD)
Niger



FEDERAL MINISTRY OF ENVIRONMENT

HEADQUARTERS, MABUSHI, ABUJA

Ref: No. FMENV/DCC/GCF/044/I/56

Date: 24th May, 2022

The Green Climate Fund ("GCF")
Songdo International Business District,
175, Art Center-daero,
Yeonsu-gu, Incheon 406-840,
Republic of Korea.

Re: Request for Issuance of a No-Objection Letter(NOL) in Support of the Renewable Energy Performance Platform(REPP) 2.0 Programme

Dear Sir/Madam,

The Ministry refers to the programme titled **Renewable Energy Performance Platform(REPP) 2.0 Programme** in Nigeria as included in the funding proposal submitted by **Camco Management Limited** to the Ministry on 6th May, 2022.

The undersigned is the duly authorized representative of Federal Ministry of Environment/Department of Climate Change, the National Designated Authority/Focal Point of Nigeria.

Pursuant to GCF decision B.08/10, the content of which the Ministry acknowledges to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

Kindly note that by communicating our no-objection, it implies that:

- (a) The government of Nigeria has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Nigeria;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

In view of the above, the Ministry confirms that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed and our no-objection applies to all projects or activities to be implemented within the scope of the programme.

Kind regards,

Mrs Iniobong Abiola-Awe

GCF Focal Person Nigeria/Director, Department of Climate Change
Federal Ministry of Environment
Nigeria



GOVERNMENT OF SIERRA LEONE
Environment Protection Agency
Ministry of the Environment
92 Dundas Street, Freetown



Ref: EPA-SL/HA/GCf/05/22/DDPPR

3rd May 2022

To:
 The Green Climate Fund ("GCF")

Dear Sir/Madam;

Re: Funding proposal for the GCF by Camco Management Limited regarding Renewable Energy Performance Platform (REPP) 2.0

We refer to the programme titled Renewable Energy Performance Platform (REPP) 2.0 in multiple countries, including Sierra Leone, which was included in the funding proposal in the concept note submitted by Camco Management Limited to us on 25th February 2022.

The undersigned is the duly authorised representative of the Environment Protection Agency, the National Designated Authority of Sierra Leone.

Pursuant to GCF decision B.08/10, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it implies that:

- (a) The Government of Sierra Leone has no objection to the programme as included in the funding proposal and as represented in the concept note;
- (b) The programme conforms with the national priorities, strategies and plans of Sierra Leone;
- (c) In accordance with the GCF's environmental and social safeguards. The programme conforms with relevant national laws and regulations, as per the GCF's environmental and social safeguards requirement.

We confirm that our national process for ascertaining no-objection to the programme, as included in the funding proposal and represented in the concept note, has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the programme's scope.

We acknowledge that this letter will be publicly available on the GCF website.

Kind regards.

Bondi Geva (PhD)

Executive Chairman

Environment Protection Agency, Sierra Leone

All communication should be addressed to the
Permanent Secretary
Telephone: (260 211) 252395
(260 211) 252394
(260 211) 252391



REPUBLIC OF ZAMBIA

MINISTRY OF GREEN ECONOMY AND ENVIRONMENT

OFFICE OF THE PERMANENT SECRETARY
CORNER OF JOHN MBITA & NATIONALIST ROADS
P.O. BOX 30147
LUSAKA, ZAMBIA

To: The Green Climate Fund ("GCF")

Lusaka, 28 September 2022

Re: Funding proposal for the GCF by the Camco Management Limited

Dear Madam, Sir,

We refer to the programme titled "*Renewable Energy Performance Platform (REPP2.0)*" in Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sao Tome, Sierra Leone and Zambia as included in the funding proposal submitted by Principle Camco Management Limited to us on 15 March 2022.

The undersigned is the duly authorized representative of Francis Mpampi, the National Designated Authority of Zambia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Zambia has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Zambia;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Francis Mpampi
National Coordinator
National Focal Point Person and National Designated Authority
Zambia

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Renewable Energy Performance Platform (REPP) 2
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Private
Accredited entity	Camco Management Limited (CAMCO)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Target countries: Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sao Tome and Principe, Sierra Leone and Zambia
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, June 7, 2023
Language(s) of disclosure	English, French, and Portuguese
Explanation on language	English, French, and Portuguese are official languages of target countries.
Link to disclosure	English: https://www.camco.fm/files/ugd/179ccb_5fb86cc261d543dfb08a86776f6a2a34.pdf French: https://www.camco.fm/files/ugd/179ccb_83c31750073444f9b2cd63d28b5c0eb0.pdf Portuguese: https://www.camco.fm/files/ugd/179ccb_4646058669904c909dc2c70d4ffeef1b.pdf

Other link(s)	https://www.camco.fm/environmental-and-social-disclosure
Remarks	An ESMS consistent with the requirements for a Category I-2 project is contained in the “Environmental and Social Management Framework”.
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity’s website	Indigenous Peoples Planning Framework Land Acquisition Plan and Resettlement Action Plan Wednesday, June 7, 2023
Language(s) of disclosure	English, French, and Portuguese
Explanation on language	English, French, and Portuguese are official languages of target countries.
Link to disclosure	https://www.camco.fm/files/ugd/179ccb_5fb86cc261d543dfb08a86776f6a2a34.pdf
Other link(s)	N/A
Remarks	An Indigenous Peoples Planning Framework titled “Indigenous Peoples Policy”, “Land Acquisition Plan” and “Resettlement Action Plan” are included in the ESMF.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, July 14, 2023
Place	Electronic and printed versions were shared with NDAs in target countries. Electronic copies were shared with the CSO and PSO active observers on 20 September 2023. Printed copies were made available for consultation at NDA offices in respective host countries, in their official language(s): <u>Cameroon:</u> Mr. Paul Tchawa Secretary General Ministry of Environment, Protection of Nature and Sustainable Development (MINEPDED) Boulevard du 20 Mai, Immeuble Ministériel No. 2, P.O Box 320, Yaoundé, Cameroon Phone : +237 696 939 011 <u>Democratic Republic of the Congo (DRC):</u> Prof. Isaac KALONDA KANYAMA Coordinateur Fonds vert pour le climat Ministère de l’Environnement Boulevard palais de la Nation 2895 Kinshasa Gombe Kinshasa Republique Democratique du Congo (RDC) Phone : +243 826 631 495 <u>Lesotho:</u> Mr Habofanoe Augustinus Makopela

Deputy Principal Secretary
Ministry of Natural Resources
Private Bag A91, Maseru 100, Lesotho
Phone: +266 59030095/ +266 22322334

Madagascar:

Mme Lovakanto Ravelomanana
Directeur du Bureau National de Coordination des
Changements Climatiques,
Ministère de l'Environnement, de l'Ecologie et des Forêts
B.P 3948, Rue Toto Radola - Antsahavola,
Antananarivo, Madagascar
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Malawi:

Mrs Tawonga Mbale-Luka
Director, Environmental Affairs Department
Environmental Affairs Department
Lingadzi House within the Ministry of Natural Resources,
Energy and Mining
Private Bag 394 Lilongwe 3,
Lilongwe, Malawi
Phone: +265 1 771 111/ +265 1 773 379

Niger:

Dr. Kamaye Maâzou
Secrétaire Exécutif
Cabinet du Premier Ministre
Conseil National de l'Environnement pour un
Développement Durable (CNEDD)
PL 17 Rue du Gouverneur Jules Brévié, BP 10193,
Niamey, Niger
Phone: +227 207 22 559/ +27 9698 7470/ +227 9022 1700

Nigeria:

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Director General
National Council on Climate Change
Department of Climate Change
Federal Ministry of Environment
Plot 444 Aguiyi Ironsi Way, Maitama Abuja, Nigeria
Phone: +234 8030570625

Sao Tome and Principe:

Dr. Fausto Policarpo Abreu das Neves
Ministério do Planeamento, Finanças e Economia Azul
Água Grande, São Tomé
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	<u>Sierra Leone:</u> Dr. Bondi Gevao Executive Chairman Environment Protection Agency 92 Dundas Street, Freetown, Sierra Leone Phone: +23230082924/ +23279049175 <u>Zambia:</u> Mr. Francis Mpampi National Coordinator - National Designated Authority Ministry of Green Economy and Environment P.O. Box 50555, Corner of John Mbita and Nationalist Road, Lusaka, Zambia Phone: +260 21 1 252625
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	TBC
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

Independent Technical Advisory Panel's assessment of FP222

Proposal name:	Renewable Energy Performance Platform (REPP 2)
Accredited entity:	Camco Management Limited (CAMCO)
Country/(ies):	Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sierra Leone and Zambia
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

Note: This assessment report is based on a review of the updated and revised proposal received from the accredited entity (AE) for the thirty-seventh meeting of the Board (B.37). It follows the assessment report based on an earlier version of the funding proposal submitted by the AE for the thirty-sixth meeting of the Board (B.36), during which it had not been endorsed by the independent Technical Advisory Panel (TAP). This revised assessment report reflects the updates and revisions the AE has implemented in response to the earlier independent TAP assessment report for B.36. It also reflects the clarification discussions held between the AE, the Secretariat and the independent TAP on the side lines of B.36 in Songdo, Korea, and the two follow-up videos calls between the parties on 9 August 2023 and 4 September 2023.

1.1 Overview

1. In its assessment, the independent TAP is guided by decision B.33/12 of the GCF Board, which explains the principles for demonstrating the impact potential of GCF-supported activities as well as other appraisal and assessment guidance in considering these investment criteria and as endorsed by the GCF Board.
2. The Renewable Energy Performance Platform 2 (REPP 2; the fund) is a new USD 250 million blended finance fund established as a *société en commandite spéciale* (SCSp) or special limited partnership in Luxembourg entitled Camco REPP SCSp and promoted by Camco Management Limited (CAMCO). The main purpose of the fund is to invest in the early-stage development and construction of on-grid as well as off-grid small-scale renewable energy projects in sub-Saharan Africa, with a focus on least developed countries. The current list of target markets for REPP 2 includes Cameroon, the Democratic Republic of the Congo, Lesotho, Madagascar, Malawi, the Niger, Nigeria, Sierra Leone, and Zambia. This list may grow, subject to further no-objection letters that the AE may receive from additional African countries.
3. REPP 2 builds on a predecessor fund, the Renewable Energy Performance Platform 1 (REPP 1), which CAMCO has been implementing since 2015 with a commitment of 88 million pounds sterling (GBP) in grant funding from the Foreign, Commonwealth and Development Office (FCDO) of the United Kingdom of Great Britain and Northern Ireland. Not all funding provided by FCDO has been drawn by REPP 1 and the fund is still active but no longer makes new investments (except possibly into REPP 2).
4. The purpose of the present funding proposal is to request GCF for funding in the form of junior equity participation in REPP 2 in the amount of USD 50 million, which CAMCO, as the AE, would invest in REPP 2 as a fiduciary for GCF through a dedicated special purpose vehicle (GCF REPP Holdings). CAMCO serves as both AE and executing entity (EE) for this funding proposal,

whereby Camco REPP 2 SCSp would appoint Camco REPP GP S.à.r.l. (wholly owned by Camco Management Limited) as general partner at first close of the fund.

5. The junior equity funded by GCF is expected to allow REPP 2 to: (i) leverage private, commercial and international financial institution (IFI) funding tranches into its capital stack (both senior equity as well as notes), based on a risk allocation where losses on the REPP 2 portfolio would first be absorbed by the junior equity (and not *pari passu* across all equity investors in REPP 2); and (ii) provide attractive commercial terms to investees due to the fact that remuneration of the junior equity tranche does not reflect risk-adjusted pricing.

6. The main purpose of REPP 2 is to drive a paradigm shift in the development of small-scale renewable energy projects in sub-Saharan Africa. REPP 2 proposes to invest in the following project types:

- (a) On-grid small-scale renewable energy generation projects (independent power producers) with an installed capacity of up to 25 MW for small hydropower, and solar power projects and 50MW for wind (categories B and C risks only) and a target allocation of 50 per cent in the REPP 2 portfolio; and
- (b) Off-grid energy ventures based on renewable energy, including mini-grid operators and solar home system (SHS) companies, with a target allocation of 40 per cent for mini-grids and 10 per cent for SHS companies in the REPP 2 portfolio.

7. REPP 2 will provide a range of financial instruments to invest in these projects, focusing on five different strategies that seek to address key gaps in the financing market:

- (a) **Early-stage growth funding for off-grid companies.** This would include hybrid equity (subordinated debt, convertible notes and other similar instruments) to support mini-grid operators and SHS companies with development expenses, thus addressing the shortage of early-stage high-risk equity in the market;
- (b) **Development-stage funding for grid-connected projects.** Similar to the above, this strategy would include hybrid equity provided to project developers to enable them to move new renewable energy projects along the development cycle, with an option to convert financing to construction loans at financial close, recognizing that developers have difficulty tapping the commercial market due to perceived and real risks in project development;
- (c) **Expansion of SHS businesses.** This would include equity as well as debt finance for SHS companies that want to grow their businesses and/or to expand into new markets and those that currently do not have access to financing in the commercial market due to their risk profile;
- (d) **Construction of mini-grid portfolios.** This would include hybrid equity as well as subordinated and senior loans to mini-grid operators to finance construction of new facilities to grow their footprint and client base. This funding would also be for mini-grid operators that cannot access commercial sources of finance due to their challenging businesses models and undeveloped/untested regulatory environments in target markets; and
- (e) **Construction finance.** Through this strategy, REPP 2 would provide hybrid equity, subordinated or senior loans to finance construction of small-scale on-grid renewable energy projects that otherwise find it challenging to mobilize the required financing from the commercial market.

8. Apart from providing financial instruments to its investees, REPP 2 will further support investees/projects through: (i) risk mitigation support (e.g. facilitating access to political risk insurance and off-taker credit enhancement/risk insurance available in the market); (ii) development of strategies to mitigate hard currency risks for investees; (iii) development of

skills and capacities of local entrepreneurs and project developers; (iv) support to environmental and social risk management by investees (e.g. providing assistance in the development of environmental and social management systems); and (v) promotion of gender equity (e.g. providing assistance in the development of investee-specific gender action plans, which will be mandatory for all projects).

9. REPP 2 also features a technical assistance facility (TAF) that will: (i) support investees and entrepreneurs with early-stage development grants to enhance their project development capabilities; (ii) work with governments/regulators in target markets to develop a conducive enabling environment for investment in small-scale renewable energy projects; and (iii) provide assistance to female entrepreneurs to help close the gender gap in the renewable energy industry. The TAF is not a subject of this request for funding from GCF; a funding application in the amount of USD 5 million for the TAF is pending with FCDO. The independent TAP commends the AE for raising funds for the TAF from other sources.

10. The proposed fund life is 20 years, with a 7-year investment period during which it would invest in approximately 37 transactions (of which 26 are expected to be innovative precedence transactions in target markets). At the level of REPP 2 itself (and using the proposed USD 100 million junior equity commitment from GCF and REPP 1/FCDO as leverage), the fund is expected to mobilize USD 150 million in additional funding from private sector investors as well as development finance institutions (DFIs) and IFIs (mobilization ratio of 1:1.5).¹ At the level of the individual assets that the fund will invest in, REPP 2 expects to mobilize a total of USD 786 million in investment, whereby approximately 50 per cent (i.e. USD 393 million) is expected to come from private sector and commercial sources (i.e. DFIs investing on commercial terms), translating into an indirect private sector/commercial funding leverage of the junior equity of roughly 1:4. (The independent TAP assumes that the remaining 50 per cent of funds mobilized at asset level are public sector (non-commercial) contributions, though the funding proposal does not provide details on the source of these funds.)

11. In terms of impact, the AE expects REPP 2 to achieve the following targets:

- (a) 330 MW of new renewable energy capacity installed (translating into 130,580 MWh of electricity produced);
- (b) 12.7 megatonnes of carbon dioxide equivalent (Mt CO₂ eq) of avoided or reduced greenhouse gas (GHG) emissions (full REPP 2 investment portfolio over technology lifetime);
- (c) New or improved access to clean energy for more than 7.7 million people;
- (d) Improved climate resilience of nearly one million beneficiaries; and
- (e) 76,801 jobs (of which 23,040 jobs will be for females).

1.2 Impact potential

Scale: Medium to high

12. The funding proposal credibly demonstrates that, if successful, REPP 2 has the potential to engender considerable mitigation, adaptation and sustainable development benefits in target countries. Overall, the theory of change, results framework and calculation/attribution

¹ The funding proposal distinguishes between three different sources of funding that REPP 2 hopes to attract to the project: (a) private sector capital (defined by independent TAP as funding from entities other than those (i) where the majority of shares or ownership in that entity (i.e. more than 50%) are held by public sector entities; or (ii) that benefit from an explicit state guarantee or other government support); (b) DFI capital invested on commercial terms, where such entities source or raise capital on their own balance sheets on a commercial basis including directly from private capital markets; and (c) IFI capital (possibly with somewhat lower return thresholds than (a) and (b), though the funding proposal does not provide further information on the precise sources/nature of IFI contributions).

methodologies proposed for assessing the overall mitigation, adaptation and sustainable development impacts are transparent and based on best practice/market standards (see annex 22 B of the funding proposal). The independent TAP was able to reconstruct the high-level impact targets provided in the funding proposal based on the background information that was provided by the AE. The funding proposal also contains reasonably detailed assessments of climate change hazards for all target countries and highlights climate change vulnerabilities (see annex 2 of the funding proposal) that support the overall climate rationale for this funding proposal.

13. Regarding the expected mitigation impact, the overall GHG emission reduction/avoidance target over the lifetime of the programme is expected to be 12,720,297 t CO₂ eq. The calculation is based on the expected pipeline of projects that the fund may finance (330 MW in total across a mix of technologies/installation sizes), the expected energy generation (812,771 MWh per year for the entire portfolio) and the relevant grid emission factors across target jurisdictions, following the clean development mechanism methodology – including AMS-IA: electricity generation by the user (version 19); AMS-ID: grid-connected renewable electricity generation (version 18); and AMS-IL: electrification of rural communities using renewable energy (version 4). In practice, the actual projects funded by REPP 2 will likely be substantially different from the (anonymized) pipeline of projects contained in the funding proposal, but the overall targets presented appear reasonable and achievable. All mitigation targets will be aligned with host-country targets and priorities. The AE has provided detailed studies of the target countries, underscoring the relevance of investment in renewable energy capacity as a key mechanism to either replace or avoid fossil fuel-based energy generation alternatives.

14. More specifically, REPP 2 is expected to realize adaptation impact via two channels:

15. First, by investing in small-scale renewable energy generation (including off-grid businesses), REPP 2 is expected to create more resilient power systems by way of diversification and decentralization:

- (a) Diversification away from large-scale hydropower generation that is increasingly subject to heightened variability in water availability due to climate change impacts; and
- (b) Increased decentralization through investment in off-grid solutions, thus reducing the reliance of energy users on power grids that are becoming increasingly unstable due to extreme weather events resulting from climate change.

16. Second, through investments in off-grid solutions (e.g., mini-grids, SHS companies), the fund expected to increase the climate resilience of people and communities (including micro, small and medium-sized enterprises) by providing them with access to clean energy as an alternative to unsustainable fossil fuel/biomass solutions. The total number of beneficiaries REPP 2 is expected to reach extends to 370,224, based on its current pipeline.

17. While the independent TAP recognizes the potential for REPP 2 to produce these positive climate outcomes under a best-case scenario, there are also various challenges that fund will have to successfully navigate to leverage its full potential.

18. One of the key factors driving the fund's potential to have impact at scale is the ability of the AE/EE to mobilize significant co-investments from other funding providers beyond the requested GCF junior equity contribution to ultimately grow the fund size to USD 250 million. In addition to raising USD 100 million in junior equity from public/catalytic investors (including the proposed USD 50 million contribution from GCF), the AE expects to raise an additional USD 100 million in senior equity and USD 50 million in notes (debt) from private sector, commercial and IFI funding sources, with a resulting leverage ratio of 1:1.5. The independent TAP notes various concerns regarding the ability of the AE/EE to mobilize sufficient funding for senior equity and notes for REPP 2, specifically:

- (a) **First-time manager and risk profile:** While the AE has experience with renewable energy investment in sub-Saharan Africa, the only real reference is the predecessor fund to REPP 2 (REPP 1), which was funded entirely by grants. (The independent TAP was informed that CAMCO is separately preparing a USD 50 million commercial and industrial fund focusing on promoting solar investments; that fund is in the process of closing.) REPP 1 has shown delayed implementation and subdued commercial returns (5.1 per cent on portfolio, which is to some extent a consequence of the strong developmental mandate that was provided to REPP 1 by its only capital provider, FCDO). Consequently, private sector investors as well as DFIs that invest on a commercial basis will consider the EE a de facto first-time manager. In that context, the size of the fund (USD 250 million) will be considered large, with an aggressive risk profile (small-scale investment in countries with ratings of “B” or below) in an overall complicated market environment (e.g., due to rising interest rate levels and deteriorating macro fundamentals in REPP 2 target markets).
- (b) **Returns for investors and liquidity options:** The return expectations REPP 2 is offering both for senior equity as well as notes will further narrow the field of interested parties. In its dialogue with the independent TAP, the AE noted that a higher hurdle rate for the senior equity would be preferable (especially with a view to enticing private sector investors) but highlighted that this could also be achieved through a refinancing of the loan book at year 12. In addition to boosting returns, a refinancing of the loan book at year 12 would provide investors (who typically would not want to lock in a funding commitment for a period of 20 years) with a liquidity option, something the independent TAP also considers crucial for the AE to attract especially private sector investors to REPP 2. The AE highlighted that there were expressions of interest from some commercial banks to participate in refinancing and highlighted that it seeks to negotiate a guarantee (potentially through Swedish International Development Cooperation Agency) to provide certainty to senior equity funders that the refinancing of the loan book will happen. The independent TAP strongly encourages the AE to prioritize discussions with potential providers of a refinancing guarantee as the independent TAP believes that, without certainty on early liquidity, the fund will have difficulty attracting investors. If a refinancing of the portfolio is successful (possibly enabled by a refinancing guarantee from the Swedish International Development Cooperation Agency or another party), REPP 2 would also make a significant contribution to secondary market development, something that is sorely lacking in much of the African continent.
- (c) **Fund life and fee structure:** The original funding proposal received for B.36 foresaw a fund life of up to 25 years, which the independent TAP deemed unduly long with a view to raising capital. While the independent TAP recognizes that an extended fund life is required to provide long-dated debt to target investees, the proposed 25-year life would have further increased the risk profile of REPP 2 for private sector as well as commercial DFI funders. It also would have created a situation where the total fee pool would become very large, representing a significant portion of capital commitments. (For clarity, the independent TAP considers the proposed fee percentages as “within” market. The challenge is that the long fund life drives the absolute volume of fees that would have to be financed out of fund returns.) In the revised proposal, the AE has reduced the fund life to 20 years, which addresses the challenge to a certain extent. The independent TAP also notes that the GCF benefits from a Most Favoured Nation clause in the term sheet that requires the fund to pass on any reduction in fee levels (e.g. because of negotiations with other public and private funders) to GCF.
19. Overall, the independent TAP believes that few investors will have the ability to consider investments of this nature, leaving a narrow funding market. In the current macro environment, even DFIs investing on a commercial basis may find it difficult to commit sizable tickets to REPP

2. The AE has provided a fundraising update to the Secretariat and the independent TAP that suggests that, despite these overall challenges, there is some decent investor interest in the fund, as well as a reasonable response from DFIs and IFIs who are currently evaluating investments in senior equity or notes for a first close of the fund with a target volume of USD 60 million. While the independent TAP commends the AE for the significant efforts it is putting into fundraising in a difficult market environment, the independent TAP also notes that there remains a risk that the AE will not be able to raise the entire USD 250 million target volume for the fund, thus making the achievement of the overall impact targets difficult if not impossible.

20. **Translating additional generation/investment in off-grid companies into actual use of clean energy.** The main investment strategy of the fund, as noted above, is to invest in on-grid and off-grid small-scale renewable energy projects. In the case of investments into on-grid renewable energy independent power producers, the positive mitigation impact appears relatively straightforward. Most power systems in the target countries currently lack sufficient generation capacity (and in extreme cases load-shedding); therefore, the additional renewable energy will be absorbed/used, thus avoiding use of fossil fuel alternatives. However, for investments in off-grid companies, the logic may not be that straightforward. The AE implies in the funding proposal that investments in mini-grid operators and SHS businesses will, by definition, translate into new connections and actual use of electricity. On the other hand, data on the ability and willingness to pay in the target markets of REPP 2 show that many households (and many microbusinesses) will find it difficult to access energy service offers due to the high initial connection charges (in particular for mini-grids, which is 40 per cent of the target allocation of REPP 2). Others, and especially vulnerable groups, will likely struggle to meet the credit profile expectations of most SHS businesses (10 per cent of the target allocation of REPP 2). If that indeed materializes, and investments don't translate into new connections and use of clean electricity, the fund will struggle not just in achieving its mitigation but also its adaptation targets.

21. This is a challenge not specific to REPP 2, and addressing it is to some extent outside its immediate sphere of influence. independent TAP notes with appreciation that the AE, in the revised funding proposal, makes a dedicated commitment (under activity 6) to work with its clients to promote access to existing schemes that provide access subsidies (e.g. Sustainable Energy for All's Universal Energy Facility), with a particular focus on vulnerable groups and micro-businesses. The independent TAP is also very much encouraged by the fact that the AE is actively engaged in the design of various minimum review guarantee schemes relevant for mini-grids, and thus has deep insight into the evolving landscape of potential guarantee providers that can then also be leveraged to its clients. The independent TAP recognizes that REPP 2, as a fund that works on commercial principles, cannot provide connection subsidies or minimum revenue guarantees from its own balance sheet.

22. Overall, the independent TAP regards the fund's impact potential as medium to high.

1.3 Paradigm shift potential

Scale: Medium

23. The main mission of REPP 2 is to deploy capital into a series of small-scale renewable energy generation projects (up to 37 transactions). The success (or impact) of the fund should thus in the first instance be judged against the quality of the portfolio that it will build, considering its mitigation, adaptation and sustainable development outcomes as well as its commercial performance.

24. However, the funding proposal suggests that REPP 2 will also offer transformational potential that goes beyond the immediate impact of its investee portfolio, principally through three channels:

- (a) First, the funding proposal highlights that REPP 2 will build local developer and business capacity, thus resulting in a larger and higher quality deal flow that will ultimately precipitate the growth of the local renewable energy industry beyond the fund and provide the basis for general industry growth and follow-on investments;
 - (b) Second, the funding proposal notes that the increase in renewable energy capacity will result in enhanced energy security in the context of increased climate hazards, thus contributing to climate-proofing the energy sector; and
 - (c) Third, according to the funding proposal, the use of clean energy produced by REPP 2 investees will trigger more income-generating activities, help improve social service delivery and thus drive overall socioeconomic development and growth.
25. The independent TAP recognizes the importance of these “transformation channels” as part of the theory of change but would like to emphasize that, in its view, the transformational impact (and thus paradigm shift) of REPP 2 in the small-scale renewable energy market of its target countries will depend to a large extent on three interrelated factors:
 - (a) The ability of REPP 2 to develop and utilize appropriate transaction models that work for small-scale investments and the challenging market contexts that the fund will operate in, as well as models that could provide the basis for replication and scaling up;
 - (b) The ability of REPP 2 to attract private sector investment (as defined in footnote 1 above) at the level of the fund on the back of a clear product offering and reasonable risk-return profile. Attracting such private sector investment is an important test of the extent to which the market understands the REPP 2 product, and a crucial basis for any scaled follow-on investments from the private sector; and
 - (c) The ability of REPP 2 to successfully leverage especially local capital providers (local commercial banks, pension fund managers, etc.) into projects that REPP 2 will finance in the on-grid or off-grid space. For the growth of the small-scale renewable energy market to go to scale and to become “self-perpetuating” (i.e., not reliant on international funding with concessional terms), the local funding market will ultimately have to come to the table.
26. On the first factor, in the funding proposal the AE emphasizes the importance of deploying tailored financing approaches to small-scale renewable energy projects in the target region (and has significant experience with this from REPP 1). In the revised proposal, the AE provides some tangible examples of its approach to “project finance light,” emphasizing the philosophy of innovation and flexibility that it will bring to the transactions it will pursue. However, there is a tension between being flexible in tailoring bespoke solutions and, at the same time, having the ambition to establish replicable transaction models in the market that could then be adopted and scaled up by others. For fundraising with private sector investors, the focus on flexibility and bespoke solutions may also be a challenging selling proposition as they typically look for a clear and efficient “product suite” to be executed. Finally, the independent TAP believes it will be difficult to refinance a portfolio of loans to projects at a later stage (as discussed above) if the transaction structures vary widely.
27. The AE has noted that the fund (based also on its experience with REPP 1) will establish specific investment standards (e.g. for its lending to SHS businesses), and will not allow a proliferation of transaction models. The independent TAP welcomes this and encourages the AE, as part of its knowledge management and outreach with the market, to share lessons learned and best practices of what types of transaction models and standardized documentation work and have potential for replication and scale. This could be of particular relevance in the work of the AE with the local funding market (as foreseen under outcome 5: contributing to stronger technology deployment, innovation and market transformation).

28. On the second factor, for reasons already highlighted above, the independent TAP believes that few private sector investors will consider investing in REPP 2. As a result, this will leave DFIs and IFIs (International Finance Corporation, African Development Bank, etc.) as the main sources of funding, with GCF then effectively providing risk mitigation and subsidy (because of the submarket hurdle rate for junior equity investors) to these DFIs/IFIs rather than to private sector investors. The AE acknowledges that reality in the revised funding proposal and notes that, due to the difficulties of raising funding (especially from the private sector) for REPP 2, it does not stipulate specific private sector fundraising targets at the level of the fund. Also, the GCF investment criteria do not prescribe any co-financing requirements or, for that matter, private sector leverage targets for GCF investments. However, the independent TAP believes that the fund's ability to attract private sector investment is an important precondition for it to effect paradigm shift. A scaling up of the scope and impact of the programme, or a replication of its approach in other regions, is only likely if the approach demonstrates its ability to leverage private sector investors into the capital structure of the fund.

29. In this context, the independent TAP welcomes the introduction of a tiered pricing structure for different investor categories in the revised and updated term sheet (annex 14 of the funding proposal) that may provide additional incentives for private sector investors to make funding available. If implemented, this would reflect the reality that private sector investors in the current macroeconomic environment will find it otherwise difficult to make a long-term funding commitment to REPP 2 without attractive returns. Even considering the risk mitigation provided by the junior equity layer, a senior preferred return of 15 per cent cannot be considered excessive. The independent TAP recommends including a clear definition of private sector investors (following the definition in footnote 1 of this assessment report) to ensure that any additional senior preferred returns accrue only to private sector investors.

30. In order to ensure that REPP 2 can unfold its full paradigm shift potential, the independent TAP recommends to the Board to include the following condition to its potential approval of this funding proposal: "Other than in relation to the first disbursement of the GCF Proceeds, delivery by the Accredited Entity to the Fund, in form and substance satisfactory to the GCF Secretariat, of written evidence confirming that Co-financing from private sources of an amount not less than ten per cent (10%) of the total capital commitments to Camco REPP SCSp (including the requested disbursement) is either already committed or expected to be committed, to Camco REPP SCSp, at the same time as the GCF Proceeds from the requested disbursement. Such private sources shall exclude co-financing which is ultimately provided by entities (i) where the majority of shares or ownership in that entity (i.e. more than 50%) are held by public sector entities; or (ii) that benefit from an explicit state guarantee or other government support agreement."

31. More broadly speaking, the mobilization of private sector funding for climate action is identified as an important objective of the GCF Private Sector Strategy. In that context, the independent TAP tries to focus on the specific private sector leverage targets (e.g., at portfolio level, or for certain categories of investments) presented in GCF funding proposals. However, writing a mobilization target into funding proposals is not the same as actual mobilization in the real world, and in the markets as we find it. We would suggest that the Secretariat explore ways to better diligence claims of private sector leverage in funding proposals, as basis for the independent TAP review. We would also suggest exploring incentive mechanisms for AEs to ensure that they industriously pursue these private sector leverage targets after having secured GCF funding. GCF is not the only funding institution grappling with this issue; many public climate finance providers are trying to develop pathways to turn billions into trillions by way of leveraging private sector capital. The independent TAP believes that, in order for GCF to successfully achieve the Private Sector Strategy's objective "to catalyse private climate finance in a manner fully aligned with a country-driven approach to meet developing countries' needs and the objectives of the USP" (updated Strategic Plan for GCF), specifying leverage targets and

introducing incentive mechanisms to drive these targets could be one powerful way to underpin the Private Sector Strategy with concrete action.

32. On the third and final factor, in the funding proposal the AE suggests that, at the level of individual assets, it expects to raise roughly USD 393 million (or 50 per cent of the total funds expected to be mobilized by REPP 2) from private sector and commercial sources. (The funding proposal is unclear about where the balance in mobilized funding would come from. The independent TAP assumes that it would come from public sector sources, such as development aid, concessional funding from development banks, and government contributions.) Considering the complexity of the asset profiles and the target markets, this relatively low share of mobilized funding from private sector and commercial sources may not come as a surprise. The estimate is based on the actual track record of REPP 1, which has mobilized USD 460 million in co-financing, with roughly 50 per cent of that coming from private sector and commercial sources. The independent TAP would strongly encourage the AE/EE to make every effort to increase the share of co-financing from local private sector sources. The independent TAP recognizes that local private sector financing is harder to obtain and more expensive (thus driving, for example, the levelized cost of electricity for small-scale on-grid renewables and other small-scale projects), but the independent TAP believes it is essential to ensuring that the local funding market engages in this market segment to achieve paradigm shift. The proposed refinancing of assets after 10–12 years via a commercial bank is an additional way to achieve the objective of involving commercial banks but, as discussed earlier, it is far from assured.

33. The independent TAP appreciates the differentiated and detailed treatment of the issues surrounding local currency financing by REPP 2 in the funding proposal. The independent TAP agrees that local currency risk can be more efficiently managed at the level of the fund than at the level of individual assets. The independent TAP also recognizes that the fund needs to balance, on the one hand, providing financing products that clients/projects can afford (thus generally favouring local currency financing) and, on the other hand, ensuring that the risk profile of REPP 2 remains attractive particularly to private sector investors (which would mean carrying as little currency mismatch on the balance sheet as possible). In actual practice, it would appear that, for a large part of the targeted portfolio, the AE/EE will transfer the currency risk to investees (as hedge/swap opportunities are limited/too expensive and the programme may not be able to carry the associated risk). The independent TAP encourages the AE to also consider partnerships with local investors in some key markets to determine whether there may be alternative ways for the fund to provide local currency solutions to projects. In addition, the AE could explore partnerships with other concessional capital providers (such as the European Commission, Nordic Development Facility, and KfW Development Bank) to raise additional local currency shares to mitigate local currency risk at level. This is being implemented very effectively in other blended finance funds such as the SANAD Fund, the European Fund for Southeast Europe, and the Green for Growth Fund. A sustainable approach to local currency investing is another key factor driving the potential for paradigm shift.

34. The EE proposes to engage with governments and regulators in key target markets with the aim to structure transactions for REPP 2 that may require government/regulatory consents/changes. The EE also intends to provide support to governments/regulators in target markets via the TAF (which is outside the scope of this proposal) to address enabling environment constraints, although the precise scope of that remains somewhat unclear. In light of this, the independent TAP acknowledges that the EE goes beyond the scope of the mandate of a typical investment manager and recognizes that structuring small-scale renewable energy transactions in the target markets will be challenging and will require consistent engagement with public sector authorities. The funding proposal argues that the “precedence” nature of the transactions that the fund will support will create a model for follow-on transactions to occur. The funding proposal, as well as the question-and-answer documentation, also points to transactions in the REPP 1 portfolio and how such model transactions with REPP 1 involvement had been instrumental in triggering additional investment activity in the respective markets

(e.g. in Burundi after the investment of REPP 1 into the Mubuga Solar Project). However, the independent TAP believes that, to drive transformational change, the fund's investments and activities would have to provide a credible pathway for follow-on investments by systematically addressing enabling environment constraints in target markets. This is not the focus of the fund (and may well be outside its sphere of influence), but the constraints ultimately limit the potential of REPP 2 to advance regulatory/legislative frameworks to systematically promote investment in small-scale renewable energy projects.

35. Overall, the independent TAP rates the potential for REPP 2 to effect paradigm shift as medium, with potential for improvement if the issues highlighted above are adequately addressed.

1.4 Sustainable development potential

Scale: High

36. The funding proposal presents a credible pathway for REPP 2 to generate significant sustainable development benefits:

- (a) **Economic co-benefits:** The funding proposal notes that the programme is expected to create more than 76,801 jobs. The impact model provided by the AE (annex 22 of the funding proposal) as well as the explanations provided in the Q+A session point to the reality that a significant share of this estimate is based on indirect job creation effects from new generation capacity supported by REPP 2 investments. The funding proposal notes that "The ex-ante estimate of the number of jobs is based on the average number of jobs required by MW [megawatt] build, multiplied by MWs financed. The job numbers are technology specific and have been driven from IRENA: Renewable Energy and Jobs, Annual Review 2021.";
- (b) **Social co-benefits:** The funding proposal highlights multiple social co-benefits resulting from programme investments, mostly due to improved access to lighting, the electrification of health facilities, access to refrigeration, etc. The independent TAP notes that those benefits (much like the economic co-benefits) only accrue if the investment in generation capacity or off-grid companies translates into more connections and actual use of clean energy by beneficiaries; and
- (c) **Gender-sensitive development co-benefits:** Finally, the funding proposal argues that the benefits from additional investment in small-scale renewable energy generation will disproportionately benefit women and girls as, at present, they tend to be responsible for sourcing alternative sources of fuel (e.g. firewood), and obtaining access to clean electricity will enable them to use their time more productively (e.g. schooling). REPP 2 will also make gender action plans mandatory for all investees, and drive compliance with the 2X gender lens investing criteria. Because the due diligence on the fund pipeline has not commenced yet, it is unclear how the projects in the pipeline currently perform against the 2X gender lens investing criteria. However, in the funding proposal, the AE states that "at the end of 2022 phase-1 of REPP had committed \$21m, which is 30%, of its funding to female-owned and led companies, which indicates that REPP 2 will similarly be able to source 2X aligned deals"; and

37. In accordance with item 7 (results management framework) of the GCF term sheet, all impacts delivered through REPP 2 are captured in the logical framework, which forms the basis of all monitoring and reporting under the programme.

38. Overall, the independent TAP rates the sustainable development impact potential of REPP 2 as high.

1.5 Needs of the recipient

Scale: High

39. The funding proposal credibly demonstrates the increasing climate change hazards afflicting the target countries, and the significant vulnerabilities of relevant populations (section 6 and annex 2 of the funding proposal). While climate change hazards and vulnerabilities vary across and within these countries, there is no doubt that all of them are significantly impacted by climate change. These impacts are aggravated by the realities of other socioeconomic development challenges in the sub-Saharan African region (as also emphasized by the Sixth Assessment Report of the United Nations Intergovernmental Panel on Climate Change).
40. While the poorest segments of society in the target markets may not be able to access clean electricity, either on-grid or off-grid, even after REPP 2 support due to economic constraints (ability to pay), there are indirect benefits that will accrue (e.g. electrification of social institutions like schools and health centres).
41. The funding requirements for renewable energy generation (on-grid and off-grid) in the region are considerable and well-documented (closing the energy access gap in sub-Saharan African countries will require an estimated annual investment of USD 28 billion up to 2030, according to the International Energy Agency). Considering slow growth, low tax revenues and stretched government budgets/high sovereign indebtedness in the target markets, the public sector will not be able to mobilize the required infrastructure investment. Multilateral development banks and other development partners are providing support but not at the scale required to close the investment gap. Local commercial sources of funding (e.g. commercial banks, pension funds) exist but typically do not finance infrastructure projects due to lack of familiarity with the asset class and/or statutory investment restrictions. The independent TAP recommends that the EE make a systematic effort in its investment activities to engage with local commercial capital providers, which would also open opportunities for leveraging local currency to projects. The independent TAP also encourages the EE to use the TAF strategically to strengthen links with local capital providers and leverage them into transactions.
42. Overall, the independent TAP rates the potential for REPP 2 to serve the needs of the recipients as high.

1.6 Country ownership

Scale: Medium to high

43. **Coherence with existing policies:** The funding proposal (annex 2) provides comprehensive evidence that REPP 2 is broadly aligned with the priorities of host governments in target markets with regard to:
- (a) Nationally determined contributions and associated implementation targets;
 - (b) National climate policies;
 - (c) National energy policies and legislative/regulatory environment for private investment in renewable energy;
 - (d) General national development policies; and
 - (e) National gender policies.
44. While the scope and depth of alignment varies across countries, the overall impact and investment focus of REPP 2 is aligned and supportive of national objectives.
45. **Capacity of the AE:** Since 2015, the AE has been implementing REPP 1, a fund similar in objectives (investment in small-scale renewable energy generation) but with some notable differences. While REPP 1 is entirely funded by grants (no commercial investor involvement in the capital structure of the fund), REPP 2 invests by committing debt or equity products (but it cannot be considered a fully commercial fund vehicle). REPP 2 also features an extended selection of target countries.

46. The independent TAP recognizes that the AE will be able to leverage many learnings from REPP 1 to REPP 2, but also views the significant changes in the capital structure (especially with expected participation of private sector and commercial investors) and the extended geography of REPP 2 as new challenges the AE must successfully manage.

47. The independent TAP also notes that REPP 2 – while predominantly a debt fund - would make investments with quasi-equity instruments. Prior experience would suggest that driving an investment portfolio consisting of different asset classes is generally challenging. Different asset classes require a variety of skill sets and, typically, equity investment involves a very different approach and mindset than debt investments. In combination with the complexity of the target markets and considering the size of the fund, the AE/EE will have to successfully navigate the resulting challenges.

48. As previously noted, the AE will also function as the EE for REPP 2. As detailed in the funding proposal and the term sheet, CAMCO would both invest any potential GCF contribution as fiduciary into the fund and would also be contracted by the fund as its investment manager under the terms of the proposed investment management agreement. In the funding proposal, the resulting potential conflicts of interest are discussed in section F (Risk Assessment and Management). During the question-and-answer session with the AE, it noted that, as a regulated manager, CAMCO has strong conflict of interest policies and procedures in place, and that it is confident that it would adequately manage potential challenges. However, it is worth pointing out that the realities of ethical walls (virtual barriers) in a small organization (~30 employees) are at least questionable. In addition, it is worth pointing out that the expected fee income from managing REPP 2 would constitute a very significant source of income for CAMCO (~30 per cent according to their estimates), which would exacerbate the potential challenge. This challenge is not specific to this funding proposal but constitutes a general issue that the GCF Secretariat would need to consider in its overall approach.

49. The independent TAP considers the alignment of REPP 2 with host country government priorities high, but assesses the AE track record as medium. Overall, the independent TAP rates the country ownership of the funding proposal as medium-high.

1.7 Efficiency and effectiveness

Scale: Medium to high

50. As also noted in the funding proposal, the estimated cost per t CO₂ eq mitigated/avoided is relatively high (USD 18.35 per t CO₂ eq). This is not surprising considering the fact that the investment focus is on small-scale renewable energy projects that typically feature higher fixed investment cost than large-scale infrastructure, resulting in higher levelized cost and thus lower efficiencies with regard to emissions avoided/reduced.

51. The long fund life (20 years) and the likely implications of this for fundraising potential with private sector and commercial investors were discussed earlier in this assessment report. As a potential junior equity investor, GCF would be the first in and last out of the fund, while private sector and commercial equity providers may find liquidity earlier (if a viable path is developed). This is a very long-term commitment; however, it is impossible to evaluate the counterfactual (what GCF could do with the funds if the fund life was shortened).

52. The independent TAP assesses the efficiency and effectiveness of the funding proposal as medium-high.

II. Overall remarks from the independent Technical Advisory Panel

53. The independent TAP endorses this funding proposal.

54. Overall, the funding proposal is ambitious and has potential to generate significant mitigation, adaptation as well as broader sustainable development benefits for the target countries through projects that will receive investments from the fund. In focusing on the small-scale segment of the renewable energy market, REPP 2 has the potential to play an instrumental role in closing the well-documented gap, especially in access to high-risk equity capital that is required for more project development (and ultimately investment) activity to occur. The focus of the fund on least developed countries, and thus on markets that have mostly seen little or no private investment thus far in small-scale renewable energy projects, is particularly welcome.

55. The independent TAP also welcomes the graduation from grant funding for the capital stack of the fund (as was the case in REPP 1) to a blended finance structure that opens the potential for private sector and commercial leverage at portfolio level. The independent TAP views this as important to help engender the emergence of a truly sustainable market for small-scale renewable energy in sub-Saharan Africa by gradually making private sector investors in particular more familiar with investing in this new asset class in jurisdictions that are unfamiliar to them. CAMCO has considerable experience in the target region and thus appears well-positioned to execute the investment mandate of REPP 2.

56. In order to ensure that REPP 2 can unfold its full paradigm shift potential, the independent TAP recommends to the Board to include the following condition to its potential approval of this funding proposal:

57. “Other than in relation to the first disbursement of the GCF Proceeds, delivery by the Accredited Entity to the Fund, in form and substance satisfactory to the GCF Secretariat, of written evidence confirming that Co-financing from private sources of an amount not less than ten per cent (10%) of the total capital commitments to Camco REPP SCSp (including the requested disbursement) is either already committed or expected to be committed, to Camco REPP SCSp, at the same time as the GCF Proceeds from the requested disbursement. Such private sources shall exclude co-financing which is ultimately provided by entities (i) where the majority of shares or ownership in that entity (i.e. more than 50%) are held by public sector entities; or (ii) that benefit from an explicit state guarantee or other government support agreement.”

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP222)

Proposal name:	Renewable Energy Performance Platform (REPP 2)
Accredited entity:	Camco Management Limited
Country/(ies):	Cameroon, Democratic Republic of Congo (DRC), Lesotho, Madagascar, Malawi, Niger, Nigeria, Sierra Leone, and Zambia
Project/Programme size:	Medium (Up to USD 250 million)

Impact potential

The iTAP assessment of the impact potential as medium-high is noted with appreciation.

The AE acknowledges the challenges highlighted by the iTAP in mobilizing significant co-investment for REPP 2. The AE is pleased to note that since the date of discussions with iTAP, the Renewable Energy Performance Platform has executed a term sheet for a junior equity interest of USD50m to REPP 2. Further positive discussions with potential investors have continued since the date of the submission of the revised FP to iTAP.

Paradigm shift potential

The iTAP assessment of the paradigm shift potential as medium is noted with appreciation.

As noted in paragraph 27, the AE is committed to sharing the lessons learnt and best practices developed during the lifetime of REPP 2, particularly in relation to workable transaction models and replicable documentation.

The AE agrees with the iTAP assessment that the leveraging of private capital is key to the paradigm shift potential of REPP 2. In relation to the proposed condition in paragraph 30 (repeated in paragraph 57), the AE strongly believes that it is involvement of the private sector by final close that is the appropriate measure of success – particularly in the current challenging fundraising environment. Accordingly, the AE proposed to iTAP a slight revision of the proposed condition as follows:

“The final disbursement of the GCF Proceeds shall be conditional on the delivery by the Accredited Entity to the Fund, in form and substance satisfactory to the GCF Secretariat, of written evidence confirming that Co-financing from private sources of an amount not less than ten per cent (10%) of the total capital commitments to Camco REPP SCSp (including the requested disbursement) is either already committed or expected to be committed, to Camco REPP SCSp, at the same time as the GCF Proceeds from the requested disbursement. Such private sources shall exclude co-financing which is ultimately provided by entities (i) where the majority of shares or ownership in that entity (i.e. more than 50%) are held by public sector entities; or (ii) that benefit from an explicit state guarantee or other government support agreement.”

This revision was not accepted by iTAP and the AE accepts the condition as proposed by iTAP.

The AE welcomes the suggestion of iTAP contained in paragraph 31.

The AE notes the encouragement of the iTAP to increase the role of local private finance sources and will seek to do so in implementing REPP 2. Similarly, the AE will actively seek to build partnerships with providers of local currency solutions (paragraph 33).

Sustainable development potential
The iTAP assessment of the sustainable development potential as high is noted with appreciation.
Needs of the recipient
The iTAP assessment of the needs of the recipient criteria as high is noted with appreciation.
Country ownership
The iTAP assessment of the country ownership criteria as medium-high is noted with appreciation.
Efficiency and effectiveness
The iTAP assessment of the efficiency and effectiveness criteria as medium-high is noted with appreciation.
Overall remarks from the independent Technical Advisory Panel: The AE welcomes the assessment report of the independent technical assessment panel ("iTAP") in response to the resubmission of the funding proposal for REPP 2.



Gender Assessment

REPP 2



10 April 2023

Author: Camco Management Limited

Version: 1

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1 Executive summary

This gender assessment explores the relationships between men and women, their access to resources and the constraints they face relative to one another in REPP 2 target countries within the context of climate change and energy access.

REPP 2's target countries have made progress in integrating the gender dimension at the political, strategic and legal levels and in some indicators REPP 2 target countries rank better than the world average. However, there are still significant gaps in achieving parity and women and girls are disproportionately affected by inadequate energy infrastructure and services, as well as unequal chances to engage in economic opportunities presented by renewable energy access. The following key barriers have been identified:

- Women face hurdles accessing modern energy
- Women's employment opportunities in renewable energy are limited
- Women's access to funding at the individual and company level is limited.

The Gender Action Plan (GAP) presented in Section 6 consists of two main actions: 1) dedicate 30% of REPP 2 funding towards companies that meet the 2X Challenge's Gender Lens Investing (2X GLI) criteria, and 2) integrate a gender equality perspective into the design, implementation and monitoring of all renewable energy activities funded by REPP 2. These activities will assist in designing products and services for the most vulnerable groups of the population, as well as overcoming the existing inequalities that exist in the renewable energy sector, which are presented in more detail in Section 4.

2 Overview of the target countries

REPP 2's investment mandate will extend towards some of the most economically and socially vulnerable countries in Africa, most of which fall into the low human development category under the 2021 Human Development Index (HDI); specifically, Democratic Republic of Congo (DRC), Madagascar, Malawi, Niger, Nigeria, Sierra Leone and Lesotho). Despite Cameroon making it into the medium human development cohort, it is still ranked 151 out of 191 countries and frequently falls below group averages on key development indicators as presented below in Table 1.¹

Table 1: Human development indicators for target countries²

Country/Indicator	Rank 2021	HDI 2021			Average annual overall HDI growth 2010-2021 (%)
		Overall	Female	Male	
World	-	0.732	0.715	0.747	0.45
Sub-Saharan Africa	-	0.547	0.519	0.572	0.77
Medium human development	-	0.636	0.586	0.666	0.81

¹ United Nations Development Programme (UNDP), 2022. [Human Development Report 2021/2022](#).

² UNDP, 2022. [Human Development Report 2021/2022](#).

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Country/Indicator	HDI 2021			Average annual overall HDI growth 2010-2021 (%)	
	Rank 2021	Overall	Female		Male
Low human development	-	0.518	0.477	0.552	0.75
Least developed countries	-	0.540	0.508	0.568	0.94
Cameroon	151	0.576	0.540	0.610	1.06
DRC	179	0.479	0.449	0.507	1.01
Lesotho	168	0.514	0.511	0.519	0.88
Madagascar	173	0.501	0.490	0.512	0.16
Malawi	169	0.512	0.502	0.519	1.06
Niger	189	0.400	0.364	0.436	1.54
Nigeria	163	0.535	0.495	0.574	0.95
Sierra Leone	181	0.477	0.452	0.506	1.01
Zambia	154	0.565	0.554	0.574	0.60

Based on the UNDP data presented in the Human Development Report 2021/2022, the majority (>50%) of the population in more than half of REPP 2 target countries lives below the International Poverty Line, which is defined as having less than USD 1.90 a day to live on.³ Low-income levels hinder the access to, and availability of, most basic services, including electricity, information, clean water, transport, healthcare and education.

Table 2: Economic wellbeing indicators⁴

Country/Indicator	Gross National Income (GNI) per capita, 2021 (2017 PPP ⁵ USD)		Population living below income poverty line 2008-2019 (%)		Gini coefficient 2010-2021
	Female	Male	National poverty line	PPP 2011 USD 1.90 a day ⁶	
World	12,241	21,210	-	-	-
Sub-Saharan Africa	2,970	4,429	43.4	45.7	-
Medium human development	2,912	9,668	-	-	-
Low human development	1,907	4,107	-	-	-
Least developed countries	1,993	3,777	-	-	-

³ At 2011 international prices calculated based on 2011 purchasing power parity (PPP) exchange rates. The International Poverty Line corresponds to the mean of the poverty lines found in the poorest 15 countries ranked by per capita consumption.

⁴ UNDP, 2022. [Human Development Report 2021/2022](#).

⁵ Purchasing power parity.

⁶ Population living below PPP \$1.90 a day: Percentage of the population living below the international poverty line of \$1.90 (in purchasing power parity terms) a day.

Country/Indicator	Gross National Income (GNI) per capita, 2021 (2017 PPP ⁵ USD)		Population living below income poverty line 2008-2019 (%)		Gini coefficient 2010-2021
	Female	Male	National poverty line	PPP 2011 USD 1.90 a day ⁶	
Cameroon	2,981	4,264	37.5	23.8	46.6
DRC	896	1,259	63.9	76.6	42.1
Lesotho	2,107	3,310	49.7	26.9	44.9
Madagascar	1,284	1,682	70.7	77.6	42.6
Malawi	1,232	1,713	51.5	70.3	38.5
Niger	936	1,535	44.5	44.5	37.3
Nigeria	3,759	5,800	46	53.5	35.1
Sierra Leone	1,453	1,789	52.9	40.1	35.7
Zambia	2,615	3,837	54.4	58.7	57.1

Indexes presented in Tables 1 and 2 place REPP 2 target countries among the poorest in the world, with women being disproportionately disadvantaged in terms of the opportunities and benefits that the countries and their economies have to offer, which results in lower values in general development indicators for women compared to men. This highlights the importance of gender-disaggregated data for the assessment of women's empowerment and economic development in the target countries.

Lack of gender-disaggregated data is especially distorting in generalised indicators such as the World Bank's International Poverty Line. Recent studies show that the way the indicator is calculated fails to consider differences between men and women.⁷ The World Bank's approach to assessing poverty relies on two main variables: consumption and income. The values for these variables are obtained through household surveys, which provide information on consumption and income at the household level, whereas statistics on the population living below the International Poverty Line seek to assess poverty at the individual level. Hence, individual indicators are calculated by dividing household-level data by the number of household members.

The World Bank recognises that this approach involves several important assumptions, such as that household resources are shared equally across family members; that family members have identical needs; that two individuals with the same per capita consumption level enjoy the same living standard; and that there are no differential costs of reaching a given welfare level per person for households of different sizes.⁸ These assumptions do not correspond to realities, as gender inequalities often make women and girls more vulnerable than men and boys to poverty. This is largely due to prevailing gender norms, the intra-household division of assets, work and responsibilities, and power relations that drive these disparities.⁹

With this in mind, REPP 2's gender assessment and respective GAP have been drawn up based on available gender-disaggregated data, which has allowed us to place women and men in the context of their respective countries and communities and to assess gender inequalities pertaining to the energy sector. The data sources

⁷ World Bank, 2018. [Gender Differences in Poverty and Household Composition through the Life-cycle: A Global Perspective](#).

⁸ World Bank, 2015. [A Measured Approach to Ending Poverty and Boosting Shared Prosperity: Concepts, Data, and the Twin Goals](#).

⁹ World Bank, 2018. [Gender Differences in Poverty and Household Composition through the Life-cycle: A Global Perspective](#).

used do not currently provide data on other gender identities and contain fragmented data on other marginalised groups with limited applicability to REPP 2.

3 Gender gap

Gender equality is measured by a number of international indexes, as presented below. Despite relative variability, African countries generally seem to have the lowest scores not only in terms of human development, but also gender equality and other related indicators.

This assessment considers the most recent indicators from UNDP Human Development Reports, World Economic Forum's Global Gender Gap Reports and OECD's Social Institutions and Gender Index Regional Reports for Africa, as well as a comprehensive body of data and qualitative assessments gathered and published by the World Bank Group. Although these sources provide a comprehensive outlook onto gender issues of the region, including REPP 2 target countries, it is recognised that these indicators and assessments do not fully address the role of women in the energy sector (as service providers, employees, and end users), since they lack sufficient country-level statistical data. Where such data is not available, the assessment herein has been made based on generalised data for the region and/or broader sectors (such as industry as opposed to energy) as well as latest available country-level data derived from other verified sources.

3.1 Indicators and indexes

Gender Development Index (GDI) measures gender inequalities in three basic dimensions of human development: health (based on life expectancy at birth), education (expected years of schooling and mean years of schooling) and command over economic resources (GNI per capita).

Gender Inequality Index (GII) measures gender inequalities in other three basic dimensions: reproductive health (maternal mortality ration and adolescent birth rate), empowerment (share of parliamentary seats held by each gender and population with at least some secondary education) and the labour market (labour force participation rate), are two main gender inequality indexes calculated by UNDP in its human development reports.¹⁰

Global Gender Gap Index (GGGI) benchmarks the current state and evolution of gender parity across four key dimensions, namely economic participation and opportunity, educational attainment, health and survival, and political empowerment).¹¹ The GGGI adopts a more holistic approach, integrating GDI and GII indicators as well as additional indicators relevant for the energy sector, such as level of enrolment in tertiary education, involvement of women as professional and technical workers, number of female legislators, senior officials and managers etc.

REPP 2 target countries perform differently in these indicators as presented below in Table 3 and 4.

GDI: Target countries have varying indicators of gender equality with two-thirds of the countries showing low levels of gender equality with absolute deviation from gender parity of more than 10%.¹² Whereas Madagascar,

¹⁰ UNDP, 2020. [Technical notes to Human Development Report 2020](#).

¹¹ World Economic Forum (WEF), 2022. [Global Gender Gap Report 2022](#).

¹² Methodology for calculating indices for the Human Development Report 2021/2022 uses absolute deviation of GDI from gender parity, $100 \cdot |GDI - 1|$ <WHAT IS THIS – A FORMULA?>, to determine each country's level of gender equality and place it into a respective GDI group. Countries with absolute deviation from gender parity of 2.5% or less are considered countries with high equality in HDI achievements between women and men and are classified as Group 1. Group 2 (medium-high equality) includes countries with 2.5-5% deviation, and Group 3 (medium equality)

Malawi and Lesotho score higher on GDI value, approaching or exceeding the world average and ranking among top-rated countries in the Sub-Saharan Africa region and the world.¹³

GII presents a different picture, where all of REPP 2 target countries experience higher level of gender inequality than the world average and some of the host countries, i.e., DRC, Niger, Nigeria, and Sierra Leone, are below the Sub-Saharan Africa average.

Table 3: Gender Development Index and Gender Inequality Index¹⁴

Country/Indicator	GDI 2021	GDI 2014	GII 2021 ¹⁵	GII 2014
World	0.958	0.924	0.465	0.449
Sub-Saharan Africa	0.907	0.872	0.569	0.575
Least developed countries	0.894	0.866	0.562	0.566
Cameroon	0.885	0.879	0.565	0.587
DRC	0.885	0.833	0.601	0.673
Lesotho	0.985	0.953	0.557	0.541
Madagascar	0.956	0.945	0.556	-
Malawi	0.968	0.907	0.554	0.611
Niger	0.835	0.729	0.611	0.713
Nigeria	0.863	0.841	0.680	-
Sierra Leone	0.893	0.814	0.633	0.650
Zambia	0.965	0.917	0.540	0.587

GGGI: Sub-Saharan Africa as a region has the sixth highest regional score - ranking ahead of Middle East and North Africa, and South Asia - and has bridged 67.9% of its gender gap.¹⁶ However, it is estimated that at the present rate it would take 98 years to close the gender gap in the region.¹⁷ At the country level, Cameroon, Madagascar and Lesotho show better results than the rest of REPP 2 target countries by virtue of high scores in the economic participation and opportunity (Cameroon and Madagascar), educational attainment (Lesotho) and

includes countries with 5-7.5% deviation. Countries with medium-low (7.5-10% deviation) and low (more than 10% deviation) equality are classified as Group 4 and Group 5, respectively.

¹³ For reference, the world average GDI of 0.958 falls within Group 2 limits (medium-high equality) with absolute deviation from gender parity of 4.2%. Group 1 countries, such as the UK and Finland, have 1.3% and 1.1% of absolute deviation, respectively. High-performing REPP 2 countries, i.e., Madagascar, Malawi and Lesotho, have absolute deviations from gender parity in HDI-attainment equal to 4.4%, 3.2% and 1.5% respectively, placing Madagascar and Malawi into the medium-high equality group and Lesotho into the high gender equality group.

¹⁴ UNDP, 2022. [Human Development Report 2021/2022](#). UNDP, 2015. [Human Development Report 2015](#).

¹⁵ Ranges from 0, where women and men fare equally, to 1, where one gender fares as poorly as possible in all measured dimensions.

¹⁶ The Global Gender Gap Report 2022 classifies countries into eight regions: North America, Europe, Latin America and the Caribbean, Central Asia, East Asia and the Pacific, Sub-Saharan Africa, Middle East and North Africa, and South Asia. North America and Europe lead all regions with 76.9% and 76.6% of the gender gap closed, respectively. Middle East and North Africa (63.4% gender gap closed) and South Asia (62.4% gender gap closed) show the lowest performance in the key dimensions of the report. By comparison, in 2006 Asia, Sub-Saharan Africa and the Middle East and North Africa had a little over 55% of their gender gaps closed (according to the [Global Gender Gap Report 2006](#)). It should be noted that regional classification and countries covered in the Global Gender Gap reports vary from year to year. Regional classifications for the Global Gender Gap Report 2022 are available in Appendix A of the report.

¹⁷ WEF, 2022. [Global Gender Gap Report 2022](#).

health and survival (Cameroon and Lesotho) dimensions (see Table 4). Other target countries remain on the lower spectrum of gender equality values.

Table 4. Global Gender Gap Index¹⁸

Country/Indicator	Global gender gap 2022		Regional gender gap rating 2021	Global gender gap 2020		Regional gender gap rating 2020
	Index	Rating		Index	Rating	
World	0.677	-	-	0.685	-	-
Sub-Saharan Africa	0.672	-	-	0.680	-	-
Cameroon	0.692	97	19	0.686	96	16
DRC	0.575	144	36	0.578	149	34
Lesotho	0.700	87	17	0.695	88	15
Madagascar	0.735	48	7	0.719	62	9
Malawi	0.632	132	30	0.664	116	23
Niger	0.635	128	29	-	-	-
Nigeria	0.639	123	27	0.635	128	27
Sierra Leone	0.672	109	22	0.668	111	21
Zambia	0.723	62	12	0.731	45	5

3.2 Health and wellbeing

Despite the progress that has been made in recent years (see Table 5), REPP 2 target countries still experience high maternal and infant mortality rates, as well as a lower life expectancy at birth compared to world averages.

Harmful practices such as gender-based violence and child marriages are also persistent across the countries, which lack sufficient legal basis for the protection for women. With the exception of DRC and Malawi, all of the target countries have legislative exemptions regarding the minimum legal age of marriage, such as the consent of parents, guardians, judges and/or courts.¹⁹ For instance, in Sierra Leone, where marriage is governed by the customary law as well as the statutory law, the Registration of Customary Marriage and Divorce Act 2007 allows for underage customary marriages if they happen with parents' consent.²⁰

Table 5. Health indicators²¹

Country//Indicator	Life expectancy at birth				Maternal mortality ratio (deaths per 100,000 live births)		Infant mortality rate (deaths per 1,000 live births)	
	2020		2014		2017	2010	2020	2014
	Female	Male	Female	Male				
World	75	71	74	70	204	248	27	32
Sub-Saharan Africa	64	60	61	58	535	626	50	58

¹⁸ WEF, 2022. [Global Gender Gap Report 2022](#). WEF, 2020. [Global Gender Gap Report 2020](#).

¹⁹ Organisation for Economic Co-operation and Development (OECD), 2021. [SIGI 2021 Regional Report for Africa, Social Institutions and Gender Index](#).

²⁰ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Sierra Leone](#).

²¹ World Bank, 2022. [Life expectancy at birth, female \(years\)](#). World Bank, 2022. [Life expectancy at birth, male \(years\)](#). World Bank, 2022. [Maternal mortality ratio \(modeled estimate, per 100,000 live births\)](#). World Bank, 2022. [Mortality rate, infant \(per 1,000 live births\)](#).

Country//Indicator	Life expectancy at birth				Maternal mortality ratio (deaths per 100,000 live births)		Infant mortality rate (deaths per 1,000 live births)	
	2020		2014		2017	2010	2020	2014
	Female	Male	Female	Male				
Cameroon	61	58	58	56	529	597	48	60
DRC	63	59	60	57	473	542	64	75
Madagascar	69	66	67	64	335	453	36	42
Malawi	68	62	64	58	349	444	29	39
Niger	64	62	61	59	509	663	46	53
Nigeria	56	54	61	52	917	978	72	80
Sierra Leone	56	54	53	51	1,120	1,360	80	96
Lesotho	58	52	53	47	544	594	70	72
Zambia	67	61	63	58	213	305	42	46

The adolescent fertility rate in the target countries also remains the highest in the world, with some countries exceeding the world average by a factor of three, as shown in Table 6 below.

Table 6. Health indicators (continued)²²

Country/Indicator	Adolescent fertility rate (births per 1,000 women ages 15-19)	
	2020	2014
World	41	45
Sub-Saharan Africa	98	108
Cameroon	99	115
DRC	119	127
Madagascar	104	116
Malawi	131	140
Niger	177	195
Nigeria	102	113
Sierra Leone	105	121
Lesotho	92	90
Zambia	115	130

As discussed in the next section, early marriages and motherhood have a detrimental effect on educational attainment among women and girls, which in turn limits the number of females capable of joining the skilled workforce or entrepreneurship sector.

²² World Bank, 2022. [Adolescent fertility rate \(births per 1,000 women ages 15-19\)](#).

3.3 Education

According to the available data, relative progress has been made in REPP 2 target countries and the rest of Sub-Saharan Africa with regards to overall access to, and the attainment of, education by both men and women. At the same time, this progress has not been significant enough for the indicators to catch up with world averages and the recorded gender gap remains persistent across all levels and fields of education.

Fewer than half of REPP 2 target countries (Cameroon, Lesotho, and Madagascar) have female literacy rates comparable to those of the male population while also closely approaching or exceeding world averages. In other countries, women are at least 20% less likely to be literate, with only 27% of Niger's adult females being able to both read and write with understanding a short simple statement about their everyday life, as presented in Table 7 below.

Table 7. Education attainment indicators

Country/Indicator	Literacy rate, adults (15+), % ²³		Mean years of schooling ²⁴			
			2021		2014	
	Female	Male	Female	Male	Female	Male
World	83 (2020)	90 (2020)	8.4	8.9	6.2	7.9
Sub-Saharan Africa	59 (2020)	72 (2020)	5.1	6.9	4.2	6.0
Cameroon	72 (2018)	83 (2018)	4.8	7.5	5.3	6.7
DRC	66 (2016)	89 (2016)	5.6	8.5	4.5	7.7
Lesotho	85 (2014)	68 (2014)	6.6	6.0	6.5	5.2
Madagascar	75 (2018)	78 (2018)	4.9	5.3	6.6	6.1
Malawi	55 (2015)	70 (2015)	4.1	4.7	3.4	5.2
Niger	27 (2018)	44 (2018)	1.7	2.8	0.8	2
Nigeria	53 (2018)	71 (2018)	6.1	8.2	4.9	7.1
Sierra Leone	35 (2018)	52 (2018)	3.5	5.8	2.2	4.0
Zambia	83 (2018)	91 (2018)	7.2	7.2	5.8	7.3

Low literacy rates have a direct correlation with educational attainment by boys and girls. Across REPP 2 target countries, 40% or fewer female adults aged 25 and older have reached (but not necessarily completed) a secondary level of education, as presented in Table 8. The actual number of adults that have ultimately completed upper secondary education is estimated at 27.5% for Sub-Saharan Africa in 2022, with a persistent gap between males (30.2%) and females (27.1%).²⁵

The Global Education Monitoring Report 2022 includes survey results of secondary school-age adolescents or their parents in Malawi, Nigeria and Sierra Leone showing their reasons for not being in school.²⁶ Among those

²³ World Bank, 2022. [Literacy rate, adult female \(% of females aged 15 and above\)](#). World Bank, 2022. [Literacy rate, adult male \(% of males aged 15 and above\)](#).

²⁴ UNDP, 2022. [Human Development Report 2021/2022](#)

²⁵ The United Nations Educational, Scientific and Cultural Organisation's (UNESCO) Institute for Statistics, 2022. [Regional and income group average estimates](#) for completion rates of different levels of education.

²⁶ UNESCO classifies the reasons for not completing secondary education or failing to ever attend school into three types: situational (life circumstances), dispositional (personal attitudes) and institutional (structural conditions).

who have never attended school, dispositional reasons are predominant, with half of those surveyed citing a lack of value or interest in education as their main reason for being out of school. Amongst those who no longer attend school, it is situational barriers - notably, the lack of resources - that keep at least 40% of adolescents from going back to school in Nigeria and Sierra Leone. Marriage and pregnancy were the reasons for approximately 20% of adolescents, mostly girls, dropping out of education in Nigeria, and about 10% in Sierra Leone and 5% in Malawi.²⁷

Table 8. Education attainment indicators (continued)²⁸

Country/Indicator	Expected years of schooling				Adults aged 25+ with at least some secondary education (%)			
	2021		2014		2021		2005-2014	
	Female	Male	Female	Male	Female	Male	Female	Male
World	12.9	12.7	12.2	12.4	64.2	70.3	54.5	65.4
Sub-Saharan Africa	10.0	10.6	9.1	10.3	31.1	44.3	22.1	31.5
Cameroon	12.4	13.8	9.5	11.2	36.8	55.0	21.3	34.9
DRC	9.6	10.1	8.6	10.8	40.3	69.1	12.8	32.4
Lesotho	12.4	11.7	11.7	10.6	27.2	24.6	21.9	19.0
Madagascar	10.2	10.1	10.2	10.5	27.3	29.8	-	-
Malawi	12.8	12.5	10.8	10.7	21.3	28.4	11.1	21.6
Niger	6.3	7.6	4.8	6.1	9.2	15.2	2.4	7.8
Nigeria	9.6	10.8	8.2	9.8	40.4	55.3	-	-
Sierra Leone	9.6	9.9	7.2	10.0	34.7	51.5	10.0	21.7
Zambia	10.9	11.0	13.0	13.9	47.1	56.8	25.8	44.0

When it comes to tertiary education, Sub-Saharan Africa is performing poorly with regard to gender parity, with women still remaining at a disadvantage, as compared to the rest of the world where gender disparity in tertiary education exists at the expense of men. The ratio of women to men that enrolled at tertiary level education for the region remained between 0.6 and 0.8 between 2000 and 2020, where the world average value increased from just under 0.7 to above 1.0 over the period.²⁹

Despite regional averages, some REPP 2 target countries have shown significant increases in the female/male tertiary education enrolment ratio, however. According to the available data, the adjusted gender parity index value has increased from 0.6 in 2000 to almost 0.9 in 2020 in Cameroon, almost reached 1.0 in Madagascar and remained at approximately 1.4 value in Lesotho, leaving men at a disadvantage.³⁰

Nevertheless, for other REPP 2 target countries, such as Niger, Malawi and DRC, female enrolment in tertiary education remains almost two times lower than male enrolment, as evidenced by data presented in Table 9.

Table 9. Tertiary education indicators³¹

²⁷ UNESCO, 2021. [Global Education Monitoring Report 2021/2: Non-state actors in education: Who chooses? Who loses?](#)

²⁸ UNDP, 2022. [Human Development Report 2021/2022](#). UNDP, 2015. [Human Development Report 2015](#).

²⁹ UNESCO, 2021. [Global Education Monitoring Report 2021/2: Non-state actors in education: Who chooses? Who loses?](#)

³⁰ UNESCO, 2021. [Global Education Monitoring Report 2021/2: Non-state actors in education: Who chooses? Who loses?](#)

³¹ WEF, 2022. [Global Gender Gap Report 2022](#).

Country/Indicator	Enrolment in tertiary education, %		STEMS attainment, %		Business, admin and law attainment, %	
	Female	Male	Female	Male	Female	Male
Cameroon	13.40	15.12	32.28	67.72	54.37	45.63
DRC	4.73	8.47	25.10	74.90	38.34	61.66
Lesotho	12.40	8.02	24.82	75.18	-	-
Madagascar	5.43	5.62	31.02	68.98	53.79	46.21
Malawi	1.29	2.17	-	-	-	-
Niger	2.82	5.87	18.03	81.97	-	-
Nigeria	10.13	14.01	-	-	-	-
Sierra Leone	-	-	-	-	-	-
Zambia	3.50	4.75	-	-	-	-

Table 10. Tertiary education indicators (continued)³²

Country/Indicator	Engineering, manufacturing and construction attainment, %		Natural sciences, mathematics and statistics attainment, %		Vocational training attainment, %	
	Female	Male	Female	Male	Female	Male
Cameroon	25.19	74.81	36.26	63.74	5.21	9.11
DRC	9.80	90.20	32.61	67.39	-	-
Lesotho	18.44	81.56	45.16	54.84	1.01	0.57
Madagascar	-	-	-	-	0.51	0.95
Malawi	-	-	-	-	-	-
Niger	7.45	92.55	20.68	79.32	0.92	1.00
Nigeria	-	-	-	-	-	-
Sierra Leone	-	-	-	-	-	-
Zambia	-	-	-	-	-	-

Out of the few women who do enrol in tertiary education in Sub-Saharan Africa, most opt for traditionally more “female” fields, such as arts and humanities or business, administration and law. According to the data collected in the Global Gender Gap Report 2022 and presented in Tables 9 and 10, all STEM specialties, as well as engineering, manufacturing, construction, natural sciences, mathematics and statistics have a significantly lower female enrolment rate compared to the male one.

Gender stereotypes and societal views on traditional gender roles in Sub-Saharan Africa have been shown to play a role in such distribution of enrolment between men and women. A 2019 IFC report, for example, found women who enrolled in technical or engineering programmes were viewed unfavourably.³³

All of the above contribute to the disadvantaged position of women when considering employment in the energy sector. Women’s low literacy rates and general lack of education plus the low STEM enrolment rates of

³² WEF, 2022. [Global Gender Gap Report 2022](#).

³³ International Finance Corporation (IFC), 2019. [Women’s participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees](#).

those few women who do sign up for tertiary programmes, means fewer women than men have sufficient qualifications for entering energy workforce in any capacity, but especially in technical roles.

3.4 Economic opportunities

3.4.1 Employment

In most REPP 2 target countries, working-aged women are less likely than men to engage in the labour market, either by working or actively looking for work, as presented in Table 11. Female unemployment is also generally higher than male unemployment, and has increased compared to 2014 indicators, likely as one of the negative consequences of the COVID-19 pandemic.

Table 11. Employment indicators

Country/Indicator	Labour force participation rate among 15+, %				Unemployment, % ³⁴			
	2021 ³⁵		2013 ³⁶		2021		2014	
	Female	Male	Female	Male	Female	Male	Female	Male
World	46.2	71.7	50.3	76.7	6.4	6.1	5.8	5.5
Sub-Saharan Africa	62.1	72.3	65.4	76.6	8.0	7.4	6.3	5.4
Cameroon	70.2	80.7	63.8	76.8	4.5	3.3	4.1	3.1
DRC	61.2	69.1	70.7	73.2	4.7	6.1	3.6	5.2
Lesotho	56.1	71.3	59.0	73.5	24.8	24.5	26.2	22.6
Madagascar	81.5	87.6	86.6	90.5	2.7	2.5	1.4	1.4
Malawi	71.6	80.0	84.6	81.5	8.9	5.2	7.1	4.8
Niger	61.7	84.3	40.0	89.7	0.7	0.8	0.4	0.6
Nigeria	47.9	59.6	48.2	63.7	8.8	10.6	5.4	3.9
Sierra Leone	56.1	55.9	65.7	69.0	4.7	6.0	3.8	5.5
Zambia	69.2	77.8	73.1	85.6	13.8	12.3	9.9	8.9

At the same time, the female labour force participation rate in REPP 2 target countries remains significantly higher than the world average of 46.2%. with the gap sometimes exceeding 30% points, such as in Madagascar where the rate is 81.5%. This corresponds to the overall trend for Sub-Saharan Africa and highlights the important role that women play in contributing towards household income. In many cases, however, as evidenced by statistical data presented in Tables 12 and 13, women's employment is characterised by high vulnerability, increased informality and lower job status compared to men.

Women tend to be engaged as contributing family workers or through informal arrangements. The share of informal employment in total employment in REPP 2 target countries is high and for women it is recorded to be higher than for men, with 89.7% of female workers employed through informal arrangements as compared to

³⁴ World Bank, 2022. [Unemployment, female \(% of female labour force\) \(modeled ILO estimate\)](#). World Bank, 2022. [Unemployment, male \(% of male labour force\) \(modeled ILO estimate\)](#).

³⁵ UNDP, 2022. [Human Development Report 2021/2022](#).

³⁶ UNDP, 2015. [Human Development Report 2015](#).

82.7% of male workers.³⁷ At the same time, available country-level data presented in Table 12 indicates that in Malawi and Sierra Leone a slightly bigger share of men are informally employed compared to women. Nevertheless, the International Labour Organisation (ILO) argues that women in the informal economy are still more disadvantaged than men, as they more often find themselves in the most vulnerable situations, for instance as domestic workers, home-based workers or contributing family workers, than their male counterparts.³⁸ For instance, in some REPP 2 target countries the gender gap in being engaged as a family worker amounts to double the percentage points for women as compared to those for men (e.g., in DRC, 26.2% of female workers contributed to the household income as family workers in 2019 compared to 5.3% of men engaged in the same type of employment). It is recorded that such distribution of labour is particularly strong in rural area, where women often work on the family's plot and contribute to the production of both cash crops and subsistence farming.³⁹

The burden of household responsibilities that fall overwhelmingly on women makes them more likely to jeopardise their jobs and careers.⁴⁰

Table 12. Employment indicators (continued)

Country/Indicator	Vulnerable employment ⁴¹ - % female/male employment, 2019 ⁴²		Contributing family workers - % female/male employment, 2019 ⁴³		Share of informal employment - % of total female/male employment ⁴⁴	
	Female	Male	Female	Male	Female	Male
World	44	43	15.5	5.7	-	-
Sub-Saharan Africa	80	67	25.0	10.6	-	-
Cameroon	81	63	15.1	6.3	95.2	86.6
DRC	88	67	26.2	5.3	97.2	86.6
Lesotho	56	40	6.9	3.2	-	-
Madagascar	87	80	52.1	18.8	95.2	91.9
Malawi	68	54	7.4	4.3	82.4	83.7
Niger	98	92	46.9	24.1	95.9	87.4
Nigeria	85	75	7.1	3.1	95.1	90.8
Sierra Leone	93	82	8.5	6.7	92.1	92.9
Zambia	82	65	35.7	9.4	93.6	81.6

³⁷ International Labour Organisation (ILO), 2018. [Women and Men in the Informal Economy: A Statistical Picture. Third Edition.](#)

³⁸ ILO, 2018. [Women and Men in the Informal Economy: A Statistical Picture. Third Edition.](#)

³⁹ OECD, 2021. [SIGI 2021 Regional Report for Africa, Social Institutions and Gender Index.](#)

⁴⁰ OECD, 2021. [SIGI 2021 Regional Report for Africa, Social Institutions and Gender Index.](#)

⁴¹ Vulnerable employment is contributing family workers and own-account workers as a % of total employment.

⁴² World Bank, 2022. [Vulnerable employment, female \(% of female employment\) \(modeled ILO estimate\).](#) World Bank, 2022. [Vulnerable employment, male \(% of male employment\) \(modeled ILO estimate\).](#)

⁴³ World Bank, 2022. [Contributing family workers, female \(% of female employment\) \(modeled ILO estimate\).](#) World Bank, 2022. [Contributing family workers, male \(% of male employment\) \(modeled ILO estimate\).](#)

⁴⁴ ILO, 2018. [Women and Men in the Informal Economy: A Statistical Picture. Third Edition.](#)

Women are also more likely to be employed in low-income jobs, especially in the agricultural sector. Based on ILO estimates, the agricultural sector dominates female employment in REPP 2 target countries. In more than half of the countries, the sector accounts for more than 50% of total female employment, reaching as much as 82% in Malawi. Nigeria is the only REPP 2 target country where distribution of labour between men and women is following general world trends, as presented in Table 13.

There is evidence to suggest that women earn less than men due in part to the high rate of informal employment and the dominance of women in the agricultural sector.⁴⁵ In addition, work in industry and services sectors is likely to require a degree of literacy and education that women in Africa and REPP 2 target countries are less likely to have access to, as evidenced by the assessment of female educational achievements in the previous section.

Table 13. Employment indicators (continued)

Country/ Indicator	Employment by sector, 2019 (% of female/male employment) ⁴⁶						Time spent on unpaid domestic chores and care work, female to male ratio, 2021 ⁴⁷
	Agriculture		Industry		Services		
	Female	Male	Female	Male	Female	Male	
World	25	28	16	27	59	45	-
Sub-Saharan Africa	53	53	8	13	40	34	-
Cameroon	48	40	11	17	41	43	2.87
DRC	72	57	3	16	25	27	-
Lesotho	40	48	12	14	48	38	2.52
Madagascar	60	68	9	9	31	23	4.85
Malawi	82	71	3	8	15	21	6.94
Niger	69	75	9	5	21	20	-
Nigeria	24	44	13	11	64	44	1.51
Sierra Leone	51	58	2	12	47	30	2.99
Zambia	55	45	4	16	41	39	-

3.4.2 Access to land

Women's ability to participate in a labour market, especially as entrepreneurs or own-account workers, as well as their ability to access financial services largely depends on ownership of assets, and real estate in particular.

Asset ownership has a multidimensional impact on the life and economic empowerment of women. The ability to administer assets and use them for productive gain purposes, the ability to use the assets as collateral for credit or as a resource base for starting a business by enhancing liquidity through rental or sale, independence

⁴⁵ OECD, 2021. [SIGI 2021 Regional Report for Africa, Social Institutions and Gender Index](#).

⁴⁶ World Bank, 2022. [Employment in agriculture, female \(% of female employment\) \(modeled ILO estimate\)](#). World Bank, 2022. [Employment in agriculture, male \(% of male employment\) \(modeled ILO estimate\)](#). World Bank, 2022. [Employment in industry, female \(% of female employment\) \(modeled ILO estimate\)](#). World Bank, 2022. [Employment in industry, male \(% of male employment\) \(modeled ILO estimate\)](#). World Bank, 2022. [Employment in services, female \(% of female employment\) \(modeled ILO estimate\)](#). World Bank, 2022. [Employment in services, male \(% of male employment\) \(modeled ILO estimate\)](#).

⁴⁷ WEF, 2021. [Global Gender Gap Report 2021](#).

from male family members and having confidence in the ability to make a living are all important prerequisites for women's economic empowerments.

Assets ownership in Africa and involves the issue of women's legal ability to own real estate. As described in Section 3.6 below, three of REPP 2 target countries (Cameroon, DRC and Niger) significantly limits women's rights to own immovable property and grants a husband all administrative rights over assets during marriage.

In practice, available statistical data, presented in Tables 14 and 15, shows that there is a significant disproportion in the amount of land owned and/or effectively managed by men and women in REPP 2 target countries. In particular, women do not have a substantial share in agricultural holdings, which amounts to making major decisions regarding resource use and exercises management control over the agricultural holding, not necessarily due to ownership rights. In addition, even though distribution of agricultural landownership by gender in certain countries is more favourable for women (e.g., 57.3% of landowners in Malawi were recorded to be women), the area of land actually owned solely by women and the value of this land is statistically smaller than the area and value of land owned solely by male landowners, as presented in Table 15.

In addition, even in the countries where women are legally granted equal rights to own and administer the property, matters related to real estate and family are in some of the host countries governed by customary law, which operates in parallel to statutory law and often disadvantages women.

Table 14. Land ownership rights⁴⁸

Country/Indicator	Distribution of agricultural holders by gender (%)		Distribution of agricultural landowners (joint or sole ownership) by gender (%)	
	Female	Male	Female	Male
DRC	8.9	91.1	-	-
Lesotho	30.8	69.2	-	-
Madagascar	15.3	84.7	-	-
Malawi	32.1	67.9	57.3	42.7
Niger	-	-	39.5	60.5
Nigeria	10.0	90.0	15.8	84.2
Zambia	19.2	80.8	-	-

Table 15. Land ownership rights (continued)⁴⁹

Country/Indicator	Distribution of land area, % of total agricultural land area			Distribution of land value, % of total agricultural land value		
	Female	Male	Joint	Female	Male	Joint
Malawi	40	42	18	39	39	23
Niger	9	62	29	5	59	36
Nigeria	4	87	9	10	78	11

3.4.3 Financial inclusion

Despite an overall increase in account ownership and access to formal financial services, vulnerable groups such as women, poor people and less educated adults continue to make up the majority of people excluded from the

⁴⁸ Food and Agriculture Organisation of the United Nations (FAO), 2022. [Gender and Land Rights Database](#).

⁴⁹ FAO, 2022. [Gender and Land Rights Database](#).

formal financial sector. Women are often excluded from formal banking services because they lack official forms of identification, do not own a mobile phone or other form of technology that enables mobile banking, and have lower financial capability.⁵⁰ In fact, nearly 60% of unbanked adults in Sub-Saharan Africa who took part in a 2022 survey said they did not have enough money to open an account in a financial institution. The other reasons for not having a mobile money account were the lack of a mobile phone and the lack of necessary documentation. These barriers are especially critical for women, who have lower economic security, are less likely to be able to obtain a mobile phone or experience a heavier documentation burden compared to men. In addition, many REPP 2 target countries lack sufficient legal protection for women willing to enter formal financial services sector, as described in Section 3.6.

This gender gap is particularly evident from the statistics, which indicate that the gap in account ownership as being between 5 to 20 percentage points for half of REPP 2 target countries, as well as generally decreased instances of women's borrowing from formal financial institutions and increased reliance on savings clubs, (see Table 16).

Table 16. Access to financial services⁵¹

Country / Indicator	Bank account ownership (%)				Source of borrowed money (%)					
	Female	Male	Urban	Rural	Formal financial institution or using mobile money account		Family or friends		Savings club	
	Female	Male	Urban	Rural	Female	Male	Female	Male	Female	Male
Cameroon	49.25	54.15	60.91	41.33	8.10	15.02	40.36	46.73	15.38	14.41
DRC	-	-	-	-	-	-	-	-	-	-
Lesotho	-	-	-	-	-	-	-	-	-	-
Madagascar	-	-	-	-	-	-	-	-	-	-
Malawi	38.14	47.75	51.32	38.74	10.10	10.00	20.73	30.21	23.47	10.89
Niger	-	-	-	-	-	-	-	-	-	-
Nigeria	34.96	55.45	53.99	33.92	6.32	7.74	41.83	47.53	4.55	4.47
Sierra Leone	24.76	33.35	38.71	22.40	5.93	5.29	37.25	40.96	14.14	14.53
Zambia	44.98	52.48	62.77	36.43	11.52	18.43	38.20	48.54	11.12	7.94

Women's limited access to banking services reduces their economic independence from male family members, as well as their ability to formally save money and administer savings at their own discretion. This results in additional barriers for women who want to engage in entrepreneurship, vocational training or any other activity that would enhance their financial standing, unless fully supported by the husband.

⁵⁰ World Bank, 2022. [The Global Findex Database 2021. Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-2019](#).

⁵¹ World Bank, 2021. [The Global Findex Database 2021](#).

3.5 Participation in decision-making and leadership

Table 17. Indicators of female participation in leadership

Country / Indicator	Share of seats held by women in parliaments, % ⁵²		Share of seats held by women in local government from 2017-2019, % ⁵³	Women in ministerial positions in 2022, % ⁵⁴	Firms with female participation in ownership, % ⁵⁵	Firms with female top managers in 2021, % ⁵⁶
	2021	2014				
World	26	22	-	-	33.1 (2020)	-
Sub-Saharan Africa	26	22	-	-	28.0 (2020)	-
Lesotho	23	27	39.0	22.2	39.1 (2016)	36.20
Cameroon	34	31	25.3	14.9	39.7 (2016)	22.90
DRC	13	11	-	17.8	15.1 (2013)	10.80
Madagascar	18	21	-	34.8	41.6 (2013)	28.20
Malawi	23	17	14.6	17.4	28.1 (2014)	14.20
Niger	26	13	15.8	15.8	14.5 (2017)	10.60
Nigeria	4	7	9.8	10.3	16.2 (2014)	13.90
Sierra Leone	12	12	18.2	13.8	18.8 (2017)	15.90
Zambia	15	11	7.1	32.26	41.5 (2019)	12.90

3.6 Legal status of women and gender stereotypes

All REPP 2 target countries have made significant improvements in promoting gender equality in law. (Based on the World Bank Group study, DRC is in the world's top 10 most improved economies since 1970 in relation to gender equality in law.)⁵⁷ Furthermore, the statutory law does not differentiate between genders in granting individual rights to a person in all of REPP 2's target countries.

Nevertheless, all target countries are characterised by a dual legal system, which allows for customary law to largely govern societal life - especially with regard to family and marital matters as well as property and economic inclusion - rendering statutory provisions on gender equality inapplicable and unenforceable.

Women are disadvantaged in a vast number of areas due to their persistent de-valuation by local customs and gender stereotypes, the effects of customary law, and also gaps in statutory law. Common restrictions that prevent economic development and empowerment of women in REPP 2 target countries are further described in Table 18 below.

⁵² World Bank, 2022. [Proportion of seats held by women in national parliaments](#).

⁵³ UNDP, 2020. [Human Development Report 2020](#).

⁵⁴ WEF, 2022. [Global Gender Gap Report 2022](#).

⁵⁵ World Bank, 2022. [Firms with female participation in ownership \(% of firms\)](#).

⁵⁶ WEF, 2021. [Global Gender Gap Report 2021](#).

⁵⁷ World Bank, 2022. [Women, Business and the Law 2022. Regional Profile. Eastern and Southern Africa](#).

Table 18. Legal restrictions for women in REPP 2 target countries

Restriction	Description
Right to movement and mobility	Cameroon, Malawi, Niger and Nigeria do not grant women the same freedom of movement as is afforded to men.⁵⁸ In Cameroon, although both genders are recognised as equals by the Constitution, the Civil Code designates the husband as the head of the household, who has the final decision on the family's domicile, contradicting the woman's constitutional rights to freedom of movement and residence within the border.⁵⁹ In addition, married women cannot obtain a passport in the same way as men in Cameroon, Madagascar, Malawi and Nigeria.⁶⁰ In Malawi and Madagascar, they are required to submit a marriage certificate with their husband's name included.⁶¹
Formal employment sector	Under Cameroon's Labour Code both men and women have equal rights to work. At the same time, the Family Code allows the husband to challenge his wife's rights to work if it is in the interests of the marriage and the children.⁶² In Nigeria, statutory law does not contain a specific prohibition of gender-based discrimination in the workplace.⁶³
Remuneration and employment in industry	Among REPP 2 target countries, only Malawi and Niger have legislation mandating equal remuneration for work of equal value.⁶⁴ Cameroon, DRC, Madagascar and Niger impose restrictions on women's ability to work in a job deemed dangerous, while Cameroon, Madagascar, Nigeria, Sierra Leone and Lesotho restrict women's right to work in industrial jobs.⁶⁵
Head of household	In Cameroon⁶⁶, DRC⁶⁷, Madagascar⁶⁸ and Niger⁶⁹, under both statutory and customary law a woman can become a head of a household only under special circumstances, thus depriving women of rights and opportunities which head of a household status grants.

⁵⁸ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. Malawi](#). World Bank, 2022. [Women, Business and the Law 2022. Niger](#). World Bank, 2022. [Women, Business and the Law 2022. Nigeria](#).

⁵⁹ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Cameroon](#).

⁶⁰ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. Malawi](#). World Bank, 2022. [Women, Business and the Law 2022. Madagascar](#). World Bank, 2022. [Women, Business and the Law 2022. Nigeria](#).

⁶¹ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Malawi](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Madagascar](#).

⁶² World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Cameroon](#).

⁶³ World Bank, 2022. [Women, Business and the Law 2022. Nigeria](#). World Bank, 2021. [Women, Business and the Law 2021. Economy Summary. Nigeria](#).

⁶⁴ Based on country data in the World Bank Group's Women, Business and the Law 2022 study.

⁶⁵ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. DRC](#). World Bank, 2022. [Women, Business and the Law 2022. Madagascar](#). World Bank, 2022. [Women, Business and the Law 2022. Niger](#). World Bank, 2022. [Women, Business and the Law 2022. Nigeria](#). World Bank, 2022. [Women, Business and the Law 2022. Sierra Leone](#).

⁶⁶ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Cameroon](#).

⁶⁷ World Bank, 2022. [Women, Business and the Law 2022. DRC](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. DRC](#).

⁶⁸ World Bank, 2022. [Women, Business and the Law 2022. Madagascar](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Madagascar](#).

⁶⁹ World Bank, 2022. [Women, Business and the Law 2022. Niger](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Niger](#).

Restriction	Description
Entrepreneurship and access to financial services	Gender-based discrimination with regards to access to credit is not prohibited in Cameroon, Madagascar, Malawi, Niger, Nigeria, and Lesotho.⁷⁰ In addition, women typically lack knowledge of legislation governing credit relationships, have limited access to financial resources and have neither credit history nor banking skills to attempt to access formal credit on their own. Opening a bank account for women in Cameroon and Niger is also more complicated for women than it is for men, as women are required to obtain their husband's permission.⁷¹
Assets	A number of restrictions on property ownership by women is in place in REPP 2 target countries. Most of them pertain to cultural norms and customary law, which forbid women to own and/or inherit property, especially land. Also, in Cameroon, Madagascar and Niger even the statutory law recognises a husband's ultimate decision-making power with regard to administering assets during marriage.⁷² Examples of the type of restrictions women face include: In Cameroon, both civil and customary legal regimes do not allow married women to own or manage land and assets. Under civil law, marital property is under the authority of the husband who may manage and make decisions without the consent of his spouse. Under customary law, only men have the right to own the land; women are often regarded as the property of the husband and are thus excluded from land ownership.⁷³ In DRC, no distinction is made in the provision of land rights between women and men under statutory law. Yet under customary law, traditional chiefs, who are largely men, hold the authority on land rights guided by patrilineal lineages. By these principles, inherited land is almost always given to sons rather than daughters and women's access to land becomes dependent on their husband. Also, given deeply embedded stereotypes surrounding the roles of women and men, a married couple's property is often registered in the husband's name, and it is more likely to see joint land titling among unrelated men than husband and wife.⁷⁴ In Zambia, where customary land accounts for the majority of land (96%) and governed by traditional authorities, only men have the right to own and power to dispose of customary land, while wife or children may only have access to it.⁷⁵

According to the 2019 Social Inclusion and Gender Index's country profiles, REPP2 target countries for which data is available have medium (DRC, Lesotho) to very high gender-based discrimination in social institutions,

⁷⁰ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. Malawi](#). World Bank, 2022. [Women, Business and the Law 2022. Madagascar](#). World Bank, 2022. [Women, Business and the Law 2022. Niger](#). World Bank, 2022. [Women, Business and the Law 2022. Nigeria](#). World Bank, 2022. [Women, Business and the Law 2022. Lesotho](#).

⁷¹ World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. Niger](#). World Bank, 2022. [Women, Business and the Law 2022. Regional Profile. Western and Central Africa](#).

⁷² World Bank, 2022. [Women, Business and the Law 2022. Cameroon](#). World Bank, 2022. [Women, Business and the Law 2022. Madagascar](#). World Bank, 2022. [Women, Business and the Law 2022. Niger](#).

⁷³ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Cameroon](#).

⁷⁴ OECD, 2019. [Social Institutions and Gender Index. Country Profile. DRC](#).

⁷⁵ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Zambia](#).

which beyond the structural and legal barriers also include norms, attitudes, stereotypes and fundamental customs and traditions restricting women's independence and economic empowerment.⁷⁶

SIGI 2021 Regional Report for Africa⁷⁷ summarises the role played by these discriminatory social institutions in women's access to employment and other economic opportunities as follows:

- **Discriminatory gender roles within the household constrain women's access to the labour market.** In addition to an increased burden on women in the form of unpaid domestic and care work, the proportion of people declaring that it is not acceptable for a woman in their family to do any paid work outside of the home is significant, ranging from 15% in Nigeria to 28% in Niger. Earning money and providing for the family is considered an integral part of masculine identity, and not seen as favourable in women.
- **Perception of women's abilities and discriminatory educational norms tend to confine women to low-status jobs in specific sectors of the economy.** In Africa, men tend to be overrepresented in "manly" sectors such as construction, mining and quarrying or transport, storage and communication. Women are predominant in sectors such as accommodation, wholesale and retail trade or household services. This is also evident from the quantitative data analysed in Section 3.4 above and takes its origins from the choice of educational path and vocational trainings. The implications for economic empowerment are far-reaching, as women in Africa seem to be largely drawn to sectors that are less profitable or where wages are lower simply because they are considered "female" sectors.
- **Men's traditional status and roles, household decision-making practices and discriminatory inheritance practices limit women's ownership of agricultural land and constrain their economic independence.** Customs and traditions related to land ownership are deeply rooted in African societies and often in practice supersede formal civil or common law in REPP 2 target countries.
- **Various discriminatory social norms and biases related to access to markets, finance, training and networks hamper women's entrepreneurship in Africa.** These norms and biases result in limited attempts by women to even enter the credit market due to self-perceived lack of creditworthiness, as well as in actual constraints over access to credit due to lack of sufficient economic resources, lower income and high informality rates in employment.

4 Gender inclusivity in the energy sector

Energy, especially electricity, is an invaluable basic commodity that the quality of people's livelihoods is indispensably dependent on. From the gender perspective, the energy sector can be viewed as an important provider of the resource base necessary for the everyday life and economic activity of either the household as

⁷⁶ OECD, 2019. [Social Institutions and Gender Index. Country Profile. Cameroon](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. DRC](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Malawi](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Madagascar](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Niger](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Nigeria](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Sierra Leone](#). OECD, 2019. [Social Institutions and Gender Index. Country Profile. Lesotho](#).

⁷⁷ OECD, 2021. [SIGI 2021 Regional Report for Africa, Social Institutions and Gender Index](#).

one social unit or of separate household members. It can also be seen as a field of economic activity with potential opportunities for women's empowerment through employment and entrepreneurship.

4.1 Women as employees

The energy sector is dominated by men. Women are underrepresented in the sector. It is also a relatively small sector in Sub-Saharan Africa and therefore not a big employer on the country- and region-wide scale compared with the world's most developed countries. The African Development Bank has estimated that in 2021, electricity, gas and water activities accounted for an estimated 1.8% of GDP in Sub-Saharan Africa.⁷⁸ Total employment in the sector constitutes a small percentage of the total employment in the target countries, ranging from 0.6% of Malawi's total workforce to 4.6% of Lesotho's, as presented in Table 19. The distribution of the already limited number of jobs on offer in the sector by gender shows significant disparities, with female employment generally not exceeding one third of total sector employment and dropping to one tenth of the sector in the case of Lesotho.

Table 19. Employment in the energy sector⁷⁹

Country/Indicator	Employment in mining and quarrying, and electricity, gas and water supply (aggregated) - 000's				Total employment in the sector – 000's	Total country workforce – 000's
	Urban		Rural			
	Female	Male	Female	Male		
Cameroon (2014)	5.7	32.2	12.8	30.4	81.1	8,434.8
DRC (2012)	13.4	152.3	64.1	452.1	681.9	27,405.3
Lesotho (2019)	2	12.9	0.8	14.2	29.9	645.4
Madagascar (2015)	-	13.9	40.7	59.4	114	11,051.8
Malawi (2013)	4.3	8.1	6.5	17.9	36.8	6,154.2
Niger (2017)	-	12	8	15.1	35.1	2,125.7
Nigeria (2019)	78.7	222.2	80.2	211.2	592.3	65,512.2
Sierra Leone (2018)	8	29.7	6.2	28.7	72.6	2,570
Zambia (2020)	9.4	57	1.2	10.7	78.4	9905.1

Despite the currently insignificant value added by the energy sector to the GDP of African countries, the renewable energy subsector has been booming. Between 2007-2019, African countries were attracting 65% of the total off-grid renewable energy investment.⁸⁰ The positive effects of such investment included not only an increase in electrification levels for rural and remote communities, but also enhanced opportunities for the economic empowerment of women.

While regional and country-level gender-disaggregated data for the energy sector remains scarce, especially in relation to the renewable energy subsector, IRENA reports that at the global level the share of women employed in the renewable energy sector (32%) is higher than in the oil and gas industry (22%).⁸¹ Renewable energy projects and companies are therefore more likely to attract women professionals than traditional energy

⁷⁸ African Development Bank (AfDB), UN Women, 2021. [Green Jobs for Women in Africa](#).

⁷⁹ ILOSTAT, 2022. [Database](#).

⁸⁰ International Renewable Energy Agency (IRENA), 2020. [Global Landscape of Renewable Energy Finance 2020](#).

⁸¹ AfDB, UN Women, 2021. [Green Jobs for Women in Africa](#). IRENA, 2020. [Global Landscape of Renewable Energy Finance 2020](#).

companies. The study conducted by the UN Women and AfDB on women's employment opportunities in the "green" economy argues that the main reason for this is the multidimensionality of renewable energy (when compared with conventional energy production), which presents more employment opportunities for women in non-STEM positions, such as finance, law, environmental sciences, economics, etc.⁸² Moreover, a range of support services, such as consulting, research and development, and the creation of permits, regulations and policies, are projected to emerge along with the sector's development.⁸³ This allows for an easier increase in female employment and economic independence without challenging the skills-related gender stereotypes that persist in REPP 2 target countries.

4.1.1 Female employment profile in renewable energy companies

The share of female employees across Sub-Saharan Africa is higher in non-technical departments, such as administration (38%), communications (34%), finance (30%) and sales (31%). At the same time, technical STEM and non-STEM departments only have a 13% and 16% share of female employees respectively, according to IFC research.⁸⁴

Women are also more likely to be employed in renewable energy companies that provide solutions in small-scale, decentralised energy systems (37% of full-time employees) compared to large-scale, on-grid, centralised energy systems (22% of full-time employees).⁸⁵

When looking at executive positions, women on average make up 30% of chief executive officers (CEOs), 16% of vice presidents and directors, and 26% of middle- and lower-level managers in renewable energy companies in Sub-Saharan Africa. Comparing CEOs across sectors, McKinsey found that 6% of African CEOs were women,⁸⁶ while a UN study found that 3.5% of women from the leading publicly traded companies in G-20 countries were women.⁸⁷ The researchers also found that there were more women on the boards of renewable energy companies in Africa (27%) than the G-20 country average (20%) and the African average (18%). By comparison, women comprised only 6% of the executive-level employees and 21% of board members in energy utilities in Sub-Saharan Africa in 2019.⁸⁸

4.1.2 Employment barriers for women in renewable energy industry

Barriers to women's participation in the renewable energy sector exist both at the recruitment stage and at career progression. At the same time, there is a notable difference in the perception of these barriers by men and women too. According to a study conducted among managers and employees of renewable energy companies in Sub-Saharan Africa, only 28% of men - compared to 40% of women - believed that women faced more barriers than men when seeking to enter the sector. The gap increased further to 30% when considering the perceived barriers to advancing women's careers in the sector, with 45% of interviewed women and only

⁸² AfDB, UN Women, 2021. [Green Jobs for Women in Africa](#).

⁸³ AfDB, UN Women, 2021. [Green Jobs for Women in Africa](#).

⁸⁴ IFC, 2019. [Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees](#).

⁸⁵ IFC, 2019. [Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees](#).

⁸⁶ McKinsey Global Institute, 2019. [The power of parity. Advancing women's equality in Africa](#).

⁸⁷ IFC, 2019. [Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees](#).

⁸⁸ International Union for Conservation of Nature (IUCN), 2019. [Energizing Equality: Unlocking the potential of women's participation in sub-Saharan African energy utilities](#).

15% of interviewed men feeling that women faced more barriers to their career advancement.⁸⁹ Table 20 presents main barriers facing women and men in the renewable energy workforce.

According to IFC research in 2019, men were less likely than women to consider societal and cultural gender norms as a barrier. Men were also less likely to believe women's household and family responsibilities and existing gender discrimination had negative effects and posed additional obstacles for women to advance their careers in renewable energy compared to men. In contrast, more men than women identified lack of skills, training opportunities and women's self-perception as the main reasons for women being less likely to receive promotion.

Table 20. Perception of the barriers facing women in the renewable energy sector⁹⁰

	Women (%)	Men (%)
Barriers to entering		
Concerns about societal and/or cultural gender norms	68	38
Lack of gender diversity targets in companies	53	15
Prevalent male-biased hiring practices	53	15
Women's self-perception	42	38
Lack of awareness of opportunities in renewable energy	32	54
Lack of STEM education or skills	26	54
Barriers to advancing		
Women's burden balancing work and family	48	29
Discrimination on the basis of gender, age, disability, sexual identity, race, etc.	48	14
Lack of mentoring opportunities	43	57
Women's self-perception	38	71
Lack of training opportunities	29	57
Lack of the required skills and qualifications	14	43

The barriers vary from country to country and each renewable energy subsector. For example, women's participation in the wind energy sector is just 21% globally and even less in Africa at 8%.⁹¹ Despite this and the fact that the majority of jobs in the sector are of a technical nature (construction and manufacturing), an IRENA survey of the sector involving 72 countries found that skills were not a barrier for women's entry and retention in the wind industry; instead it was social norms and perceptions, followed by a lack of suitable corporate policies, including work-life balance, care solutions, etc. that were the barriers.⁹²

A USAID-funded study provides another example, having in 2020 identified significant gaps in female representation in solar home system (SHS) market research, production and most field positions (i.e.,

⁸⁹ IFC, 2019. [Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees.](#)

⁹⁰ IFC, 2019. [Women's participation in the renewable energy workforce in Sub-Saharan Africa. Identifying Barriers and Opportunities for Women as Leaders and Employees.](#)

⁹¹ AfDB, UN Women, 2021. [Green Jobs for Women in Africa.](#)

⁹² IRENA, 2020. [Wind energy: A gender perspective.](#)

technicians who provide installation and after-sales support).⁹³ And a 2021 study of training institutions in Nigeria found evidence that the gender gap started before women and men joined the workforce, referencing the lack of gender diversity among students.⁹⁴

Based on the literature review, the following have been identified as the key barriers for women's employment and progression in general, including in Sub-Saharan Africa's renewable energy sector:

- **Workplace policies and practices**, including:
 - The absence of effective recruitment and promotion policies with specific gender diversity targets, which would exclude gender biases;
 - The lack of access to field work, which impedes women's career advancement by limiting their opportunities to increase their understanding of renewable energy value chain and technical matters;
 - Gender pay gap;
 - Sexual harassment and violence in the workplace; and
 - The lack of access to mentoring and training opportunities.
- **Social norms and gender stereotypes**, which affect girls and boys from an early age, often impeding their opportunities for future employment in any sector, not just renewable energy. Such norms and stereotypes relate to traditional gender roles in society and at the household level and are linked to low school and university enrolment rates, the widespread misconception of women not being suited for technical roles, the unfavourable treatment of female employment, and the increased burden of unpaid household duties and responsibilities.
- **Legal barriers**, which relate to restrictions of women's involvement in the work force. Such restrictions exist through general limitations such as the right of a husband to make decisions on his wife's employment, which is afforded to men in Cameroon, or indirectly through lack of prohibition of gender-based discrimination in employment, as is the case of Nigeria. Labour legislation in a selection of countries in Africa, including REPP 2 target countries, such as Cameroon, DRC, Madagascar, Nigeria, Sierra Leone and Lesotho, also contains more specific restrictions on women's employment in dangerous and industrial jobs, which constitute core business functions in the renewable energy sector. It is argued that denial of access for women to opportunities in these jobs impedes their ability to reach leadership positions.

4.2 Women as energy users

Interconnections between the empowerment of women and the energy sector exist not only in employment opportunities afforded to women in the sector, but also in electricity access and its impact on the everyday life of women.

It is widely recognised that where physical energy access is translated into actual use of energy services, it can help reduce women's poverty by: (i) saving time (by substituting for manual labour) and enhancing convenience and comfort; (ii) reducing the use of polluting energy forms such as wood fuel and kerosene; (iii) creating opportunities for new income generation options and for enhancing existing ones; (iv) building social capital in education and health, a precondition for women's empowerment; and (v) providing access to information and entertainment.

⁹³ Deloitte, 2020. [Nigeria Power Sector Program. Gender Mainstreaming in the Solar Home System Value Chain.](#)

⁹⁴ Deloitte, 2021. [Power Africa. Nigeria Power Sector Program. Social Inclusion and Diversity in Nigeria's Power Sector.](#)

However, in Africa access to electricity is limited due to the absence of sufficient energy infrastructure, and even where such infrastructure is physically available the stability of the grid is often vulnerable. Furthermore, marginalised groups, including women and poor people, are often hindered in their use of modern clean energy by lack of finance, appliances, information, and training or education.⁹⁵

According to a World Bank report, Sub-Saharan Africa remained the world's least electrified region in 2020 and accounted for 77% of the global population lacking access to electricity. More than one third of REPP 2 target countries are among the top 20 countries with the largest deficit in energy access: Nigeria (92 million people), DRC (72 million), Niger (20 million) and Madagascar (18 million), with Nigeria and DRC being world leaders in the total number of unserved people.

While availability of country-level gender-disaggregated data on electricity access in Sub-Saharan Africa is limited and it is difficult to assess the extent to which lack of electricity access can be attributed to women in male-headed households, studies have nonetheless shown that women-owned households and enterprises find it harder to access modern energy.⁹⁶ The reasons for existing disparities in access can be attributed to the traditional distribution of gender roles, as well as differences in access to finance. Funding schemes for electricity connections that take gender into account have been known to significantly increase overall connection rates.⁹⁷

Country-level studies have been performed for two REPP 2 target countries – Niger and Zambia – yielding different results for each. In Niger, female-headed households were less likely to have electricity access, either through grid-connection or off-grid solutions, as presented in Table 21. In Zambia, female-headed households were more likely to have electricity access through grid connection in both rural and urban areas, while male-headed households were more likely to utilize off-grid solutions. At the same time, more female-headed households (93.1%) did not have electricity access at all in Zambian rural areas compared to male-headed households (86.7%) but were slightly more likely to have electricity access in urban areas than male-headed households, as shown in Table 23 below.

Table 21. Electricity access by gender of the household head in Niger⁹⁸

Niger	No electricity (%)		Grid access (%)		Off-grid access (%)	
	Female	Male	Female	Male	Female	Male
Nationwide	84.4	80.2	15.9	15.78	0.7	4.0
Urban	59.2	49.5	38.2	46.1	2.6	4.4
Rural	93.0	88.8	7.0	7.3	0	3.9

Table 23. Electricity access by gender of the household head in Zambia⁹⁹

Zambia	No electricity (%)		Grid access (%)		Off-grid access (%)	
	Female	Male	Female	Male	Female	Male
Nationwide	57.9	57.5	40.7	36.8	1.5	5.7
Urban	23.7	24.1	75.5	74.5	0.8	1.4
Rural	93.1	86.7	4.7	3.9	2.1	9.4

⁹⁵ World Bank, 2017. [Energy Access and Gender: Getting the Right Balance](#).

⁹⁶ IRENA, 2022. [Renewable Energy Market Analysis. Africa and Its Regions](#).

⁹⁷ IRENA, 2022. [Renewable Energy Market Analysis. Africa and Its Regions](#).

⁹⁸ World Bank, 2020. [Niger. Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework](#).

⁹⁹ World Bank, 2019. [Zambia. Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework](#).

The primary reasons for the lack of electricity connectivity also differed between the countries. In Niger, for both female- and male-headed households, the main reason for not being connected to the grid was that the grid was too far away or not available (82.4% and 87.4%, respectively), as presented in Table 24. This is equally an issue for female- and male-headed households in rural areas. However, it is more likely that female-headed households will be unable to pay for the connection cost compared with their male counterparts (8.8% versus 6.2%). This is especially true in urban areas, where 31.7% of female-headed households reported the connection cost as the main issue, compared with 23.6% of male-headed households.¹⁰⁰

In Zambia, both male-headed households and female-headed households identified “grid too far or unavailable” as the main reason they were not connected to the grid. At the same time, as presented in Table 26 below, financial burden of interconnection cost was a bigger barrier for female-headed households than for male-headed households.

Table 24. Niger. Barriers to gaining access to the electricity grid in Niger¹⁰¹

	Grid too far	Expensive connection cost	Complicated administrative procedure	Monthly fee too expensive	Renting, landlord's decision	Applied, waiting for connection	Other
Female	82.4	8.8	2.7	1.6	1.5	0.6	2.4
Male	87.4	6.2	3.7	0.4	1.2	0.6	0.6

Table 26. Zambia. Barriers to gaining access to the electricity grid in Zambia¹⁰²

	Connection cost too expensive	Grid too far	Renting, landlord's decision	Other
Female	42.1	48.8	5.2	3.8
Male	36.7	53.7	6.2	3.4

Regarding cooking technologies, higher disparities were present in Niger, where female-headed households were slightly more likely to use three-stone stoves and less likely to use traditional stoves¹⁰³ compared to male-headed households, while there was no gender gap in the use of LPG stoves. In Zambia, male-headed households and female-headed households used similar cooking technologies.

At the same time, in both countries women aged 15 and older spent considerably longer cooking or in the cooking area (more than two hours per day in Niger and on average 10.6 hours per week in Zambia) than either men, girls, or boys. Women are thus much more likely to be affected by indoor air pollution.

The above statistics supports aggregated global and regional findings that limited electricity access and the reliance on solid fuels for cooking and other household energy-demanding chores – as well as informal economic sector activities – has a higher toll on women, contributing to their “time poverty” and health issues.¹⁰⁴

¹⁰⁰ World Bank, 2020. [Niger. Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework.](#)

¹⁰¹ World Bank, 2020. [Niger. Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework.](#)

¹⁰² World Bank, 2019. [Zambia. Beyond Connections: Energy Access Diagnostic Report Based on the Multi-Tier Framework.](#)

¹⁰³ For classification of cookstoves in Niger, a traditional stove is defined as basic wood stoves, charcoal burning stoves, and mineral coal stoves fall within this category. The pot sits for the most part on the fuel, has a low combustion temperature due to poor insulation, and a lot of cold excess primary air because of too many openings.

¹⁰⁴ UNDP, 2016. [Policy Brief. Gender and sustainable energy.](#)

Increasing both genders' access to modern energy services has the capacity to (i) help women save time by substituting manual labour and reducing workloads, (ii) reduce indoor air pollution and enable other health benefits for women and children, (iii) improve education by enabling children to study and women to have greater flexibility in the organisation of their work patterns, (iv) boost income generation, and (v) offer social benefits such as increased funding of households' well-being and the education of children, including girls, as a result of the increased income of female members of the household, who reinvest most of their income in their families.¹⁰⁵

4.3 Limited funding to women-owned/led companies

Africa has a higher share of female-owned businesses than other regions, yet women entrepreneurs still face multiple challenges accessing the finance and support they need to grow their businesses.¹⁰⁶ There is an estimated USD 42 billion financing gap for African women entrepreneurs across business value chains.¹⁰⁷ In 2021, less than 1% of funding raised by start-ups in Africa went to female-only teams, and only 18% to teams that received funding were considered gender-diverse.¹⁰⁸

Based on Global Banking Alliance's review of various government and private sector financing models, there are only a few examples of programmes geared specifically to increase access for women owned micro, small and medium enterprises. In 2017, only 16-20% of women in West, Central, East and Southern Africa had access to long-term financing from financial institutions.¹⁰⁹

This is further confirmed by research undertaken by the Investment Manager, which suggests that currently there are no dedicated funding facilities that would directly target: 1) African women entrepreneurs in the clean energy space or 2) businesses that specifically address women's needs through their business model design.

¹⁰⁵ World Bank, 2017. [Energy Access and Gender: Getting the Right Balance](#).

¹⁰⁶ Global Entrepreneurship Monitoring, 2019. [Women's Entrepreneurship Report](#).

¹⁰⁷ AfDB, 2017. [Unleashing women's entrepreneurship through strategic partnerships](#).

¹⁰⁸ Africa: The Big Deal.

¹⁰⁹ AfDB, 2017. [Unleashing women's entrepreneurship through strategic partnerships](#).

5 Gender Action Plan

REPP 2's investment mandate extends towards some of the most economically and socially vulnerable countries in Africa, with the majority (>50%) of the population living below the International Poverty Line in more than half of the target countries. The REPP 2 Gender Action Plan (GAP) summarises the actions that need to be implemented at the facility level to ensure gender equality considerations are integrated at the investment level.

As discussed above, gender gaps exist across key aspects of society – from economics and politics to health, education and involvement in the energy sector. Women in Sub-Saharan Africa find it hard to access modern energy, yet they are often burdened with domestic chores that prevent them from pursuing education or income-generating activities. Access to energy can help individuals, households and business to complete certain time-consuming tasks in more efficient manner. Access to electricity may free up time from fuel collection and household tasks for longer periods of study time or other productive use of time, decreasing gender gaps in key educational and economic areas. Also access to energy, through access to lights, information, mobile phones, are linked to increase perceived feeling of safety and security.¹¹⁰

Furthermore, there is limited funding available to female-owned or female-led businesses in the region's renewable energy sector. REPP 2 is taking direct action to increase the number of renewable energy companies that are aligned with 2X Challenge's Gender Lens Investing (2X GLI) criteria by allocating a portion (minim 30%) of the Programme's funding to these companies. And since it is imperative to designing and deliver products and services that better serve women's needs, REPP 2 also works with its investees to integrate gender equality perspectives into the design, implementation, monitoring and assessment of all of their renewable energy initiatives.

We believe that funding women, empowers them to be invested and build companies, teams, products and markets they see transformative for the generations to come. This is a meaningful step to overcome the gender equality gap.

5.1 Actions

Two main actions have been identified to bridge the gender equality gap:

Action 1: Integrate a gender equality perspective into the design, implementation, monitoring and assessment of all renewable energy activities financed by REPP 2. This means requesting all investees to put in place sub-project level GAPs (Activity 9: Non-financial support to promote gender equity) with actionable targets to improve gender equality within their operations and project implementation will positively impact the industry and its beneficiaries. This involves:

1. Analysing gender equality in the investee company, country of operation, and targeted sector;
2. Establishing an investee-specific GAP by identifying gender performance indicators and gender-disaggregated targets against an established baseline; and
3. Integrating the outcomes of (1) and (2) into a monitoring plan.

REPP 2 will provide investee companies with supporting in assessing gender baseline and establishing action plan. Such support includes:

¹¹⁰ Supported by an initial project-level data collection conducted in Goma DRC:
https://static1.squarespace.com/static/5cc327c55acb9b00019d816a/t/63d16fc81da3f71c853b105a/1674670025093/Assessing+the+Peace+effects+of+PRECS_Final_Goma_June+2022_updated.pdf

The GAP submitted by investee companies together with the Environmental and Social Management System (ESMS) prior investment reaching financial close. Therefore, both ESMS and GAP are conditions precedent prior funding for construction funding (activity 4 and 5) and expansion funding (activity 3).

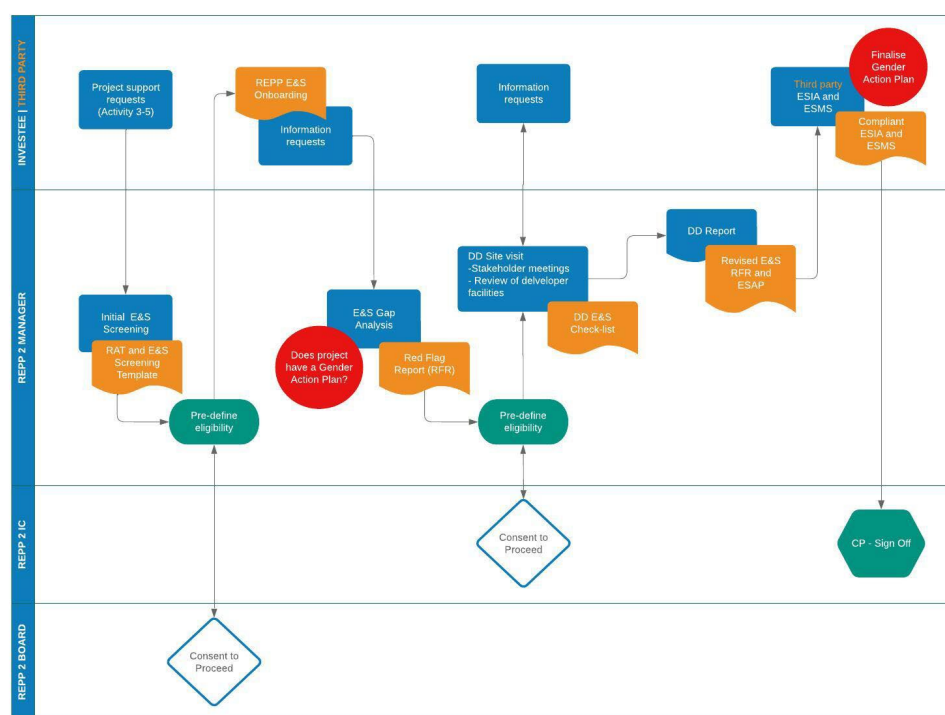


Figure 2. Gender sensitivity screening for construction and expansion funding.

Action 2: Dedicate a target of 30% of all funding towards companies that meet 2X GLI criteria. These are companies that:

- Are founded by a woman or women or are majority-owned by women; and / or
- Have at least 30% women in senior management; and / or
- Provide products or services designed to disproportionately benefit women; and / or
- Have a workforce that is at least 30% women, together with a progressive human resource policy.

The funding can be in from of early-stage growth funding for off-grid companies (activity 1), development-stage funding for grid-connected projects (activity 2), finance for expansion of SHS businesses (activity 3), finance for construction of mini-grid portfolios (activity 4) or gap funding for construction of grid-connected projects (activity 5) and can be in from of debt, equity or hybrid instruments, subject to the company's funding needs.

The above actions are further defined in Table 25 along with indicators that will be monitored and the expected targets. For ex-ante target setting the simple assumption has been made that 30 % 2X GLI aligned funding will lead to 30 % of female owned and led companies benefiting, 30 % of jobs being dedicated to women and 30 % end beneficiaries at household and business level to be women in line with the 2X GLI Criteria. This is anticipated to be a conservative assumption as at household level female to male ratio is typically 50 -50.

In line with REPP2 monitoring plan, gender indicators, actions and results are monitored and reviewed and where relevant this action plan or its underlying actions and targets are updated accordingly.

Table 25. REPP 2 Gender Action Plan

Activities	Indicators and targets	Timeline	Responsibilities	Costs
<u>Impact statement:</u> REPP 2 will aim to create a paradigm shift by leveraging private sector funding to promote investments in small-scale renewable energy projects in LDCs and other sub-Saharan African countries. As result of this, local populations of some of the world's most vulnerable communities, including women and girls, will have improved access to clean energy, improved access to climate resilient solutions and information, and the Host Countries will transition towards low-emission resilient development pathways, meeting their NDC targets. Accordingly, with increased investment, the GHG emissions from electricity generation will be reduced, climate resilience of the energy infrastructure is enhanced, and climate resilience of vulnerable people and communities will be improved. <u>Outcome statements:</u> Outcome statement 3: Improved climate resilience for vulnerable people and communities, through access to clean energy solution which improves resilience and builds gender equity. Outcome statement 4: Increased private sector investment in renewable energy projects that enables local developers, builds gender equity and drives a market for smaller-scale renewables. Outcome statement 5: Contributing to stronger technology deployment, innovation and market transformation				
<u>Outputs statements:</u> Output 1: Increased commercial viability All REPP 2 activities, including early-stage growth and development funding, gap funding for construction and support for market development combine to demonstrate the potential for technical and commercial viability of smaller-scale renewable energy businesses and projects including female owned and led businesses in Host Countries and across the region. Output 2: Increased renewable energy capacity and generation Early-stage and construction funding from REPP 2 ensures that new renewable energy capacity and generation is realised. Renewable energy capacity and generation financed by REPP 2 contributes directly to improved access to clean electricity for women and men and host-country priorities as outlined in their policy frameworks as well as their Nationally-Determined Contributions and National Adaptation Plans. Output 3: Improved skills and capacity REPP 2 support for project developers establish a more active and commercially-viable market for smaller-scale renewables, which is better-able to attract both public and private finance in future. These impacts accrue to Host Countries and more broadly throughout sub-Saharan Africa, with a demonstration impact for other emerging markets and with developers being better placed to expand to those markets. REPP 2 will directly contribute to improved gender equity through a focus on gender mainstreaming within investee projects and businesses.				
Action 1: Integration of a gender equity perspective into the design, implementation, monitoring and assessment of all renewable energy activities.				
Action 1 is aligned with the below activities:	Number of investee companies with an approved gender action plan (GAP (estimated at 100% ex ante):	Targets ¹¹¹ are presented for:	Investment Manager will be responsible for reviewing and approving	Estimated USD 10,000 - 35,000 per project for the gender analysis and

¹¹¹ Annual targets are presented in annex 22 in the REPP2 targets tab, country specific targets (also for each year) are presented in annex 22 in the underlying country tabs.

Activities	Indicators and targets	Timeline	Responsibilities	Costs
- Activity 1: Provide early-stage growth funding for off-grid companies (to fund GAP development, where relevant) - Activity 2: Provide development funding for grid-connected projects (to fund GAP development, where relevant) - Activity 8: Improve environmental and social risk mitigation - Activity 9: Non-financial support to promote gender equity (sub-project level GAPs)	15 companies 37 companies The following additional evidence of its implementation must be provided ex-post: - Number of investee company employees who have completed gender awareness and skills training. - Number of investee companies who have designed products and services specifically to support women and girls, and/or other vulnerable groups. Number of ESIA processes designed and implemented to consult with a range of vulnerable groups incl. women and girls (estimated at 100% ex ante): 15 companies 37 companies	mid-term (4 years) and - final term (7 years) of the fund. The first target value provided under each indicator if for mid-term, the second for final term.	the GAPs prepared by investees. Investees will be responsible for the implementation of GAPs and gender-disaggregated monitoring and reporting.	implementation of action plan. Estimated USD 25,000-75,000 per project for the ESIA (which will include a gender expert).
Action 2: A target of 30% of the underlying activity funding will be dedicated towards companies that meet the 2X GLI criteria				
Action 2 is implemented through following activities: - Activity 1: Provide early-stage growth funding for off-grid companies - Activity 2: Provide development funding for grid-connected projects - Activity 3: Finance expansion of SHS businesses - Activity 4: Finance construction of mini-grid portfolios - Activity 5: Provide gap funding for construction of grid-connected projects	Beneficiaries reached, disaggregated by gender: - Total beneficiaries: 291,939 - Total female beneficiaries: 116,350 - Total beneficiaries: 729,847 - Total female beneficiaries: 290,875 Beneficiaries with improved access to lighting: - Total beneficiaries: 1,490,336 - Total female beneficiaries: 447,101 - Total beneficiaries: 3,725,840 - Total female beneficiaries: 1,117,752 Beneficiaries with improved access to information: - Total beneficiaries: 730,250 - Total female beneficiaries: 219,075 - Total beneficiaries: 1,825,625 - Total female beneficiaries: 547,687	Targets¹¹² are presented for: - mid-term (4 years) and - final term (7 years) of the fund. The first target value provided under each indicator is for mid-term, the second for final term.	The REPP 2 General Partner will be asked to approve a funding target in line with 2X GLI criteria. Investment Manager will be responsible for committing funding in line with 2X GLI criteria. Investees will be responsible for gender disaggregated monitoring and reporting.	REPP 2 staff and resources

¹¹² Annual targets are presented in annex 22 in the REPP2 targets tab, country specific targets (also for each year) are presented in annex 22 in the underlying country tabs.

Activities	Indicators and targets	Timeline	Responsibilities	Costs
	Beneficiaries adopting improved livelihoods and/or new climate resilient livelihood options, through resilient PUE options: - PUE customers: 11,061 - PUE female customers: 3,318 - PUE customers: 27,652 - PUE female customers: 8,295 USD funding committed to 2X GLI criteria-aligned companies: - USD 28 million - USD 70 million Number of jobs disaggregated by gender: - 9,216 female jobs - 23,040 female jobs			

Appendix 1

Table 26. List of organisations working locally with issues regarding GBV and SAH.

Following is a list of organisations which are in different ways working with issues regarding GBV and gender equality. The list includes international organisations as well as businesses that only work on a regional level. The primary objective of each organisation is to teach and inform about inequality, as well as give information on how to approach the subject. The organisations are suggested to be contacted by investee companies to assist with project level gender baseline setting, stakeholder engagement as well as establishing gender sensitive grievance mechanism. This list is not exclusive, and investee may consider partnering with a different organisation working on gender equality issues in the project community.

Country	Organisations working on gender equality and GBV issues	URL or contact details
Cameroon	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN) http://www.anppcan.org/	http://www.anppcan.org/
	International Republican Institute https://www.iri.org/wdn/?utm_source=www.wdn.org&path=%2F	https://www.iri.org/wdn/?utm_source=www.wdn.org&path=%2F
	UN Women https://africa.unwomen.org/en/news-and-events/stories/2018/11/gender-based-violence-in-cameroon	https://africa.unwomen.org/en/news-and-events/stories/2018/11/gender-based-violence-in-cameroon
	WFAC Cameroon https://wfaccameroon.org/about-wfac/	https://wfaccameroon.org/about-wfac/
DRC	Peace Women http://peacewomen.org/content/drc-reducing-gender-based-violence-drc-through-behaviour-change-strategies	http://peacewomen.org/content/drc-reducing-gender-based-violence-drc-through-behaviour-change-strategies
	UN Women https://africa.unwomen.org/en/news-and-events/stories/2020/09/drc-takes-a-step-towards-zero-tolerance-against-gender-based-violence	https://africa.unwomen.org/en/news-and-events/stories/2020/09/drc-takes-a-step-towards-zero-tolerance-against-gender-based-violence

Country	Organisations working on gender equality and GBV issues	URL or contact details
Lesotho	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN)	http://www.anppcan.org/ http://www.anppcan.org/
	Together for Girls https://www.togetherforgirls.org/en/countries/lesotho	https://www.togetherforgirls.org/en/countries/lesotho
	UN Women https://africa.unwomen.org/en/where-we-are/eastern-and-southern-africa/south-africa-mco	https://africa.unwomen.org/en/where-we-are/eastern-and-southern-africa/south-africa-mco
Madagascar	UNICEF https://www.unicef.org/madagascar/en	https://www.unicef.org/madagascar/en
	Women Strong International https://www.womenstrong.org/	https://www.womenstrong.org/
Malawi	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN)	http://www.anppcan.org/
	Together for Girls https://www.togetherforgirls.org/en/countries/malawi	https://www.togetherforgirls.org/en/countries/malawi
	UNICEF	https://www.unicef.org/malawi/
	UN Women https://africa.unwomen.org/en	https://africa.unwomen.org/en https://www.unicef.org/malawi/
Niger	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN)	http://www.anppcan.org/
	UN Women https://africa.unwomen.org/en	https://africa.unwomen.org/en
Nigeria	ActionAid https://nigeria.actionaid.org/	https://nigeria.actionaid.org/
	Project Alert on Violence Against Women https://projectalertnig.org/	https://projectalertnig.org/
	Women at Risk International Foundation (WARIF) https://warifng.org/	https://warifng.org/
	Women Empowerment and Legal Aid (WELA) https://welaonline.org/	https://welaonline.org/

Country	Organisations working on gender equality and GBV issues	URL or contact details
Sierra Leone	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN)	http://www.anppcan.org/
	Men's Association for Gender Equality https://www.menengagesierraleone.org/	https://www.menengagesierraleone.org/
	UNFPA https://sierraleone.unfpa.org/en	https://sierraleone.unfpa.org/en https://sierraleone.unfpa.org/en
	UN Women	https://africa.unwomen.org/en
Zambia	African Network for the Prevention against Child Abuse and Neglect (ANPPCAN)	http://www.anppcan.org/ Telephone: +260 972006717
	Girl Empowerment Alliance for Change https://www.facebook.com/GEAC2007/	Email: girl empowermentalliance4change@gmail.com
	Together for Girls https://www.togetherforgirls.org/en/countries/zambia UNICEF https://www.unicef.org/zambia/	https://www.togetherforgirls.org/en/countries/zambia https://www.unicef.org/zambia/

REPP Academy 2021

Session 8:

**Aligning mini-grid development with the 2X Challenge
Gender Lens Investment criteria (GLI)**

Inka Schomer

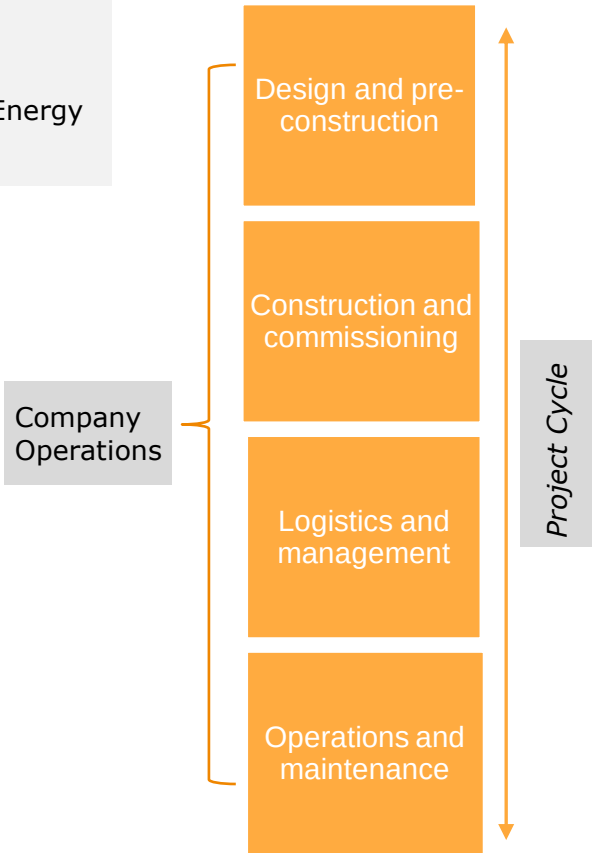
Wednesday 13 October, 12:15-13:30

Overarching Questions on the Table

Themes:

- Women’s Employment
- Understanding Customers and Productive Uses of Energy
- Risk Management: Gender-Based Violence (GBV)

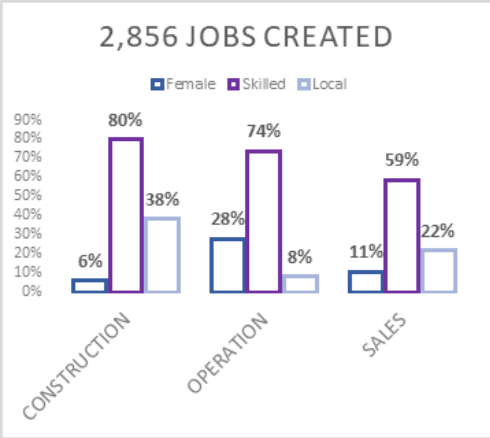
- Do you feel like you have enough knowledge about gender equality and how it links to mini-grids and your company?
- Do you think mini-grid companies, like your own, can play a role in addressing gender equality issues? Please reflect on any ideas and thoughts.
- Does your company have any existing commitments to gender equality? If so, what are they?
- Have you considered monitoring gender equality at the company level or community/site level? If so, what parameters have you considered? If not, why not?
- Do you collect any sex-disaggregated data? If yes, what data do you collect? In what areas would you like to expand your data collection efforts?
- Apart from reporting to lenders/investors, what have you used this type of data for?



Women's Employment

Women have an under-recognized role as energy consumers, producers and as energy entrepreneurs. This provides a good foundation for their engagement in mini-grid operations. The business case for women's employment can be made both at the HQ and the community level:

- **Performance:** Firms with greater diversity and presence of female leaders have exhibited stronger performance and return on investments.
- **Innovation:** Companies with a larger footprint of female employees are likely to be more innovative (and bring those ideas to market). Women score as well as men in abilities linked to innovation (e.g. motivating others and inspiring their team).
- **Improved Service Delivery:** Workforce diversity roots a company in its customer base and can improve service quality.
- **Safer Operating Environment:** Diverse teams can enhance the safety culture and ensure protocols are followed.
- **Community buy-in:** Enhanced community engagement and employment can create greater social license to operate.



Mrinda's community mini-grid deployment in India created approximately 986 jobs. These jobs include 180 direct full-time jobs, 131 full-time equivalent jobs from contractual work, and 675 productive-use jobs through additional entrepreneurial activities.



Opportunities to enhance talent pool and enhance equality



Women's Employment



Education

- Increase girls and women's exposure to the energy sector (e.g. on-site internship)
- Mentoring and connection to role-models
- Changing the study environment, and looking at bias in curricular and in teaching
- Working with parents



School-to-work transition

- Designing vocational programs
- Engaging women through school-to-work transition career choices and through exposure to the sector
- Job advertisements: looking at language and company image
- Engaging women's professional networks
- Assess bias in recruitment process and engaging men
- Setting quotas



Retention

- Addressing biases in workplace and work culture (e.g. policies)
- Accommodating care responsibilities (e.g. travel distances, working hours)
- Paid family leave
- Offering appropriate workplace facilities
- Address gender wage gaps
- Sexual harassment



Advancement

- Standardizing job performance reviews and skills assessments
- Offering effective career progression training
- Ensuring women leaders have sponsors
- Addressing and discussing discrimination

Image sources: Kalahari GeoEnergy, LTDVItalia Burundi, Microsoft Stock Images



managed by  camco
CLEAN ENERGY



Women's Employment

- What recruitment strategies have you considered for your company?
- What are the kinds of gender gaps you have observed (e.g. who is employed in technical roles or in leadership?)
- Have you noticed any cultural or societal norms that exist about the role of women versus men at the project site? If so, are you able to share any details and your observations?

Additional questions:

- Does your company promote a positive working environment for women? What would enhance the work environment for women?
- Do you have any company policies or benefits that are designed to enhance women's employment (e.g. sexual harassment policies, flexible working hours, safer transport, upskilling opportunities etc.)?
- What could your organization be doing to encourage women to work at your company? Have you engaged employees on the issue? What have they said?
- What type of interventions that were mentioned above could work best at your company internally to address the bottlenecks to women's employment?
- Are women perceived to have the right skill sets? What could be done to address skills gaps, if any?
- What could you do differently to provide opportunities at the community level?



Resources

- ICRW. 2021. Off-Grid Energy Gender Opportunities Explorer
- IRENA. 2019. Renewable Energy. A Gender Perspective
- GWNET. 2020. Women for Sustainable Energy Strategies to Foster Women's Talent for Transformational Change
- IFC. 2018. A Toolkit of Actions and Strategies for Oil, Gas, and Mining Companies: Increasing Gender Diversity from the Workforce to the Boardroom
- Schomer, I and Hammond, A. 2020. Stepping Up Women's STEM Careers in Infrastructure



Understanding Customers and Productive Uses of Energy

Mini-grid companies have an inherent interest in local consumers and the utilisation of locally-generated power.

With access to energy services, local entrepreneurs, micro-enterprises, and farmers can generate local jobs and income, and enhance livelihoods. However, given inherent gender-gaps at the community level, specific attention needs to be paid to ensure productive use of energy gains are equitably distributed.

Barriers to equitable distribution of benefits include:

- Women have inequitable access to resources, such as land, collateral, and finance, indicating a need for tailored approaches (e.g. mobile money, subsidies, ready boards).
- Women are involved in different enterprises and access to livelihoods given cultural expectations and norms.
- Cultural norms, limited participation in community and household decision-making.
- Women have fewer opportunities to attend trainings and have less access to information given different mobility patterns.
- Unequal labor burden in terms of household and care work (unpaid) and fuelwood collection which leads to time poverty

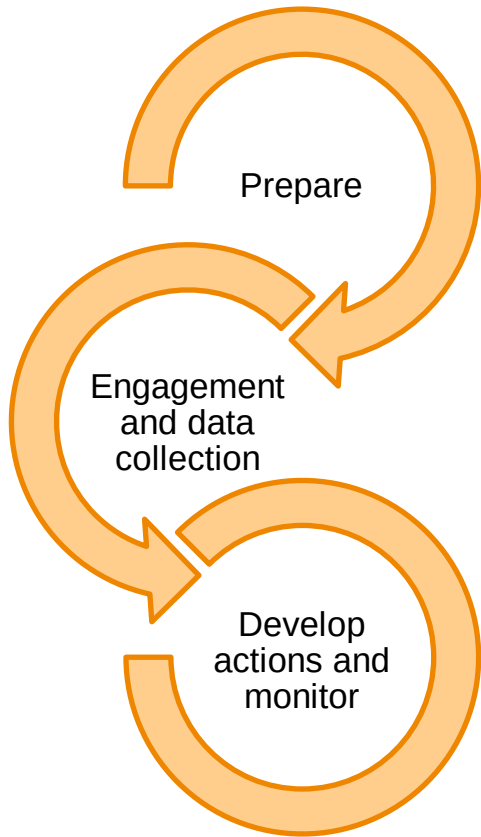
A deeper understanding of the women's segment can help a company increase its customer base and market size, enhance loyalty and manage risks (e.g. payment, community ownership).



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Understanding Customers and Productive-Uses of Energy



- Staffing and budgeting for understanding customers
- Who to engage (e.g. female famers, business owners, male partners, service providers, local officials, and partners)
- How to engage (e.g. FGD, interviews, community mapping, and roundtable (including tools))
- What to engage on (e.g. energy needs and value chains, willingness to pay, access to finance etc.)
- When to engage (e.g. times, location, partner consent, season etc.)

- Finalize specific research approach, methods, and tools to undertake data collection
- Test and finalize data collection tools (questionnaires, surveys) to collect data
- Areas you may consider collecting data on include:

- Women’s role in key sectors and value-chains (e.g. agribusiness) and in the households
- Women’s energy needs to do current tasks and reduce labor/costs and meet energy use aspirations
- Investigate affordability constraints and ability to pay (e.g. seasonal income with crop harvests etc.)
- Access to finance (for connection, services and appliances adoption)
- Access to information and business development opportunities
- Identify key stakeholders that women trust or that could be engaged to form partnerships

- Identification of key gender gaps with regards to demand and supply gives company detailed description of possible issues to address
- Verify findings at the company level through workshops/discussions
- Commit to company action, set targets, allocate budget, and monitor progress



Understanding Customers and Productive Uses of Energy

- Could you share any observations or experiences on engaging male versus female customers (household consumers, enterprises or farmers) in service provision?
- Who has access to energy technologies and services? What kind of businesses are men versus women involved in at the project sites you have worked at?
- Gender gaps are often apparent in productive uses of energy. What actions could close this gap at the mini-grid site level?

Additional questions:

- Do you incorporate a different approach in engaging female and male customers in your business model? If so, how? If not, why not?
- In your outreach to new customers, have you developed different approaches to understand and target women and men in, for example, marketing, information, or customer feedback?
- What changes could be made around the requirement to gain access to energy services (e.g. the upfront connection fee/down payment, documentation, price point and timeline for payment for connection?)
- Do you currently partner with any other stakeholders (e.g. female-leaders, women's groups, NGO's etc.) to enhance outreach to female customers?



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Customer Engagement and Productive Uses of Energy

Resources

- Green-Energy Help Desk. 2020. Module 2: Mini-grid Community Engagement
- World Bank. 2019. Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers
- USAID. 2018. Guide to Community Engagement for Power Projects in Kenya
- Eales. A. 2018. Social Impacts of Mini-grids: Towards an Evaluation Methodology
- World Bank. 2017. Mini-Grids and Gender Equality: Inclusive Design, Better Development Outcomes
- Aggarwal, V. et al. 2014. Rural Energy Alternatives in India: Opportunities in Financing and Community Engagement for Renewable Energy Microgrid Projects



Risk Management: Gender-based Violence

Gender-based violence (GBV) is an umbrella term for any harmful act perpetrated against a person based on socially ascribed (i.e. gender) differences between males and females. It includes acts that inflict physical, sexual, or mental harm or suffering, threats of such acts, coercion, and other deprivations of liberty. These acts can occur in public or in private spaces. Common forms of GBV include **sexual violence** (rape, attempted rape, unwanted touching, sexual exploitation, and sexual harassment), **intimate partner violence** (also called domestic violence, including physical, emotional, sexual, and economic abuse), **forced and early marriage, and female genital mutilation**. (IASC 2015).

Violence against women is a manifestation of historically unequal power relations between men and women, which have led to domination over, and discrimination against, women by men and to the prevention of the full advancement of women (UN Declaration on the Elimination of Violence Against Women 1993).

Workplace Sexual Harassment	Sexual Exploitation and Abuse (SEA)
• Unwanted sexual advances • Request for sexual favors • Sexual physical contact	• Exploitation of a vulnerable position, differential power or trust for sexual purpose • Actual or threatened sexual physical intrusion

Source: WBG Good Practice Note- Environmental & Social Framework for IPF Operations: Addressing Sexual Exploitation and Abuse and Sexual Harassment (SEA/SH) in Investment Project Financing Involving Major Civil Works



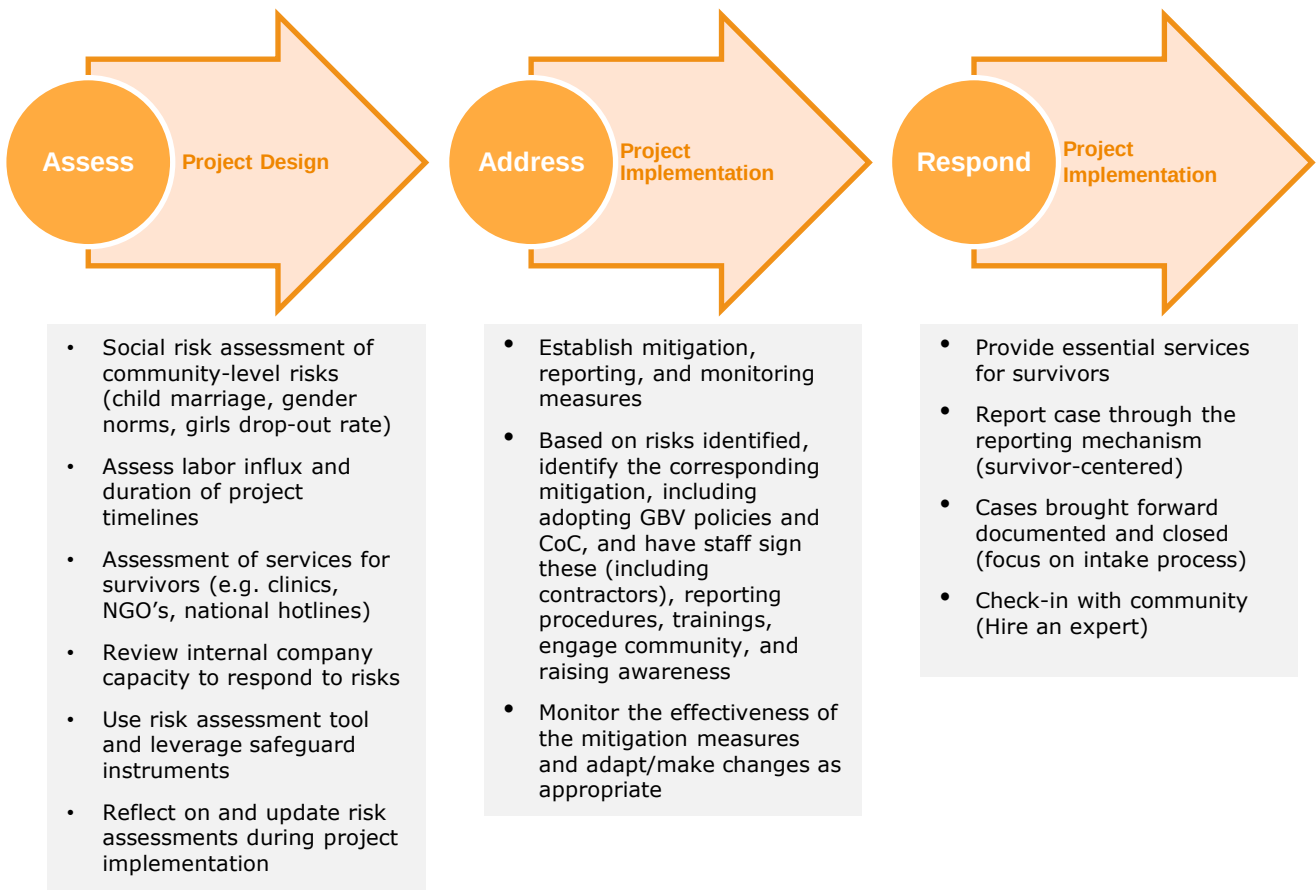
Company and Project-Site Risk



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Risk Management: Gender-based Violence



Risk Management: Gender-Based Violence

- *Can you share any reflections on the possible GBV risks involved in mini-grid construction and operation? Do you think the risks of GBV are increased?*

Additional questions:

- How long will construction take? What is the level of labor hired for this project? Are employees transient and working away from families?
- Have discussions taken place with community and local groups that focus on for the rights of women, children, and adolescents at the project site?
- Are community members likely to provide goods or services to workers and contractors (e.g. entertainment venues, food, materials)?
- Are there any schools near the project site? If yes. What age are students?
- Have you as a company put in place any GBV mitigation and response measures?
- What will be the supervision arrangement at site during construction? How hard is supervision (e.g. remote, in a FCV location etc.)?
- Is there anybody with a background in GBV that could assists with the GBV related work in your company?



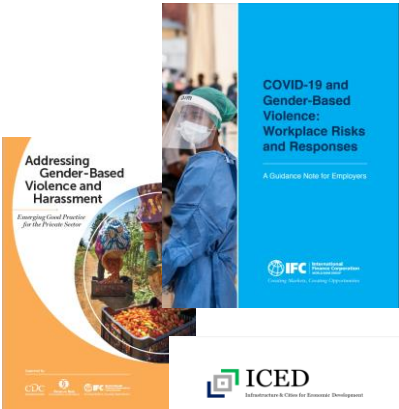
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Risk Management: Gender-Based Violence

Resources

- IFC. 2020. COVID-19 and Gender-Based Violence Workplace Risks and Responses
- EBRD, IFC and CDC Group, Addressing Gender-Based Violence and Harassment: Emerging Good Practice for the Private Sector. 2020. Addressing Gender-Based Violence and Harassment Emerging Good Practice for the Private Sector
- Infrastructure and Cities for Economic Development's (ICED). 2019. Sexual Exploitation, Abuse and Harassment in Infrastructure Tool
- World Bank. 2019. Good Practice Note: Environmental & Social Framework for IPF Operations: Addressing Sexual Exploitation and Abuse and Sexual Harassment (SEA/SH) in Investment Project Financing Involving Major Civil Works
- USAID. 2015. Toolkit for Integrating GBV Prevention and Response into USAID Energy and Infrastructure Projects



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